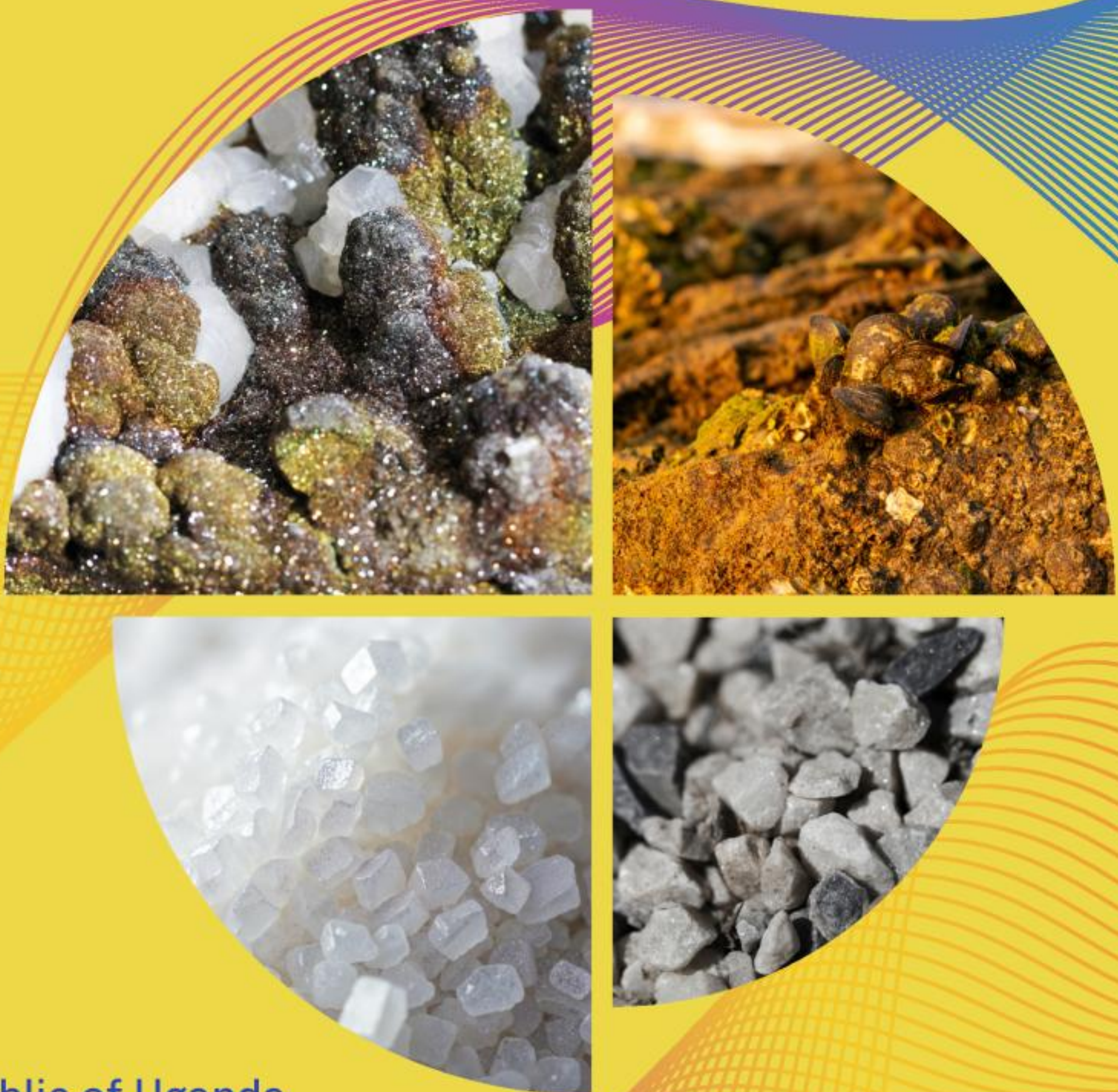
National Planning Authority

PHOSPHATES VALUE CHAIN ANALYSIS REPORT



Republic of Uganda

Government of the Republic of Uganda

Sustainable Extractives Industry Development Programme of the Fourth National Development Plan (NDPIV)

Value Chain Analysis Report for Phosphates

January 2026

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List of Abbreviations and Acronyms

Abbreviation	Full Meaning
ANRM	Agronomic Nutrient Requirement Model
BoU	Bank of Uganda
CAPEX	Capital Expenditure
CIF	Cost, Insurance and Freight
COMESA	Common Market for Eastern and Southern Africa
CSA	Climate-Smart Agriculture
DAP	Di-Ammonium Phosphate
DC	Development Committee
DCP	Dicalcium Phosphate
EDPM	Econometric Demand Projection Model
EAC	East African Community
EIRR	Economic Internal Rate of Return
ENPV	Economic Net Present Value
ESG	Environmental, Social and Governance
ESIA	Environmental and Social Impact Assessment
EU	European Union
FIRR	Financial Internal Rate of Return
FNPV	Financial Net Present Value
FOB	Free on Board
Foskor	Foskor (South Africa)
FX	Foreign Exchange
GBV	Gender-Based Violence
GHG	Greenhouse Gas
IRR	Internal Rate of Return
JV	Joint Venture
JPMC	Jordan Phosphate Mines Company
MAAIF	Ministry of Agriculture, Animal Industry and Fisheries

Abbreviation	Full Meaning
MAP	Mono-Ammonium Phosphate
MDAs	Ministries, Departments and Agencies
MEMD	Ministry of Energy and Mineral Development
MoFPED	Ministry of Finance, Planning and Economic Development
MoWT	Ministry of Works and Transport
MCP	Monocalcium Phosphate
NDP IV	Fourth National Development Plan
NEMA	National Environment Management Authority
NMC	National Mining Company
NPA	National Planning Authority
NPF	National Planning Framework
NPV	Net Present Value
NPK	Nitrogen–Phosphorus–Potassium Fertiliser
OCP	Office Chérifien des Phosphates (Morocco)
OPEX	Operating Expenditure
PA	Phosphoric Acid
P_2O_5	Phosphorus Pentoxide
PFS	Pre-Feasibility Study
PIM	Public Investment Management
PPP	Public–Private Partnership
QA/QC	Quality Assurance / Quality Control
R&D	Research and Development
ROM	Run-of-Mine
SA	Sulphuric Acid
SSP	Single Super Phosphate
STPP	Sodium Tripolyphosphate
TSP	Triple Super Phosphate
UIA	Uganda Investment Authority
UNBS	Uganda National Bureau of Standards

Abbreviation	Full Meaning
V2040	Uganda Vision 2040
VCA	Value Chain Analysis
WDI	World Development Indicators
ZLD	Zero Liquid Discharge

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Executive Summary

Purpose and Context

The Government of Uganda commissioned this Value Chain Analysis (VCA) to assess the **strategic, commercial, and policy viability of developing a domestic phosphate-to-fertiliser industry anchored on the Sukulu phosphate resource**. The study responds to Uganda's persistent dependence on imported fertilisers, rising exposure to global price volatility, and low fertiliser application rates that constrain agricultural productivity, food security, and income growth.

The analysis is aligned with the **Fourth National Development Plan (NDP IV)**, Uganda Vision 2040, and the Tenfold Growth Strategy, all of which prioritise **agro-industrialisation, import substitution, value addition to natural resources, and private-sector-led industrial development**. The VCA is a **pre-investment diagnostic**, intended to inform **pre-feasibility and feasibility studies**, which then produce detailed engineering designs or financial structuring.

Key Problem and Strategic Opportunity

Uganda's agricultural sector is characterised by:

- ✓ low and uneven fertiliser use,
- ✓ high dependence on imported phosphate fertilisers,
- ✓ significant price volatility transmitted from global markets,
- ✓ long and costly logistics chains that inflate farm-gate prices.

At the same time, Uganda possesses a **known phosphate resource at Sukulu (Tororo)** that has remained underdeveloped for decades. Previous approaches focused narrowly on mining or export options and did not address the **full phosphate-to-fertiliser value chain**, limiting development impact.

The core opportunity identified by this study is to **transform Sukulu from a dormant mineral deposit into an integrated agro-industrial platform**, supplying fertilisers to the domestic and regional markets, stabilising prices, supporting agricultural productivity, and catalysing broader industrial development.

Scope and Methodological Approach

The VCA adopts a **mixed-methods, multi-layered analytical approach**, combining:

A. **Value chain mapping** of upstream (mining and beneficiation), midstream (chemical processing and manufacturing), and downstream (distribution and farm-level use) segments;

B. **Market and demand analysis**, using:

   - ✓ an **Agronomic Nutrient Requirement Model (ANRM)** to estimate technical phosphorus needs, and

- ✓ an **Econometric Demand Projection Model (EDPM)** to estimate market uptake based on income, credit, and price dynamics;

A. **Cost structure and competitiveness analysis**, focusing on mining-to-market cost build-ups, delivered-cost comparisons with imports, and product-level profitability;

B. **Benchmarking and case studies** from Morocco, Jordan, South Africa, India, and China;

C. **Constraint, risk, and opportunity diagnostics**, covering policy, institutional, environmental, social, gender, and climate-resilience dimensions;

D. **Scenario and strategic entry-point analysis**, leading to the identification of a preferred development pathway.

The methodology is bounded to avoid overlap with pre-feasibility and feasibility activities such as detailed plant engineering, financial modelling, and ESIA preparation.

Market Demand and Commercial Rationale

The combined ANRM and EDPM analyses provide a robust demand foundation for domestic phosphate fertiliser production.

- The **ANRM** confirms a **large agronomic phosphorus requirement**, reflecting Uganda's crop mix, soil characteristics, and yield targets.
- The **EDPM** shows that, even under conservative assumptions, **market-based fertiliser demand grows steadily to 2030 and 2040**, driven by income growth, improved access to credit, and price responsiveness.

The key insight is that **Uganda has sufficient and growing domestic demand to anchor local production**, provided fertilisers are competitively priced and reliably supplied. Domestic production therefore addresses both **supply-side constraints** and **demand realisation challenges**.

Cost Competitiveness and Value Addition

A central finding of the VCA is that **competitiveness in a landlocked economy is determined by delivered cost-to-market, not ex-factory cost**.

Imported fertilisers incur:

- ✓ ocean freight,
- ✓ port handling and demurrage,
- ✓ long-distance inland transport,
- ✓ inventory financing costs.

These layers typically add **30–50 percent** to international fertiliser prices by the time products reach Ugandan farmers.

Domestic production at Sukulu offers a structural advantage by:

- ✓ eliminating international shipping and port congestion costs,

- ✓ shortening inland logistics,
- ✓ enabling bulk handling and clustered utilities.

The analysis confirms strong competitiveness for:

- ✓ **SSP and TSP**, where bulkiness makes imports particularly uncompetitive;
- ✓ **DAP and blended NPKs**, as scale and integration improve.

Benchmarking and International Lessons

International experience reinforces the study's findings:

A. **Morocco (OCP)** demonstrates the power of full integration and delivered-cost focus.

B. **Jordan (JPMC)** shows the limits of export-led mining in the absence of strong domestic demand.

C. **South Africa (Foskor)** highlights beneficiation as a necessary but insufficient step without downstream integration.

D. **India** illustrates how blending and policy-enabled demand can rapidly scale fertiliser use.

E. **China** confirms that **industrial clustering** reduces costs, improves compliance, and builds resilient ecosystems.

The consistent lesson is that **phased integration, clustering, and policy coherence outperform export-only or fragmented models**.

Development Scenarios Assessed

Three development scenarios were analysed:

1. **Scenario 1: Phased Integrated Fertiliser Cluster (Preferred).** Mining, beneficiation, sulphuric acid, phosphoric acid, fertiliser manufacturing, blending, and circular-economy systems developed in phases.
2. **Scenario 2: Beneficiation and Superphosphate-First (Transitional).** Mining, beneficiation, sulphuric acid, and SSP/TSP production as an initial step, with an upgrade path to Scenario 1.
3. **Scenario 3: Mining and Export-Led (Non-Preferred).** Mining with limited beneficiation and bulk export of phosphate rock.

Preferred Scenario and Strategic Direction

The VCA clearly identifies **Scenario 1: Phased Integrated Fertiliser Cluster** as the preferred development pathway.

This scenario:

- ✓ aligns best with EDPM-confirmed demand growth,
- ✓ maximises domestic value addition and employment,
- ✓ offers the strongest delivered-cost competitiveness,

- ✓ provides a platform for regional leadership and industrial upgrading,
- ✓ allows risk to be managed through **phasing and modular expansion**.

Scenario 2 is retained as a **credible transitional option** where capital mobilisation or risk appetite requires staging, provided it is contractually locked into an upgrade pathway. Scenario 3 is retained only as a reference baseline and not as a development anchor.

Key Constraints and Risks

The study identifies binding constraints across the value chain:

A. **Upstream:** mine readiness, beneficiation capacity, infrastructure gaps;

B. **Midstream:** capital intensity of chemical processing, dependence on imported sulphur and ammonia, energy reliability;

C. **Downstream:** fragmented distribution, affordability constraints, weak quality enforcement;

D. **Policy and institutions:** fragmented mandates, unclear roles, limited coordination;

E. **Environmental and social:** water use, waste management, community engagement;

F. **Gender and climate:** unequal fertiliser access, climate-sensitive demand.

These are **manageable risks**, but only through coordinated intervention.

Opportunities and Strategic Entry Points

The Sukulu value chain presents multiple entry points:

A. **Upstream:** mining, beneficiation, shared infrastructure;

B. **Midstream:** sulphuric and phosphoric acid, fertiliser manufacturing, blending;

C. **Downstream:** storage, distribution, feed-grade and industrial phosphates;

D. **Cross-cutting:** PPPs, regional exports, circular economy solutions.

These opportunities reinforce each other when developed as an integrated cluster.

Policy and Institutional Implications

Successful development requires:

- ✓ a clear anchoring role for the **National Mining Company** upstream,
- ✓ structured PPP and JV arrangements downstream,
- ✓ strong fertiliser quality regulation,
- ✓ demand-side support (extension, credit facilitation),
- ✓ cluster-level governance and shared infrastructure management.

Policy stability and coordination are as important as capital investment.

Concluding Statement

The Value Chain Analysis confirms that Sukulu is **not merely a mining opportunity**, but a **strategic platform for fertiliser import substitution, agro-industrial transformation,**

and regional industrial leadership. When developed through a phased, integrated, and institutionally coordinated approach, Sukulu can support productivity growth, stabilise fertiliser markets, mobilise private investment, and advance Uganda’s long-term green and inclusive development objectives.

1 Introduction to Value Chain Analysis (VCA)

1.1 Introduction

Uganda's ambition to establish a competitive, integrated phosphate fertiliser industry requires a clear understanding of how the current phosphate-to-fertiliser system operates, the actors involved, the markets served, and the constraints that influence performance across the value chain. The **Value Chain Analysis (VCA)** is therefore the foundational diagnostic study that precedes both the **Pre-Feasibility** and **Feasibility Studies**, providing the analytical base for determining whether and how Uganda should move forward with developing a phosphate-based fertiliser plant in Sukulu–Tororo.

This introductory chapter outlines the purpose, scope, relevance, and methodological framework of the VCA, setting the stage for an in-depth assessment of Uganda's fertiliser value chain and its potential contribution to industrialisation, agricultural transformation, and economic growth.

1.2 Purpose and Objectives of the VCA

The primary purpose of the VCA is to **diagnose the current performance, structure, market dynamics, and competitiveness** of the phosphate-to-fertiliser value chain in Uganda and the wider region. As the initial analytical step, the VCA provides critical evidence for decision-making before detailed engineering, financial modelling, and economic appraisal are undertaken.

The VCA seeks to:

A. Map the **entire phosphate value chain** from extraction to final product use;
B. Identify **actors, roles, and governance structures** across the chain;
C. Assess **market demand, supply gaps, pricing trends**, and regional competitive positioning;
D. Analyse the **cost structure and competitiveness** of different phosphate-based products;
E. Examine **policy, regulatory, and institutional frameworks** influencing industry development;
F. Benchmark Uganda against **global phosphate producers** to draw actionable lessons;
G. Identify **binding bottlenecks, system inefficiencies, and capability gaps**;
H. Highlight **investment opportunities, industrial linkages, and strategic entry points** for public and private actors.

The VCA therefore provides the **strategic justification and direction** for whether Uganda should invest in local phosphate beneficiation and fertiliser manufacturing—and in what form.

1.3 Analytical Scope

The Value Chain Analysis (VCA) is designed to be **sufficiently comprehensive to capture the full phosphate-to-fertiliser ecosystem**, while remaining **clearly bounded** to avoid duplication with activities undertaken at the pre-feasibility and feasibility stages. The scope focuses on identifying structural linkages, constraints, opportunities, and strategic options that inform downstream technical and financial analysis.

1.3.1 Thematic Scope

The VCA examines the phosphate value chain across three core segments, complemented by cross-cutting thematic issues:

Upstream segment: Phosphate mining, beneficiation, supporting infrastructure, energy supply, and logistics systems that determine the availability, quality, and cost of phosphate feedstock.

Midstream segment: Processing and manufacturing activities, including sulphuric acid and phosphoric acid production, fertiliser granulation, blending, product differentiation, and readiness for downstream diversification.

Downstream segment: Storage, wholesale and retail distribution, last-mile delivery, fertiliser marketing, extension services, and farm-level application that influence demand realisation and market uptake.

Cross-cutting issues: Policy and regulatory frameworks, trade and market access, institutional arrangements, gender and equity considerations, climate resilience, circular-economy opportunities, and readiness for PPP and private-sector participation.

1.3.2 Geographic Scope

The geographic scope of the VCA spans three interrelated levels:

National scope: Uganda-wide analysis, with particular focus on Tororo/Sukulu as the phosphate resource base and major fertiliser consumption zones across the country.

Regional scope: Assessment of regional market dynamics within the East African Community (EAC) and COMESA, including trade flows, demand gaps, and competitive positioning.

Global scope: Review of major global phosphate producers, exporters, and suppliers whose cost structures, pricing, and logistics influence Uganda's competitiveness.

1.3.3 Temporal Scope

The analysis is anchored in both historical evidence and forward-looking projections:

Historical period: Review of supply, price, and demand trends from **2010 to 2024**, providing context for structural market behaviour and competitiveness.

Prospective period: Demand and market projections for **2030 and 2040**, aligned with the Fourth National Development Plan (NDP IV) and Uganda Vision 2040.

1.4 Importance of VCA as First Step Before Feasibility Study

The VCA is an **initial step** and is essential for ensuring that subsequent investment appraisal stages are grounded in accurate, context-specific, and market-validated evidence.

The VCA is critical because:

- ✓ It **defines the problem**, market failure, and opportunity space that justify Government intervention;
- ✓ It identifies the **value chain structure and bottlenecks** that the project must address;
- ✓ It informs the **strategic options analysis** in the Pre-Feasibility Study;
- ✓ It determines the **scale and type of plant** most appropriate to Uganda's demand, resource quality, and competitiveness;
- ✓ It provides the baseline information required for **technical engineering, financial modelling, and economic appraisal** during feasibility;
- ✓ It ensures that project design is **evidence-driven rather than assumption-driven**.

Without the VCA, the feasibility study risks being misaligned with market realities, value chain constraints, institutional capacity, and Uganda's economic development objectives.

1.5 Alignment with National Frameworks

The VCA is directly aligned with Uganda's national development priorities and sectoral strategies.

1.5.1 Fourth National Development Plan (NDP IV)

The VCA supports NDP IV goals by:

- Promoting **industrialisation and mineral beneficiation**;
- Facilitating **import substitution** of fertilisers and industrial chemicals;
- Strengthening Uganda's **manufacturing base**;

- Enhancing **agricultural productivity** through improved access to fertilisers;
- Anchoring industrial growth in **strategic resource utilisation**.

1.5.2 NDPIV Agro-Industrialisation Programme

The VCA addresses key constraints to agricultural productivity:

- Low fertiliser use (<5%),
- High import dependency,
- Limited distribution systems,
- Fragmented market information.

The study helps identify how domestic phosphate fertiliser production can support the programme's objectives.

1.5.3 Tenfold Growth Strategy

The VCA contributes to the Tenfold Strategy by:

- Identifying opportunities to develop **industrial clusters** around phosphate processing;
- Assessing Uganda's potential to become a **regional fertiliser hub**;
- Exploring value-added pathways such as SSP, TSP, DAP, MAP, NPKs, and industrial chemicals;
- Supporting **export growth engines** and regional competitiveness.

1.6 Methodological Overview

The Value Chain Analysis (VCA) applies a mixed-methods, multi-layered diagnostic approach that combines quantitative modelling with qualitative and systems-based analysis. The methodology is designed to move progressively from market and resource fundamentals, through cost and competitiveness diagnostics, to investment-ready strategic options, without encroaching on the detailed engineering and financial modelling reserved for the pre-feasibility and feasibility stages.

The approach integrates economic reasoning, agronomic logic, industrial systems analysis, and international benchmarking, ensuring that recommendations are evidence-based, context-specific, and policy-relevant.

A. Value Chain Mapping

The study undertakes detailed mapping of the phosphate-to-fertiliser value chain, covering:

- core functions (mining, beneficiation, processing, manufacturing, distribution),
- key actors (public institutions, private firms, service providers),

- physical and financial flows,
- governance and coordination arrangements.

This mapping identifies structural gaps, coordination failures, and leverage points across the upstream, midstream, and downstream segments.

B. Market and Demand Analysis

Market demand is analysed using a dual-track approach:

- **Agronomic Nutrient Requirement Model (ANRM):** Estimates the technical phosphorus requirement based on crop areas, soil characteristics, and recommended nutrient application rates.
- **Econometric Demand Projection Model (EDPM):** Estimates realistic, market-based fertiliser uptake by modelling historical fertiliser use against income, credit, and price proxies, with projections to 2030 and 2040.

This combined approach distinguishes between *potential demand* and *realised demand*, providing a robust basis for plant sizing and phasing.

C. Cost Structure and Competitiveness Analysis

The VCA analyses competitiveness along the full cost chain, including:

- mining and beneficiation cost build-ups,
- chemical processing and manufacturing costs,
- logistics, storage, and distribution costs,
- import-parity and delivered-cost comparisons.

This analysis focuses on **delivered cost-to-market**, which is the decisive competitiveness metric for a landlocked economy, and evaluates product-level profitability and sensitivity to key cost drivers.

D. Benchmarking and International Case Studies

The study benchmarks Uganda's phosphate value chain against selected comparator countries representing different development models:

- ✓ **Morocco** (fully integrated industrial model),
- ✓ **Jordan** (mining-to-export model),
- ✓ **South Africa** (beneficiation-driven model),
- ✓ **India** (blending- and policy-driven demand model),
- ✓ **China** (large-scale industrial clustering model).

Benchmarking focuses on technology choices, logistics configuration, industrial structure, policy frameworks, and sequencing strategies, drawing lessons that are adaptable to Uganda's context rather than replicating scale.

E. Constraints and Opportunity Diagnostics

The VCA applies structured diagnostic tools to identify binding constraints and latent opportunities, including:

- SWOT analysis to assess internal strengths and weaknesses,
- PESTLE analysis to capture policy, institutional, and macro-level factors,
- problem-tree analysis to trace root causes of underperformance across the value chain.

This diagnostic work underpins the identification of priority intervention areas and risk mitigation measures.

F. Strategic Entry Point and Scenario Analysis

Building on the diagnostics, the study identifies:

- ✓ strategic entry points for public and private investment,
- ✓ feasible development pathways across upstream, midstream, and downstream segments,
- ✓ alternative investment scenarios with different risk–return profiles.

Three development scenarios are defined and assessed, culminating in a **preferred phased integrated fertiliser cluster model**, which is carried forward into the pre-feasibility options analysis.

The methodology is deliberately **iterative and integrative**:

- ✓ demand modelling informs plant configuration and phasing,
- ✓ cost and competitiveness analysis filters feasible product lines,
- ✓ benchmarking validates strategic choices,
- ✓ constraints analysis shapes policy and institutional recommendations.

This integration ensures that the VCA does not produce isolated findings, but a **coherent, decision-oriented narrative** that directly supports the transition to technical and financial modelling.

2 Overview of Phosphates and Their Applications

2.1 Chemistry and Classification of Phosphates

Phosphate minerals form the foundation of the global fertiliser industry. Their chemical composition, mineralogy, and physical characteristics determine the efficiency of beneficiation, the suitability for different processing routes, and the quality of downstream fertiliser and industrial products. Understanding the types of phosphate rocks and their chemical properties is therefore essential for assessing Uganda's potential to produce phosphate fertilisers and other value-added products competitively.

2.1.1 Types of Phosphate Rocks

Phosphate rock used in fertiliser and industrial applications is typically classified according to its **geological origin, mineral composition, and ore characteristics**. Globally, three major categories exist:

2.1.1.1 Sedimentary Phosphate Deposits

- Represent more than **80% of global reserves**.
- Formed through marine sedimentation processes.
- Typically rich in **carbonate-fluorapatite (CFA)**.
- Found in Morocco, Tunisia, Jordan, Egypt, China, and parts of East Africa.

Characteristics:

- Medium to high P_2O_5 content (typically **12%–35%**).
- Higher levels of impurities such as **carbonates, clays, silicates, and organic matter**.
- Generally easier to mine due to shallow depth and softer rock.

Relevance to Uganda: The Sukulu deposit is a **carbonatite-associated phosphate**, which shares characteristics with sedimentary types but with distinct mineralogy (apatite in carbonatite matrix). Processing routes must therefore be customised for its mineral profile.

2.1.1.2 Igneous Phosphate Deposits

- Account for 15–20% of global reserves.
- Associated with **alkaline and ultramafic complexes** (e.g., South Africa, Russia, Finland).

- Dominated by **apatite** in hard rock formations.

Characteristics:

- Generally high-purity ore with fewer deleterious elements.
- Lower P_2O_5 grades (typically **4%–12%** before beneficiation).
- Require intensive grinding, flotation, and magnetic separation.

Implications: While more expensive to process, they produce very high-grade concentrates (>38% P_2O_5).

2.1.1.3 Biogenic (Organic) Phosphate Deposits

- Derived from guano or bone material.
- Small-scale and rarely used for industrial fertiliser production.
- Low relevance to large-scale industrial projects.

2.1.2 Nutrient Components (P_2O_5, Impurities, Mineralogy)

The economic value and processing behaviour of phosphate rock is primarily determined by:

2.1.2.1 P_2O_5 Content (Phosphorus Pentoxide)

P_2O_5 is the industry-standard measure of phosphate nutrient content.

- **High-grade ore:** >30% P_2O_5
- **Medium-grade ore:** 20–30% P_2O_5
- **Low-grade ore:** <20% P_2O_5

Sukulu data (historical studies):

- Beneficiable grades: **12%–14% P_2O_5** (raw ore)
- Historical single superphosphate feedstock: **41%–42% P_2O_5** (beneficiated sample)

P_2O_5 concentration determines:

- The suitability of ore for **direct acidulation**;
- Whether **granular fertilisers** (DAP, MAP, TSP) can be produced economically;
- The beneficiation intensity required to upgrade ore.

2.1.2.2 Key Impurities Affecting Processing

A. **Carbonates ($CaCO_3$, $MgCO_3$):**

  ✓ High carbonate content consumes acid during processing.

- ✓ Increases acidulation cost for phosphoric acid and SSP/TSP.

B. **Silica (SiO_2):**
   - ✓ Causes abrasion during grinding; reduces flotation efficiency.

C. **Iron and Aluminium Oxides (Fe_2O_3, Al_2O_3):**
   - ✓ React with phosphoric acid, forming sludge (“fluorophosphates”).
   - ✓ Reduce yield and increase waste generation.

D. **Organic Matter:**
   - ✓ Interferes with flotation; requires washing or calcination.

E. **Heavy Metals (e.g., cadmium, uranium):**
   - ✓ Raise environmental and regulatory concerns.
   - ✓ Require mitigation depending on product specifications (especially for export markets).

F. **Dolomite ($CaMg(CO_3)_2$):**
   - ✓ Highly undesirable for DAP and TSP production.

Sukulu ore characteristics: Carbonatite-hosted ores typically include **apatite**, magnetite, pyrochlore, and fluorite. Beneficiability depends heavily on mineral liberation at fine grind sizes.

2.1.2.3 Mineralogy and Its Processing Implications

Phosphate rocks are predominantly composed of apatite-group minerals:

- **Fluorapatite:** $Ca_{10}(PO_4)_6F_2$
- **Hydroxyapatite:** $Ca_{10}(PO_4)_6(OH)_2$
- **Carbonate-fluorapatite (CFA):** $Ca_{10}(PO_4,CO_3)_6F_2$

Mineralogical factors influencing processing:

1. **Grain size and dissemination:**

   Fine-grained apatite requires finer grinding but improves liberation.

2. **Association with gangue minerals:**

   Apatite interlocked with magnetite, silicates, or carbonates affects flotation efficiency.

3. **Reactivity to acid:**

   CFA reacts more readily with sulphuric acid than igneous apatite—beneficial for producing SSP/TSP.

4. **Fluorine content:**

   Influences emissions control during acidulation.

2.1.3 Beneficiation Implications

Beneficiation upgrades raw ore to a suitable grade for fertiliser or industrial production. Processing requirements depend on ore mineralogy and P_2O_5 grade.

2.1.3.1 Common Beneficiation Techniques

1. **Crushing and Grinding**
    - Required for liberation of apatite.
    - Carbonatite-hosted ores often grind well but may require fine milling.
2. **Washing and Desliming.** Removes clays and fine silts that interfere with flotation.
3. **Flotation.**
    - Most widely used method for upgrading low- to medium-grade ores.
    - Reagents (collectors, depressants, frothers) must be tailored to the specific mineralogy.
4. **Magnetic Separation.** Useful when ore contains magnetite (common in carbonatite complexes like Sukulu).
5. **Calcination (less common).** Used to remove organic matter or dolomite; energy-intensive.

2.1.3.2 Beneficiation Challenges Specific to Sukulu-Type Ores

- **Moderate P_2O_5 (~12–14%)** requires flotation to achieve commercially viable concentrates (>30% P_2O_5).
- **Complex mineralogy** (apatite within carbonatite matrix) may require multi-stage flotation.
- **High carbonate content** increases reagent consumption.
- **Presence of magnetite** may require prior magnetic separation to reduce contamination.

2.1.3.3 Implications for Downstream Processing

The beneficiation outcome determines:

- Whether the ore can support **acidulation** (phosphoric acid, SSP/TSP production).
- Whether it is suitable for **high-analysis fertilisers** like DAP or MAP.
- The **cost of production**, driven by acid consumption and processing efficiency.
- The **type of fertiliser plant configuration** recommended at feasibility stage.

In summary, the chemistry and classification of phosphate rock strongly influence technical feasibility, processing economics, and competitiveness—key considerations for Uganda's phosphate fertiliser project.

2.2 Industrial and Agricultural Functions of Phosphates

Phosphates play a central role in global agriculture, livestock nutrition, and a wide range of industrial processes. Their unique chemical properties make them indispensable inputs for soil fertility, plant metabolism, animal growth, and various manufacturing industries. Understanding these functions is fundamental to evaluating the economic potential of Uganda's phosphate deposits and the possible range of downstream products that can be manufactured locally.

2.2.1 Role in Soil Fertility and Crop Nutrition

Phosphorus (P), primarily supplied through phosphate fertilisers, is one of the three essential macronutrients required for plant growth—alongside nitrogen (N) and potassium (K). Unlike nitrogen and potassium, phosphorus is relatively immobile in the soil and easily bound to iron, aluminium, and calcium compounds, making it unavailable to plants. As a result, soils in tropical regions, including Uganda, are often severely depleted of available phosphorus.

Key roles of phosphates in soil fertility include:

- ✓ **Restoring nutrient balance:** Replenishes phosphorus removed by crop harvests and soil erosion.
- ✓ **Improving soil microbial activity:** Enhances beneficial microorganisms involved in nutrient cycling.
- ✓ **Supporting root establishment:** Phosphorus availability is crucial during seedling establishment.
- ✓ **Promoting efficient fertiliser use:** Balanced nutrients improve the uptake and utilisation of nitrogen and potassium.

In Uganda, where fertiliser use remains below 5% and many soils are highly weathered and acidic, phosphate application is critical to improving productivity and closing the national yield gap.

2.2.2 Essential Nutrient Functions for Plant Growth

At the physiological level, phosphorus supports a wide range of essential plant processes. It is a structural component of nucleic acids (DNA, RNA), energy molecules (ATP, ADP), and cell membranes.

Phosphates contribute to:

2.2.2.1 Energy Transfer (ATP metabolism)

- Phosphorus forms the backbone of **adenosine triphosphate (ATP)**, the molecule responsible for storing and transferring energy within plant cells.
- Drives cell division, photosynthesis, and nutrient assimilation.

2.2.2.2 Root Development

- Stimulates early root growth, promoting deeper and more extensive rooting systems.
- Improves a plant's ability to access water and other nutrients.
- Critical in early growth stages of cereals, legumes, tubers, and fruit crops.

2.2.2.3 Flowering and Reproductive Development

- Enhances flowering, fruit set, and grain formation.
- Essential for reproductive success and final yield.

2.2.2.4 Yield Formation and Crop Quality

- Supports seed development, starch formation, oil synthesis, and protein content.
- Improves uniformity and accelerates maturity, reducing vulnerability to drought and pests.

Deficiency symptoms include stunted growth, purpling of leaves, delayed maturity, and significant yield loss.

2.2.3 Functions in Livestock and Poultry Feed

Phosphates are also essential for the growth and health of livestock and poultry. They provide bioavailable phosphorus and calcium, which are critical for skeletal development, metabolic functions, and reproductive performance.

Key phosphate feed products include:

2.2.3.1 Dicalcium Phosphate (DCP)

- Produced by reacting phosphate rock with hydrochloric acid or phosphoric acid.
- Contains **18–21% phosphorus** and **23–28% calcium**.
- Commonly used in poultry, dairy, cattle, and pig feed.

Functions:

- Supports bone and teeth formation.
- Prevents rickets and bone disorders.
- Enhances feed conversion efficiency.
- Improves fertility and milk production.

2.2.3.2 Monocalcium Phosphate (MCP)

- Higher bioavailability than DCP.
- Contains **21–22% phosphorus** and **16–17% calcium**.
- Preferred in poultry and pig feed because of better absorption.

Functions:

- Improves egg shell quality in layers.
- Enhances weight gain and feed efficiency.
- Supports intestinal enzyme activity.

2.2.3.3 Defluorinated Phosphate (DFP)

- Safe feed-grade alternative when raw ore has high fluorine content.

Importance for Uganda: With a growing livestock and poultry sector, domestic production of feed-grade phosphates presents a major opportunity for import substitution.

2.2.4 Industrial Functions of Phosphates

Beyond agriculture, phosphates are critical raw materials in multiple industrial sectors due to their chemical versatility, buffering capacity, sequestration properties, and heat resistance.

2.2.4.1 Detergent and Cleaning Industry

- **Sodium tripolyphosphate (STPP)** is widely used as a water softener and detergent builder.
- Enhances cleaning efficiency by binding calcium and magnesium ions.
- Large-volume global demand.

2.2.4.2 Ceramics and Glass Manufacturing

- Phosphates improve strength, binding, and melting properties.
- Used in refractory materials for furnaces.

2.2.4.3 Metallurgy

- Phosphates act as corrosion inhibitors, fluxing agents, and metal surface treatment chemicals.
- Enable anti-rust coatings and improve metal finishing.

2.2.4.4 Water Treatment

- Polyphosphates prevent scale formation and corrosion in water distribution systems.
- Widely used by water utilities and industrial process plants.

2.2.4.5 Pharmaceutical and Food Industries

- Used as:
    - pH stabilisers
    - Buffering agents
    - Emulsifiers
    - Nutrient supplements
- Common in medicinal syrups, vaccines, baked products, processed meats, and beverages.

2.2.4.6 Specialty Chemical Applications

- Flame retardants
- Resin modifiers
- Adhesives

These industrial applications offer potential horizontal diversification for Uganda, especially within industrial parks.

2.2.5 Environmental and Renewable Energy Functions

In emerging green technologies and environmental applications, phosphates play an increasingly important role.

2.2.5.1 Battery Manufacturing

- **Lithium Iron Phosphate ($LiFePO_4$)** batteries are gaining global prominence:
    - Safer and more stable than lithium-ion alternatives.

- Longer cycle life.
- Lower thermal runaway risk.

Growing interest in electric mobility and renewable storage drives demand.

2.2.5.2 Water Purification and Environmental Remediation

- Certain phosphate compounds remove heavy metals through precipitation.
- Used in:
  - Industrial wastewater treatment
  - Municipal water purification
  - Soil remediation of lead and arsenic contamination

2.2.5.3 Carbon Capture and Mineralisation

- Phosphate minerals can participate in carbon sequestration pathways, supporting climate mitigation initiatives.

2.2.5.4 Circular Economy Opportunities

- Reuse of phosphogypsum for construction materials.
- Recovery of phosphate from wastewater streams (struvite formation).

These emerging applications highlight phosphates' strategic importance beyond fertilisers and underscore the potential for Uganda to integrate into future mineral value chains.

2.3 Products Manufactured from Phosphates

2.3.1 Mining and Beneficiation Outputs

Mining and beneficiation represent the **first stages of value addition** in the phosphate value chain. The quality and form of products generated at this stage directly determine the technical options, cost structures, and competitiveness of downstream fertiliser and industrial production. For Uganda, understanding these outputs is crucial for establishing the appropriate processing configuration and investment focus for the Sukulu phosphate project.

2.3.1.1 Beneficiated Phosphate Concentrate

Beneficiated phosphate concentrate is the primary product generated after ore processing and mineral upgrading. Beneficiation aims to increase the phosphate content (P_2O_5 grade) while reducing impurities such as carbonates, silica, iron oxides, aluminium oxides, dolomite, and organic matter.

Key Characteristics

- **Typical P_2O_5 grade:** 28% – 40% depending on ore type and beneficiation efficiency.
- **Physical form:** Usually a fine-grained or pelletised concentrate.
- **Mineralogy:** Predominantly apatite (fluorapatite or hydroxyapatite) with significantly reduced gangue minerals.

Production Process

Beneficiation typically involves:

1. **Crushing and Grinding** – to liberate apatite grains.
2. **Washing and Desliming** – to remove clays and slimes that reduce flotation efficiency.
3. **Flotation** – the main upgrading step using collectors, frothers, and depressants.
4. **Magnetic Separation** – when associated with magnetite-bearing ores, such as in carbonatite-hosted deposits like Sukulu.
5. **Drying and/or Agglomeration** – to produce a stable concentrate.

End Uses

Beneficiated concentrates form the **primary raw material** for:

- **Phosphoric acid production** (via acidulation with sulphuric acid).
- **SSP (Single Super Phosphate)** and **TSP (Triple Super Phosphate)** production.
- **High-analysis fertilisers** such as **DAP (Di-ammonium Phosphate)** and **MAP (Mono-ammonium Phosphate)**.
- **Blending into NPK formulations**.
- **Industrial applications** where high-purity feedstock is required.

Strategic Implications for Uganda

- Beneficiated concentrate is essential for **medium- and high-analysis fertiliser production**, enabling Uganda to enter the competitive fertiliser market.
- Quality of concentrate determines **acid consumption**, which significantly affects production costs.
- Establishing a beneficiation plant catalyses upstream mining, midstream processing, and downstream fertiliser manufacturing clusters.

2.3.1.2 Direct Application Phosphate Rock (DAPR)

Direct Application Phosphate Rock (DAPR) refers to **phosphate rock used without chemical processing or acidulation**. Instead, it is milled and applied directly to soils, primarily in agriculture.

Key Characteristics

- **Typical P_2O_5 grade:** 12% – 30%
- **Form:** Finely ground rock powder
- **Solubility:** Low solubility; effectiveness depends heavily on rock reactivity

Suitability Criteria

DAPR effectiveness is governed by:

1. **Reactivity of rock**
    - Highly reactive carbonate-fluorapatite rocks dissolve better in acidic soils.
    - Igneous rocks are usually poorly reactive.
2. **Soil type**
    - Best results in **acidic soils (pH < 5.5)** prevalent in many tropical regions, including parts of Uganda and the Great Lakes.
3. **Cropping systems**
    - Most effective for long-term soil fertility improvement rather than quick-release nutrient supply.

Advantages

- Lower production cost due to the absence of acidulation.
- Suitable for smallholder farming systems in low-input agriculture.
- Improves soil phosphorus reserves over time.

Limitations

- Slow nutrient release makes it **unsuitable for high-yield commercial agriculture**.
- Ineffective in neutral or alkaline soils.
- Not suitable for formulating compound fertilisers or high-analysis blends.

End Uses

- **Direct soil application** in acidic soils.
- **Organic and regenerative agriculture systems** where gradual nutrient release is acceptable.
- Blending into **low-analysis fertiliser products** in certain contexts.

Relevance to Uganda

- Uganda has widespread **acidic Ferralsols and Acrisols**, suitable for DAPR application in some regions.
- DAPR may complement—but not replace—manufactured phosphate fertilisers required for commercial agriculture.
- It offers a potential **low-cost fertiliser input** for smallholder farmers where immediate yield response is not the primary objective.

2.3.1.3 Strategic Importance of Mining and Beneficiation Outputs

Both beneficiated concentrate and DAPR represent **entry points for Uganda's industrialisation pathway**:

- ✓ They set the foundation for **value addition**, influencing capital investment, technology choices, and downstream product diversification.
- ✓ Beneficiated concentrate supports a **fully integrated fertiliser complex**, while DAPR offers a **low-cost, broad-based soil fertility option**.
- ✓ Understanding these outputs informs the technical and economic feasibility of producing different fertiliser types (SSP, TSP, DAP, MAP, NPKs) and non-fertiliser phosphate products (DCP, MCP, STPP, industrial chemicals).

2.3.2 Acidulation Products

Acidulation is a critical step in the phosphate value chain, transforming beneficiated phosphate rock into a range of downstream fertiliser and industrial products. The two most important outputs of this stage are **phosphoric acid (PA)**—the basis for nearly all high-analysis phosphate fertilisers—and **sulphuric acid (SA)**, the primary reagent used in PA production. Their quality, production cost, and plant configuration significantly influence the competitiveness of any phosphate-based industry.

2.3.2.1 Phosphoric Acid (PA)

(Industrial Grade and Food Grade)

Phosphoric acid is the central intermediate in the manufacture of phosphate fertilisers, animal feed phosphates, and various industrial chemicals. Globally, more than **90% of phosphoric acid produced is used in fertiliser manufacturing**.

1. Types of Phosphoric Acid

Phosphoric acid produced through acidulation is primarily classified into:

A. Industrial Grade (Merchant Grade Acid – MGA)

- P_2O_5 concentration typically **26–30%**.

- Contains impurities from the ore and process conditions.
- Used for:
    - Production of fertilisers such as **DAP**, **MAP**, **TSP**, and **NPKs**.
    - Manufacturing of **feed-grade phosphates** (DCP, MCP).
    - Various industrial processes where ultra-high purity is not required.

B. Food Grade / Purified Phosphoric Acid

- P_2O_5 concentration typically **54–85%**, depending on concentration method.
- Produced via further purification (solvent extraction, ion exchange, crystallisation).
- High purity and low heavy-metal content are essential.
- Applications include:
    - Beverages and soft drinks
    - Pharmaceuticals
    - Food processing
    - Water treatment

2. Production Process

Phosphoric acid is usually manufactured via the **wet process**, where beneficiated phosphate rock reacts with sulphuric acid:

$$Ca_5(PO_4)_3F + 5H_2SO_4 + 10H_2O \rightarrow 3H_3PO_4 + 5CaSO_4 \cdot 2H_2O + HF$$

Inputs:

- Beneficiated phosphate concentrate
- Sulphuric acid
- Water and process heat

Co-products:

- **Phosphogypsum ($CaSO_4 \cdot 2H_2O$)**
- Fluorine compounds
- Heat and steam

3. Applications of Phosphoric Acid

A. Fertilisers

- **DAP (Diammonium Phosphate)**
- **MAP (Monoammonium Phosphate)**
- **TSP (Triple Super Phosphate)**
- Compound **NPK fertilisers**

These are critical for modern agriculture and high-yield cropping systems.

B. Animal Feed Phosphates

- Dicalcium phosphate (DCP)
- Monocalcium phosphate (MCP)

C. Industrial Applications

- Metal treatment and rust prevention
- Detergent and chemical manufacturing
- Catalysts and flame retardants

4. Strategic Importance for Uganda

- Phosphoric acid production is the **gateway to high-value fertiliser manufacturing**.
- A domestic PA plant enables Uganda to produce DAP, MAP, and TSP competitively.
- PA availability supports downstream industries—feed production, detergent chemicals, food processing.
- Reduces dependence on imported phosphoric acid and derived fertilisers, contributing to **import substitution and industrial diversification**.

2.3.2.2 Sulphuric Acid (SA)

(Input for PA Production)

Sulphuric acid is the **largest-volume inorganic chemical** produced globally and is the primary reagent required to dissolve phosphate rock during the acidulation process. SA production capacity is an essential determinant of a country's ability to manufacture phosphate fertilisers at scale.

1. Production Process

Sulphuric acid is produced via the **Contact Process**, involving:

1. **Burning sulphur or pyrite** to form sulphur dioxide (SO_2).
2. Oxidation of SO_2 to sulphur trioxide (SO_3) using a vanadium pentoxide catalyst.
3. Absorption of SO_3 in water to form H_2SO_4.

Many modern fertiliser complexes use **sulphur-burning plants** due to high purity, reliability, and lower maintenance compared with pyrite-roasting.

2. Characteristics Relevant to Fertiliser Production

- Typically produced at **98% concentration** for phosphoric acid plants.
- Energy-intensive process—generates **significant heat**, which can be captured for steam and power generation.
- Quality and continuity of SA supply strongly influence production costs of PA.

3. Industrial and Agricultural Uses

While a substantial share of sulphuric acid globally is used in fertiliser manufacturing, it also supports:

- Petrochemical industry
- Metal processing and leaching
- Detergents and industrial chemicals
- Textile and pulp & paper industries
- Water treatment systems

4. Ratio of SA to PA in Fertiliser Manufacturing

To manufacture phosphoric acid by the wet process:

1 tonne of P_2O_5 requires approximately 3 tonnes of sulphuric acid

This ratio means that a fertiliser complex must have **stable, large-scale SA production** or reliable imports.

5. Strategic Importance for Uganda

- SA production is indispensable for establishing a **self-sufficient integrated fertiliser complex** at Sukulu.
- A domestic SA plant reduces the need for costly imports and ensures operational reliability.
- Surplus SA can support emerging industrial clusters—battery manufacturing, detergents, metal refining, and water treatment.
- Energy recovered from SA production (steam/power) can reduce operating costs of the fertiliser plant.

2.3.2.3 Implications for Uganda's Phosphate Value Chain

1. **Phosphoric acid is the central intermediate** for almost all phosphate fertilisers; therefore, Uganda must build PA capacity to become competitive in fertiliser production.
2. **Sulphuric acid availability** determines the scale and feasibility of PA production.
3. Together, PA and SA form the backbone of an **integrated fertiliser industry** capable of producing:
    - SSP, TSP
    - DAP, MAP
    - NPKs
    - Feed phosphates and industrial chemicals
4. Control over these acidulation products positions Uganda for **regional leadership** in fertiliser supply within the EAC and COMESA regions.

2.3.3 Fertiliser Products

Phosphate-based fertilisers are the backbone of modern agriculture, supplying phosphorus—an essential macronutrient required for energy transfer, root development, flowering, and yield formation. From basic single superphosphates to high-analysis ammoniated products and balanced NPK blends, these fertilisers play a pivotal role in enhancing soil fertility and increasing agricultural productivity.

This section describes the major fertiliser products that can be manufactured from beneficiated phosphate rock or phosphoric acid, their characteristics, uses, and strategic implications for Uganda.

2.3.3.1 Single Super Phosphate (SSP)

Single Super Phosphate is one of the earliest manufactured phosphate fertilisers and remains important in developing regions due to its affordability and suitability for acidic soils.

Key Characteristics

- **P_2O_5 content:** 16–20% (available form)
- **Composition:** Calcium dihydrogen phosphate + gypsum
- **Physical form:** Powdered or granulated
- **Production Process**

Produced by reacting **phosphate rock with sulphuric acid**:

$$PR + H_2SO_4 \rightarrow SSP\ (Ca(H_2PO_4)_2 + CaSO_4)$$

Advantages

- Low-cost fertiliser for smallholder farmers
- Supplies both **phosphorus and sulphur** (important in East African soils)
- Performs well in acidic soils common in Uganda

Applications

- Cereals, legumes, oilseeds, sugar cane, pasture crops
- Integrated soil fertility management systems

Relevance for Uganda

Uganda's acidic, phosphorus-deficient soils make SSP a strong fit for widespread smallholder use.

Its production requires only SA and beneficiated rock—simpler than other fertilisers.

2.3.3.2 C.2 Triple Super Phosphate (TSP)

TSP is a highly concentrated phosphate fertiliser with minimal sulphur content.

Key Characteristics

- **P_2O_5 content:** 44–48%
- **Form:** Granular, highly concentrated P source
- **Production Process**

Made by reacting **beneficiated phosphate rock with phosphoric acid**:

$$PR + H_3PO_4 \rightarrow TSP$$

Advantages

- High nutrient density reduces transport and application costs
- Suitable for soils already containing sufficient sulphur
- Excellent starter fertiliser for root crops and legumes

Applications

- Vegetables, fruit crops, plantation crops, legumes
- Base fertiliser in NPK blending operations

Relevance for Uganda

TSP can supply domestic blending plants and reduce dependence on imported P-rich fertilisers.

2.3.3.3 Di-Ammonium Phosphate (DAP)

DAP is the world's most widely used phosphate fertiliser, valued for its high P content and balanced nitrogen contribution.

Key Characteristics

- **P_2O_5 content:** 46%
- **N content:** 18%
- **Physical form:** Free-flowing granules

Production Process

Reacting **phosphoric acid with ammonia**:

$$H_3PO_4 + 2NH_3 \rightarrow (NH_4)_2HPO_4$$

Advantages

- Supplies both **phosphorus and nitrogen**
- High solubility ensures quick nutrient availability
- Excellent for basal application

Applications

- Maize, rice, wheat, potato, sugarcane, pulses
- Fertiliser blends (NPK)

Strategic Importance

DAP is highly demanded across East Africa. Local DAP production positions Uganda as a **regional fertiliser supplier**.

2.3.3.4 Mono-Ammonium Phosphate (MAP)

MAP is another high-analysis ammoniated phosphate fertiliser with a higher phosphorus-to-nitrogen ratio compared to DAP.

Key Characteristics

- **P_2O_5 content:** 48–52%
- **N content:** 10–12%
- **Production Process**

Reacting **phosphoric acid with ammonia**, but at different ratios than DAP:

$$NH_3 + H_3PO_4 \rightarrow NH_4H_2PO_4$$

Advantages

- Highly efficient P source with lower pH impact than DAP
- Ideal for acidic or marginal soils
- Reduces ammonia volatilisation

Applications

- Plantation crops (tea, coffee), horticulture, legumes
- Precision agriculture and fertigation systems

Relevance for Uganda

MAP provides a flexible input for both field crops and high-value horticulture sectors.

2.3.3.5 NPK Formulations (Various Grades)

NPK fertilisers combine Nitrogen (N), Phosphorus (P), and Potassium (K) in various proportions tailored to specific crops and soil conditions. They are essential for balanced crop nutrition.

Typical NPK Grades

- 17-17-17
- 20-10-10
- 10-26-26
- 15-15-15
- 23-21-0+4S
- Custom blends for horticulture, cereals, legumes, and plantation crops

Production Methods

1. **Blended NPK:** Dry mixing of granular N, P, and K components
2. **Compound NPK:** Chemical granulation where nutrients are fused in each granule
3. **Bulk blending:** Flexible regional blending systems for localised demand

Advantages

- Provides balanced nutrition
- Improves nutrient-use efficiency
- Supports crop-specific fertiliser recommendations

Applications

- Widely used for maize, beans, bananas, rice, vegetables, sugarcane, tea, and coffee

Importance for Uganda

- NPKs account for a large share of fertiliser used in Uganda
- Domestic phosphate-based inputs support regional blending industries
- NPK production enhances value addition and industrial diversification

2.3.3.6 Compound Fertilisers

Compound fertilisers are granulated products in which each granule contains a specified proportion of N, P, and K—ensuring uniform nutrient distribution.

Key Characteristics

- Higher quality than simple blends
- Lower risk of segregation during transport and application
- Suitable for mechanised and precision application

Production

Produced via **granulation processes**, often using phosphoric acid, ammonia, and potash simultaneously.

Advantages

- Superior agronomic performance
- High nutrient density
- Consistent nutrient delivery

Applications

- Commercial agriculture
- Tea, sugarcane, tobacco, horticulture
- Export-oriented crops requiring high nutrient efficiency

Strategic Importance

Producing compound fertilisers domestically would significantly reduce fertiliser imports, support large-scale farming, and increase Uganda's regional competitiveness.

2.3.3.7 Strategic Implications of Fertiliser Products for Uganda

- **SSP** supports smallholder productivity and acidic soils.
- **TSP, DAP, MAP** enable Uganda to enter regional value chains and reduce reliance on imports.
- **NPKs and compound fertilisers** support mainstream commercial agriculture and agro-industrialisation.
- A diversified fertiliser portfolio enhances **market resilience**, supports **import substitution**, and drives **industrialisation**.

2.3.4 Animal Feed Products

Phosphate-based feed supplements are essential components of balanced nutrition for livestock and poultry. They supply biologically available phosphorus and calcium—two minerals critical for skeletal development, metabolic efficiency, reproductive performance, and overall animal health. As Uganda's livestock and poultry industries expand, demand for high-quality feed phosphates is rising, creating a significant opportunity for domestic production linked to the phosphate value chain.

The two most widely used feed phosphates globally—and the most relevant for Ugandan markets—are **Dicalcium Phosphate (DCP)** and **Monocalcium Phosphate (MCP)**.

2.3.4.1 Dicalcium Phosphate (DCP)

1. Overview

Dicalcium Phosphate is one of the most commonly used feed-grade phosphates, supplying both phosphorus and calcium in bioavailable form. It is widely used across poultry, dairy, cattle, piggery, and aquaculture sectors.

2. Chemical Composition

- **Phosphorus content:** 18–21%
- **Calcium content:** 23–28%
- **Chemical formula:** $CaHPO_4 \cdot 2H_2O$

3. Production Process

DCP is typically produced from:

- **Phosphoric acid + limestone reaction**, or
- **Hydrochloric acid digestion of phosphate rock**, followed by purification.

Key steps include:

1. Acid digestion of phosphate rock
2. Precipitation of dicalcium phosphate
3. Filtration, drying, and granulation

4. Functions and Benefits in Animal Nutrition

- Supports **bone and skeletal development**
- Prevents rickets and osteoporosis in animals
- Improves **feed conversion efficiency**
- Enhances **reproductive performance** and fertility
- Crucial for **milk production** in dairy animals
- Supports enzyme activation and nutrient metabolism

5. Applications

- Poultry (broilers and layers)
- Pigs (weaners, growers, finishers)
- Dairy and beef cattle
- Sheep and goats
- Fish feed in aquaculture systems

6. Relevance for Uganda

Uganda imports all DCP used in the feed industry. Domestic production would:

- Lower feed costs
- Improve availability and quality of feed supplements
- Strengthen the livestock value chain
- Provide a key downstream use for locally produced phosphoric acid

2.3.4.2 Monocalcium Phosphate (MCP)

1. Overview

MCP is a higher-grade, more bioavailable feed phosphate compared to DCP. It is preferred for poultry and pig production due to superior digestibility and rapid absorption.

2. Chemical Composition

- **Phosphorus content:** 21–22%
- **Calcium content:** 16–17%

- **Chemical formula:** $Ca(H_2PO_4)_2 \cdot H_2O$

3. Production Process

Manufactured by reacting **purified phosphoric acid** with limestone or calcium carbonate in controlled conditions to produce a soluble calcium phosphate.

Key process features:

- High-purity phosphoric acid is required
- Lower fluorine content than DCP
- More energy-efficient digestion

4. Functions and Benefits in Animal Nutrition

- Provides highly bioavailable phosphorus
- Enhances **growth rate** and feed efficiency in broilers and pigs
- Improves **egg shell quality** in layers
- Supports metabolic functions, enzyme activation, and DNA synthesis
- Reduces environmental phosphorus excretion due to improved absorption

5. Applications

- High-performance poultry systems
- Pig farming operations
- Aquaculture feed formulations
- Young animal diets requiring easily digestible phosphorus

6. Relevance for Uganda

Uganda's modern poultry and piggery sectors are rapidly expanding. Domestic MCP production would:

- Support high-growth livestock value chains
- Reduce reliance on expensive imported feed inputs
- Improve competitiveness of Ugandan livestock producers
- Position Uganda as a potential supplier within the EAC region

2.3.4.3 Strategic Implications for Uganda's Phosphate Value Chain

1. **Feed phosphates diversify revenue streams** beyond fertiliser markets.
2. They support Uganda's **livestock intensification** goals under NDP IV.
3. DCP and MCP production strengthens **backward and forward linkages**:
    - Backward: mining → beneficiation → phosphoric acid
    - Forward: feed mills → livestock → food processing

4. Feed phosphate production is a **high-value, high-demand sector** with strong regional market potential.
5. Successful production requires **high-quality phosphoric acid**, making it fully compatible with an integrated phosphate industrial complex.

2.3.5 Industrial Products

Phosphate-derived industrial chemicals form a significant segment of global phosphate consumption outside agriculture and livestock. Their versatility in chemical reactions, buffering, sequestration, and flame-retardant properties makes them indispensable across manufacturing, food processing, pharmaceuticals, detergents, water treatment, and technology industries.

For Uganda, the development of an integrated phosphate chemical complex presents opportunities to diversify beyond fertiliser production, reduce import dependency, and stimulate industrial cluster development.

2.3.5.1 Sodium Tripolyphosphate (STPP) – Detergent Industry

1. Overview

STPP is one of the most important inorganic chemicals used in modern detergent and cleaning products. It functions as a **builder**, improving detergent efficiency by softening water and dispersing soils.

2. Chemical Composition

- Formula: $\mathbf{Na_5P_3O_{10}}$
- Produced from phosphoric acid and sodium carbonate or sodium hydroxide.
- **3. Key Industrial Functions**
- Water softening through sequestration of calcium and magnesium ions
- Preventing dirt redeposition on fabrics
- Enhancing surfactant efficiency in detergents
- Used in household detergents, industrial cleaners, and dishwashing tablets

4. Relevance for Uganda

Uganda imports significant volumes of detergents and cleaning products. Domestic STPP production would:

- Support local detergent manufacturers
- Reduce import bills
- Anchor new industries in chemical manufacturing zones

- Create backward linkages for phosphoric acid utilisation

2.3.5.2 Metal Surface Treatment Chemicals

1. Overview

Phosphate chemicals such as zinc phosphates and manganese phosphates are widely used for **metal pre-treatment**, corrosion protection, and coating processes.

2. Functions

- Enhances adhesion of paints and coatings
- Prevents rust and corrosion
- Provides protective phosphate layers on steel and iron
- Used in automotive, electronics, construction, and fabrication industries

3. Strategic Value

A domestic supply of metal-treatment chemicals supports:

- Automotive assembly
- Steel rolling mills
- Engineering and fabrication sectors

This is consistent with Uganda's industrialisation agenda under NDP IV.

2.3.5.3 Flame Retardants

1. Overview

Certain phosphate compounds (e.g., ammonium polyphosphates, red phosphorus derivatives) act as **flame retardants** in polymers, textiles, electronics, and construction materials.

2. Applications

- Plastics and polyurethane foams
- Electrical and electronic equipment
- Furniture and upholstery
- Construction materials (gypsum boards, insulation foam)

3. Mechanism

Phosphate-based flame retardants function by:

- Promoting char formation
- Minimising flammable gas release
- Reducing heat generation

4. Economic Potential

Global demand for phosphorus-based flame retardants is rising due to:

- Increasing fire-safety regulations
- Transition away from halogenated flame retardants

Uganda could target regional markets through specialty chemical production.

2.3.5.4 Ceramic Binders

1. Overview

Phosphate-based binders (e.g., aluminium phosphate, magnesium phosphate) improve the structural integrity and workability of ceramic and refractory materials.

2. Uses

- High-temperature ceramics
- Refractory linings for furnaces and kilns
- Adhesives in the glass and construction industry
- Industrial floor coatings

3. Benefits

- High bonding strength
- Thermal stability
- Resistance to chemical corrosion

4. Strategic Fit

Linkages with Uganda's construction, cement, and ceramics industries create additional demand for phosphate chemicals.

2.3.5.5 Pharmaceuticals

1. Overview

Purified phosphoric acid and specialised phosphate salts are extensively used in pharmaceutical formulations.

2. Functions

- Buffering agent in oral medications
- Acidulant in syrups and suspensions
- pH regulator in intravenous solutions
- Component in dental cements and restorative materials

3. Market Opportunities

Local availability of purified phosphoric acid supports:

- Growth of Uganda’s pharmaceutical manufacturing sector
- Reduced import reliance for key excipients
- Regional export potential for medical-grade chemicals

2.3.5.6 Food Additives (E-number Phosphates)

1. Overview

Food-grade phosphate additives (E338–E343, E450–E452) are widely used in the food and beverage industry.

2. Applications

- Emulsification in dairy products
- Moisture retention in meats and processed foods
- Leavening in baked goods
- Acidification in beverages
- Stabilisation in sauces and instant products

3. Importance for Uganda

As Uganda’s agro-processing sector grows, demand for food-grade phosphates will rise. Domestic production enables:

- Reduced manufacturing costs
- Improved quality control
- Integration of agro-industrial value chains

4. Typical Products

- Sodium phosphates (E339)
- Potassium phosphates (E340)
- Calcium phosphates (E341)
- Pyrophosphates and polyphosphates (E450–E452)

2.3.5.7 Water Purification Chemicals

1. Overview

Polyphosphates and orthophosphates are used in water treatment to prevent scale, corrosion, and heavy metal contamination.

2. Functions

- Corrosion inhibition in municipal water pipes
- Scale prevention in boilers and industrial equipment

- Removal of lead and copper ions
- Stabilisation of iron and manganese in groundwater

3. Relevance for Uganda

With expanding urban centres and industrial zones, water utilities increasingly require:

- Corrosion-control chemicals
- Scale inhibitors
- Heavy-metal removal agents

These products create synergies with national water service providers and private industrial plants.

2.3.5.8 Strategic Implications of Industrial Phosphate Products

1. **Diversification & Value Addition:** Industrial phosphate chemicals expand Uganda's export and manufacturing portfolio beyond fertilisers.
2. **Import Substitution:** Uganda currently imports nearly all industrial phosphate chemicals, representing a major cost-saving opportunity.
3. **Industrial Cluster Development:** PA-based industrial chemicals anchor downstream industries in detergents, ceramics, pharmaceuticals, and metallurgy.
4. **Regional Market Potential:** EAC and COMESA countries have growing demand for water treatment chemicals, STPP, and flame retardants.
5. **Integration with Fertiliser Complex:** Shared inputs (mainly phosphoric acid and sulphuric acid) facilitate economies of scale and operational integration.

2.4 Downstream Markets for Phosphate-Based Products

Phosphate minerals and their derivative products feed into a wide range of downstream markets that are fundamental to national development, industrialisation, and food security. Understanding these markets is essential for assessing the economic potential and industrial linkages of a domestic phosphate industry in Uganda.

The major downstream markets include agriculture, livestock feed, agro-processing, and industrial chemical clusters, each offering unique growth and value addition opportunities.

2.4.1 Agricultural Fertiliser Market

A. Overview

The agricultural sector is the largest global consumer of phosphate-based products, accounting for more than **80% of total phosphate demand**. In Uganda, low soil fertility—particularly phosphorus deficiency—is a key constraint to agricultural productivity.

B. Market Drivers

- Rapid population growth demanding increased food production
- Soil nutrient depletion due to continuous cultivation
- Transition toward commercial agriculture and higher-yield hybrid varieties
- Government programmes promoting fertiliser adoption (e.g., e-voucher systems, input subsidies)

C. Key Phosphate Fertilisers Used

- SSP, TSP
- DAP, MAP
- NPK blends and compound fertilisers

D. Market Potential in Uganda

- Fertiliser use in Uganda remains **<5 kg/ha**, among the lowest globally
- NDP IV targets a significant increase in fertiliser application
- Large potential market in both smallholder and commercial farming
- Regional export opportunities in EAC and COMESA

E. Strategic Fit

A domestic fertiliser industry directly supports:

- Food security enhancement
- Yield improvement and agro-industrial expansion
- Import substitution and forex savings
- Regional supply integration

2.4.2 Livestock Feed Industry

A. Overview

Demand for poultry, pork, dairy, and beef is rising rapidly due to urbanisation, income growth, and dietary shifts. Feed phosphates such as **DCP and MCP** are essential additives in livestock and poultry diets.

B. Market Drivers

- Increased commercialisation of poultry and piggery
- Expansion of dairy value chains
- Growth of aquaculture in Uganda and the region

- Rising demand for compound feed formulations

C. Key Products Used

- Dicalcium phosphate (DCP)
- Monocalcium phosphate (MCP)
- Defluorinated phosphate (DFP)

D. Market Potential

- Uganda imports nearly all feed-grade phosphates
- Feed costs account for **60–70%** of livestock production costs
- Local production would dramatically lower costs and improve competitiveness

E. Strategic Fit

A domestic supply of feed phosphates strengthens:

- Livestock productivity
- Feed mills and agro-industrial clusters
- Regional trade potential in the Great Lakes and East Africa

2.4.3 Agro-Processing and Food Manufacturing

A. Overview

Food-grade phosphates serve numerous functions in processing, stabilisation, preservation, and product formulation. As Uganda's agro-processing sector expands, demand for these products increases.

B. Key Food and Beverage Uses

- pH stabilisation in beverages
- Emulsifiers in dairy and cheese production
- Moisture retention in meats and processed foods
- Leavening agents in bakery products
- Nutrient fortification in cereals and infant foods

C. Market Drivers

- Growth of Uganda's formal food manufacturing sector
- Increasing domestic and regional demand for processed foods
- Expansion of supermarkets and modern food distribution systems
- Rising export opportunities within EAC and COMESA

D. Strategic Fit

Local production of food-grade phosphates enables:

- Import substitution for critical food additives
- Enhanced competitiveness of agro-processors
- Assurance of quality and supply stability

2.4.4 Industrial Clusters (Detergents, Ceramics, Steel, Water Treatment)

Phosphate chemicals feed into a multitude of non-agricultural industrial clusters, each with strong potential for development in Uganda's industrial parks.

1. Detergent and Cleaning Products

- STPP and related phosphates are essential in detergent formulations
- Uganda imports large volumes of detergents annually
- Domestic STPP supports local manufacturing and reduces imports

2. Ceramics and Construction Materials

- Phosphate binders improve the strength and thermal stability of ceramics
- Growing demand from construction, tiling, and furnace lining industries

3. Steel and Metal Surface Treatment

- Phosphate chemicals prevent corrosion and improve paint adhesion
- Industrial users include steel rolling mills, automotive assembly, and machinery fabrication

4. Water Treatment and Environmental Services

- Phosphate-based corrosion inhibitors and scale preventives are used in:
  - Municipal water systems
  - Industrial boilers and cooling towers
  - Wastewater treatment plants.

E. Industrial Market Potential

- Uganda's industrialisation agenda is strengthening demand for chemicals
- Industrial phosphate products support emerging industrial parks (e.g., Mbale, Namanve, Tororo)
- Regional industrial markets (Kenya, Tanzania, Rwanda) offer export opportunities

F. Strategic Fit

Developing industrial-grade phosphate products promotes:

- Multi-sector diversification
- Local manufacturing ecosystem development
- Higher domestic value addition

- Technology transfer and skills upgrading

2.4.5 Overall Strategic Importance of Downstream Markets

Developing downstream markets for phosphate-based products enables Uganda to:

- Transform a natural resource into multiple industrial value chains
- Strengthen agriculture, livestock, food processing, and manufacturing
- Reduce import dependency for fertilisers and chemicals
- Stimulate industrial park development and backward/forward linkages
- Create a platform for regional export leadership

Phosphate-based products thus represent a unique opportunity to drive **broad-based economic transformation** consistent with NDP IV, the Tenfold Strategy, and Uganda's industrialisation ambitions.

2.5 Implications for Uganda's Industrialisation

Uganda's phosphate deposits present a rare opportunity to catalyse large-scale industrial transformation. Phosphates are not only essential for fertiliser production but also form the backbone of numerous industrial chemicals, feed additives, construction materials, and specialised manufacturing processes. Harnessing these opportunities can significantly advance Uganda's structural transformation agenda, deepen industrial linkages, and expand regional trade competitiveness.

The implications for Uganda's industrialisation are discussed under three pillars: **vertical diversification**, **horizontal diversification**, and **import substitution**.

2.5.1 2.5.1 Opportunities for Vertical and Horizontal Diversification

2.5.1.1 A. Vertical Diversification

Vertical diversification refers to adding value along the phosphate value chain—from mining to high-value finished products.

Uganda's phosphate resources enable a progression through the following industrial layers:

1. **Upstream**
    - Mining and beneficiation
    - Phosphate concentrate production
2. **Midstream**

    - Sulphuric acid and phosphoric acid production
    - Fertiliser manufacturing: SSP, TSP, DAP, MAP
    - Feed-grade phosphates (DCP, MCP)
3. **Downstream (High Value)**
    - Industrial chemicals (STPP, sodium phosphates, polyphosphates)
    - Pharmaceuticals and food additives
    - Water treatment chemicals
    - Ceramic binders, flame retardants, metal surface treatments

Each downstream step has **increasing value addition**, deeper technology requirements, and stronger industrial linkages. Vertical integration improves margins, ensures quality control, stabilises supply chains, and maximises economic benefits from Uganda's mineral base.

2.5.1.2 Horizontal Diversification

Horizontal diversification expands production into related sectors that use phosphate-based chemicals as inputs.

Key sectors include:

- **Agro-processing:** Food additives, stabilisers, emulsifiers
- **Livestock feed manufacturing:** DCP/MCP for poultry, dairy, piggery, and aquaculture
- **Chemical manufacturing:** Detergents, industrial cleaners
- **Construction materials:** Phosphogypsum-based boards, ceramics
- **Metallurgy:** Anti-corrosion chemicals, metal pre-treatment
- **Green technologies:** Battery materials, water purification reagents

Horizontal diversification reduces reliance on agricultural fertiliser demand alone, diversifies Uganda's export basket, and supports the growth of multiple industrial clusters.

2.5.2 2.5.2 Potential for Downstream Value-Added Industries

Developing a phosphate-based industrial ecosystem in Uganda has transformative potential for downstream value chains. Key opportunities include:

A. Agro-Industrial Value Chains

- NPK blending for specific soils/crops
- Fortification of agricultural inputs
- Integration with seed, pesticide, and extension service markets
- Enhanced crop productivity for agro-processing industries (tea, coffee, sugar, palm oil)

B. Livestock and Poultry Industry

- Reduced cost of feed supplements, improving competitiveness
- Strengthening Uganda's dairy, poultry, piggery, and beef industries
- Supporting emerging aquaculture feed manufacturing

C. Chemical Manufacturing Clusters

- Detergents and cleaning chemicals
- Solvents and industrial reagents
- Corrosion inhibitors and speciality metal treatments
- Surfactants and plasticiser chemicals

D. Construction and Engineering Materials

- Ceramic binders
- Flame retardants
- Phosphogypsum-based construction products
- Industrial coatings and adhesives

E. Food and Pharmaceutical Industries

- Production of food-grade phosphates (E-number additives)
- Manufacturing of pharmaceutical intermediates and excipients
- Strengthening Uganda's pharmaceutical manufacturing capacity

F. Green and Sustainable Technologies

- Battery materials (lithium iron phosphate)
- Water purification and sanitation products
- Environmental remediation agents (phosphate stabilisers)

Each of these industries contributes to job creation, technology diffusion, and the strategic development of regional export capabilities.

2.5.3 Import Substitution Opportunities Across Multiple Sectors

Uganda is currently highly import-dependent on phosphate-based products. Local production would unlock significant import substitution potential.

A. Fertiliser Sector

- Uganda imports nearly all phosphate fertilisers (DAP, MAP, TSP, NPKs).
- Import substitution saves foreign exchange and stabilises domestic supply.
- Regional sales to Kenya, Tanzania, Rwanda, Burundi, South Sudan, and DRC create export-led growth.

B. Livestock Feed Sector

- DCP and MCP imports contribute heavily to high feed costs.
- Local production reduces livestock input prices by **10–20%**, improving competitiveness.

C. Industrial Chemicals

The following products are fully imported yet widely used across industries:

- STPP (detergents)
- Metal surface treatment chemicals
- Ceramic binders
- Industrial phosphates for food processing and water treatment

Local production would:

- Reduce costs for manufacturers
- Support import substitution in detergent and industrial chemical markets
- Promote growth of domestic industrial parks

D. Pharmaceuticals and Food Processing

Uganda imports:

- Food-grade phosphates
- pH regulators
- Pharmaceutical excipients and additives

Domestic availability strengthens:

- Uganda's pharmaceutical manufacturing
- The agro-processing industry
- Compliance with quality and supply chain standards

E. Potential Foreign Exchange Savings

A fully integrated phosphate industry could save Uganda **tens of millions of dollars annually**, while generating export earnings from high-value derivatives.

2.5.4 Implications for Uganda's Industrial Transformation

Developing the phosphate value chain offers Uganda a unique opportunity to:

- Move from raw mineral exports to **high-value industrial production**
- Build competitive **fertiliser, feed, food processing, and chemical industries**
- Strengthen agricultural productivity and food security
- Integrate into **regional industrial markets** as a supplier

- Support the creation of **industrial clusters** in Tororo, Mbale, and Namanve
- Position the country as a **strategic manufacturing hub** in EAC and COMESA

Phosphates, therefore, present a strong foundation for achieving Uganda’s structural transformation goals under **NDP IV**, the **Tenfold Strategy**, and the **Agro-Industrialisation Programme**.

3 Mapping of the Phosphate-to-Fertiliser Value Chain

3.1 Global Value Chain Overview

The global phosphate-to-fertiliser value chain is a complex, highly integrated system that spans mining, chemical processing, fertiliser manufacturing, logistics, industrial utilisation, and agricultural application. A few countries with large, high-quality phosphate rock reserves and strong industrial infrastructure dominate it. Understanding this global landscape is crucial for identifying Uganda's competitive advantages, potential risks, and strategic positioning opportunities.

3.1.1 Structure of the Global Phosphate Value Chain

The global value chain can be divided into four main stages:

1. Upstream: Phosphate Ore Extraction and Beneficiation

- Major activities: mining, crushing, washing, flotation, drying
- Outputs: beneficiated phosphate concentrate (28–40% P_2O_5)
- Leading producing countries: **Morocco, China, USA, Jordan, Saudi Arabia, Russia, Egypt**
- Morocco (OCP) controls over **70% of global reserves**, shaping global supply and pricing dynamics.

Key determinants of competitiveness:

- Ore grade and mineralogy
- Mining costs
- Beneficiation efficiency and technology

2. Midstream: Acidulation and Fertiliser Manufacturing

This segment includes conversion of phosphate rock into phosphoric acid and production of various fertilisers.

a) Acidulation

- **Phosphoric acid (PA)** production (wet-process)
- Requires large volumes of sulphuric acid (SA)
- Co-generation of phosphogypsum as by-product

b) Fertiliser Production

- **Single Superphosphate (SSP)**
- **Triple Superphosphate (TSP)**
- **Di-ammonium Phosphate (DAP)**

- **Mono-ammonium Phosphate (MAP)**
- **NPK formulations** (compound or blended)

Global DAP/MAP leaders:

- Morocco (OCP), USA (Mosaic), Saudi Arabia (Ma'aden), Russia (PhosAgro), China

3. Downstream: Distribution and Logistics

Phosphate fertilisers are bulky commodities that require efficient logistics:

- Ports, bulk terminals, and conveyor systems
- Rail and road transport networks
- Storage facilities for acids, fertilisers, and raw materials
- National and regional distributor networks
- Last-mile delivery to agro-dealers and farmers

High logistics costs significantly influence delivered fertiliser prices, particularly in landlocked countries such as Uganda.

4. End Use: Agricultural, Industrial, and Feed Markets

Globally:

- **80–85%** of phosphate products are used in fertiliser manufacturing
- **5–10%** in animal feed (DCP/MCP)
- **5–10%** in industrial chemicals (STPP, food additives, pharmaceuticals)

Agriculture is the dominant end-use, particularly in emerging markets where fertiliser consumption is increasing to support food security and intensification.

3.1.2 Key Global Players and Market Dynamics

1. Morocco (OCP)

- Controls **70–75%** of global reserves
- Fully integrated value chain: rock → acid → fertilisers → industrial chemicals
- Major exporter of DAP/MAP, phosphoric acid, and rock

2. China

- Largest producer and consumer of phosphates
- Highly diversified industrial phosphate chemicals industry
- Influences global price trends through export quotas and domestic policy shifts

3. United States (Mosaic)

- Major DAP/MAP producer
- Highly mechanised mining operations

- Significant exporter to Latin America and Asia

4. Middle East (Saudi Arabia, Jordan)

- Modern, large-scale integrated complexes (e.g., Ma'aden)
- Strategically located near major shipping corridors
- Low-cost inputs (energy, sulphur) give them a competitive edge

3.1.3 Key Global Trends Affecting the Value Chain

1. Increasing Fertiliser Demand

- Population growth and food demand
- Expansion of commercial agriculture in Africa and Asia
- Greater emphasis on balanced fertilisation (NPK use rising)

2. Supply Concentration and Price Volatility

- Heavy concentration of reserves in Morocco and China
- Export restrictions and geopolitical risks impact global prices
- Price spikes in 2008, 2021–2022 demonstrated vulnerability for import-dependent economies

3. Shift Toward High-Analysis Fertilisers

- DAP, MAP, TSP preferred due to lower transport costs per nutrient
- NPK blends increasingly tailored to soil and crop requirements

4. Sustainability and Environmental Regulations

- Pressure to reduce fluorine emissions and manage phosphogypsum
- Adoption of circular economy practices (waste recycling, recovery of phosphate from wastewater)

5. Growth of Industrial and Feed Phosphate markets

- Expanded demand for DCP/MCP with commercial livestock growth
- Rising use of phosphates in detergents, water treatment, and food industries

3.1.4 Strategic Lessons from the Global Value Chain

1. **Full integration improves competitiveness.** Countries controlling mining → beneficiation → acids → fertilisers achieve lower production costs.
2. **Scale matters.** Integrated complexes operating at 1–3 million tonnes per year are most cost-efficient.

3. **Logistics are critical.** Transport accounts for up to **40%** of landed fertiliser costs in landlocked regions.
4. **Product diversification stabilises revenue.** Industrial phosphates, feed phosphates, and fertilisers create multiple revenue streams.
5. **Strong state or quasi-state institutions facilitate long-term planning.** OCP (Morocco), Ma'aden (Saudi Arabia), JPMC (Jordan) demonstrate the role of coordinated investment.

3.1.5 Relevance for Uganda

Understanding the global value chain reveals:

- Uganda must build **midstream processing capacity** (acidulation + fertiliser production) to capture value.
- Beneficiation alone is insufficient for competitiveness.
- Diversification into **NPK blends, SSP/TSP, DAP/MAP, DCP/MCP, and STPP** is viable with the right infrastructure.
- Strategic positioning as a **regional supplier** to EAC and COMESA markets is achievable.
- Building efficient logistics (rail, bulk handling, storage) is essential to reducing production and distribution costs.

3.2 Uganda's Current Value Chain Performance

Uganda's phosphate-to-fertiliser value chain remains nascent and fragmented, characterised by heavy import dependency, limited domestic processing capacity, and underdeveloped distribution systems. While the country hosts strategic phosphate deposits at Sukulu in Tororo, these resources have not yet been fully transformed into a competitive fertiliser or industrial chemical industry. Mapping the current performance of the value chain reveals both the constraints and opportunities that shape Uganda's potential to build a vertically integrated phosphate sector.

3.2.1 Upstream: Mining and Beneficiation

1. Resource Availability

- Uganda's primary known phosphate resource is the **Sukulu carbonatite complex** in Tororo.
- Historical and recent geological assessments estimate **approximately 200+ million tonnes** of phosphate ore, with raw ore grades around **12–14% P_2O_5**.

- This grade requires beneficiation before acidulation or high-analysis fertiliser production.

2. Current Mining Activity

- Mining activity has intermittently begun but remains sub-optimal due to:
  - Limited beneficiation infrastructure
  - Challenges with ore handling and processing
  - Incomplete downstream integration
  - Governance, licensing, and operational coordination issues

3. Beneficiation Capacity

- Uganda **does not yet have a fully operational mineral beneficiation plant** producing high-grade phosphate concentrate suitable for phosphoric acid production.
- Existing beneficiation attempts have been limited in scale, technology, and consistency.

Performance Summary (Upstream)

Table 3- 1: Performance Summary (Upstream)

Indicator	Current Status	Implication
Ore quality	Medium grade (12–14% P_2O_5)	Requires significant beneficiation
Beneficiation capacity	Limited	Bottleneck for midstream processing
Operational continuity	Unstable	Weakens reliability for downstream investors
Infrastructure	Improving (Tororo industrial zone, power access)	Potential for cluster development

3.2.2 Midstream: Acidulation and Fertiliser Manufacturing

Uganda currently lacks industrial-scale chemical processing capacity to convert phosphate rock into phosphoric acid and related fertilisers.

1. Phosphoric Acid Production

- **No wet-process phosphoric acid (PA) plant** is operational in Uganda.
- All PA required for DAP, MAP, TSP, DCP, or MCP must be imported if fertilisers were to be manufactured domestically.

2. Sulphuric Acid Production

- Current sulphuric acid production exists on a small scale for industrial use but **not at the scale required** for phosphoric acid manufacturing (~3 tonnes of SA per tonne of P_2O_5).
- Fertiliser-grade SA production capacity is absent.

3. Fertiliser Manufacturing and Blending

- Uganda has **NPK blending facilities**, but these rely entirely on imported:
  - DAP
  - MAP
  - MOP
  - Urea
- No domestic production of SSP, TSP, DAP, MAP, or compound fertilisers occurs at an industrial scale.

Midstream Performance Summary

Table 3- 2: Midstream Performance

Segment	**Status**	**Key Limitation**
PA production	Non-existent	Limits ability to make high-value fertilisers
SA production	Small-scale	Insufficient for acidulation
Fertiliser production	Limited to blending	100% dependent on imported intermediates
Technology	Fragmented	No integrated chemical complex

Uganda's midstream gap is the **most binding bottleneck** preventing value addition from phosphate resources.

3.2.3 Downstream: Distribution, Marketing, and End-Use Systems

1. Fertiliser Distribution System

- Fertilisers are distributed mainly through:
  - Private agro-dealers
  - Cooperatives
  - NGO-supported channels
- The system is **fragmented, price-sensitive, and dominated by imports**.

2. Agricultural Fertiliser Use

- Uganda has one of the lowest fertiliser application rates globally: **<5 kg/ha**, compared to the African average of 20 kg/ha and global average >130 kg/ha.
- Low adoption is driven by:
  - High retail prices due to transport and import dependency
  - Limited access to credit
  - Weak extension systems
  - Low farmer awareness of fertiliser benefits

3. Feed and Industrial Markets

- Uganda imports all feed-grade phosphates (DCP, MCP).
- Industrial phosphate use (detergents, ceramics, water treatment) relies entirely on imports.
- No domestic production of industrial phosphate chemicals exists.

Downstream Performance Summary

Table 3- 3: Downstream Performance Summary

Sector	Current Status	Implication
Fertiliser use	Very low	High potential for market expansion
Feed industry	Fully import-dependent	Opportunity for DCP/MCP production
Industrial uses	Fully import-dependent	Lost economic value
Market structure	Fragmented	Inefficient distribution and high markups

3.2.4 Logistics and Infrastructure Performance

1. Transport

- Fertilisers reach Uganda via **Mombasa** and **Dar es Salaam** ports.
- Transport costs—fuel, handling, road wear—significantly increase landed prices.
- Lack of bulk handling and rail transport infrastructure is a major constraint.

2. Power and Water

- Tororo benefits from growing industrial infrastructure and grid access, but:
  - Sulphuric acid plants require reliable, high-capacity power

- Phosphoric acid plants require continuous water and energy supply

3. Industrial Parks

- Tororo, Mbale, and Namanve industrial parks provide potential sites for cluster development.
- Infrastructure gaps still need addressing: wastewater management, shared utilities, chemical handling facilities.

3.2.5 Policy, Institutional, and Regulatory Performance

Strengths

- Fertiliser Policy (2016) provides a framework for quality control
- Mining Act (2022) modernises mineral governance
- Alignment with NDP IV's agro-industrialisation and mineral beneficiation agenda
- Government commitment to import substitution and local content

Weaknesses

- Limited institutional coordination between:
  - MEMD
  - MAAIF
  - NEMA
  - UIA
  - Private sector
- Unclear roles in phosphate sector development
- Gaps in investment promotion and long-term PPP structuring
- Standards enforcement challenges (fertiliser quality, counterfeit inputs)

3.2.6 Summary of Uganda's Value Chain Performance

Table 3- 4: Summary of Uganda's Value Chain Performance

Stage	Current Performance	Key Opportunity
Upstream	Medium-grade ore, limited beneficiation	Build beneficiation capacity
Midstream	No acidulation or fertiliser manufacturing	Develop integrated SA + PA + fertiliser complex

Stage	Current Performance	Key Opportunity
Downstream	Fragmented markets, high import dependency	Fertiliser, feed phosphate, and industrial chemical production
Logistics	High transport costs	Invest in bulk handling and rail connectivity
Institutional	Policy partially supportive	Build strong institutional coordination and governance

3.2.7 Implications for Feasibility and Strategic Positioning

Uganda's current value chain performance reveals:

- **High potential but unrealised value** due to missing midstream infrastructure
- A significant opportunity to become a **regional fertiliser and industrial phosphate hub**
- The need for a **fully integrated phosphate industrial complex** (beneficiation → PA → SA → fertilisers → feed/industrial products)
- Strong justification for Government intervention under the DC Guidelines due to market failures and strategic national importance

3.3 Actor Mapping

Actor mapping identifies the key organisations and stakeholders involved in, influencing, or affected by the phosphate-to-fertiliser value chain. With the **National Mining Company (NMC)** now operational and mandated to take a **lead role in strategic mineral development**, including phosphate mining and beneficiation, the structure of upstream governance has evolved significantly.

The NMC's leadership has implications for coordination, investment alignment, PPP structuring, and long-term sector planning.

Below is the updated actor mapping.

3.3.1 Upstream Actors (Mining and Beneficiation)

These actors are responsible for geological assessment, mining, beneficiation, and initial value addition. Under the new institutional landscape, **NMC leads this segment**.

1. National Mining Company (NMC) – *Lead Upstream Actor*

NMC is the Government of Uganda's commercial mining arm with the mandate to:

- Lead and/or participate in **strategic mineral development**, including phosphates;
- Enter Joint Ventures (JVs) and PPPs with private investors;
- Mobilise capital for mining and beneficiation projects;
- Ensure national ownership in strategic minerals;
- Promote responsible mining, community management, and environmental compliance;
- Oversee the development of beneficiation plants for phosphate concentrate.

Role in the phosphate value chain:

- Principal custodian of the Sukulu phosphate deposit;
- Lead developer of mining and beneficiation operations;
- Key negotiator and shareholder in partnerships with private investors;
- Ensures strategic alignment with national industrialisation goals under NDP IV.

2. Directorate of Geological Survey and Mines (DGSM), MEMD

Role:

- Regulatory oversight, licensing, compliance enforcement;
- Resource mapping and geological data provision;
- Supervisory support to NMC where required.

3. Private Mining Companies / JV Partners

Potential partners include:

- International mining investors;
- Fertiliser companies seeking backward integration;
- Local investors.

Role:

- Provide capital, technology, and operational expertise for mining and beneficiation;
- Enter into JVs with NMC for large-scale extraction and processing.

4. Local Governments (Tororo District)

Role:

- Land access facilitation;
- Community engagement;
- Local infrastructure support;
- Coordination of local development benefits.

3.3.2 Midstream Actors (Acidulation, Fertiliser and Chemical Manufacturing)

1. Fertiliser and Chemical Manufacturers

Potential investors in:

- Sulphuric acid plants
- Phosphoric acid plants
- DAP, MAP, SSP, TSP production facilities
- NPK compound/blending plants
- Feed phosphate and industrial chemical lines

These entities will likely operate through **PPP or JV models with NMC**, given upstream integration requirements.

2. Industrial Parks and Clusters

- Tororo Industrial Zone
- Mbale Industrial Park
- Kampala/Namanve Industrial Park

Role: Hosting midstream processing facilities with shared utilities.

3.3.3 Downstream Actors (Distribution and End Use)

1. Agro-Dealers and Fertiliser Distributors

- National input distributors
- Farmer cooperatives
- Large-scale aggregators

2. Farmers (Smallholder & Commercial). Primary end-users of fertilisers.

3. Feed Manufacturers. Consumers of DCP and MCP.

4. Industrial Users. Consumers of:

- STPP
- Water treatment phosphates
- Ceramic binders
- Metal surface treatment chemicals

5. Regional Export Markets (EAC/COMESA). Potential buyers of Ugandan fertilisers and industrial phosphate chemicals.

3.3.4 Support Service Actors

1. Transport and Logistics Providers

- URC (rail)
- Road transporters
- Inland container depots
- Port operators (Mombasa, Dar es Salaam)

2. Financial and Investment Institutions

- MoFPED, DFIs, commercial banks
- Development partners
- Private equity funds

3. Standards and Research Institutions

- UNBS
- NARO
- Makerere University, Busitema University
- Private laboratories

3.3.5 Regulatory, Policy, and Oversight Institutions

1. MEMD (Policy Lead for Mining). Provides oversight for NMC and the mining sector regulation.

2. MAAIF (Lead for Fertiliser Regulation). Quality control, extension services, and agricultural policy.

3. NEMA. Environmental and social safeguards, ESIA approvals.

4. NPA. Strategic guidance, project appraisal, and alignment with national development frameworks.

5. UIA. Investor facilitation, licensing, industrial park oversight.

6. URA. Customs procedures, tax incentives, import/export regulation.

7. UNBS. Fertiliser and chemical quality standards enforcement.

3.3.6 Relevance of NMC Leadership for the Value Chain

The operationalisation of the National Mining Company fundamentally strengthens the upstream segment by:

- Ensuring **state participation** and strategic oversight of phosphate resource development;

- Providing **institutional stability** for long-term mining operations;
- Reducing investment risk for private partners;
- Making integrated mining–to–fertiliser complexes more bankable;
- Enabling Uganda to capture greater **value addition** rather than exporting raw materials;
- Improving coordination between mining and midstream manufacturing operations.

In effect, NMC's leadership is a **critical enabler** for establishing a competitive, vertically integrated phosphate industrial cluster in Uganda.

3.4 Functional Mapping

Functional mapping outlines **what functions are performed** at each stage of the phosphate-to-fertiliser value chain, **who performs them**, and **how effectively these functions are executed**.

This provides a clear picture of systemic strengths and weaknesses, enabling evidence-based planning for value chain upgrading and industrialisation.

The major function groups include:

1. Upstream functions
2. Midstream functions
3. Downstream functions
4. Cross-cutting enabling functions

Each function affects the efficiency, competitiveness, and sustainability of the entire value chain.

3.4.1 Upstream Functions (Mining, Beneficiation, Resource Management)

These functions determine the quantity, quality, and reliability of phosphate ore supply for downstream processing.

1. Mineral Resource Exploration & Geological Mapping

Function:

- Geological surveys
- Resource estimation
- Drilling programmes and sampling

Actors:

- **National Mining Company (NMC)** – Lead
- DGSM (regulatory and data oversight)

- Private exploration partners

Current Status:

- Geological data exists but requires modernisation and detailed beneficiation trials.

2. Mining and Ore Extraction

Function:

- Overburden removal
- Ore extraction (drilling, blasting, excavation)
- Ore handling and stockpiling

Actors:

- **NMC (lead operator)**
- Private mining contractors under NMC supervision

Current Status:

- Mining activities have begun intermittently but lack stable beneficiation linkage.

3. Beneficiation (Upgrading Ore to Concentrate)

Function:

- Crushing, grinding
- Flotation, magnetic separation
- Removal of impurities (silica, carbonates)
- Drying and concentrate handling

Actors:

- **NMC-led beneficiation operations**
- Potential JV partners for mineral processing

Current Status:

- No fully operational industrial beneficiation plant yet; major bottleneck.

4. Mining Logistics & Infrastructure

Function:

- On-site haulage
- Stockpile management
- Internal roads, conveyors, water systems

Actors:

- NMC
- Tororo District Local Government
- Logistics contractors

Current Gaps:

- Infrastructure requires modernisation to handle high-volume operations.

3.4.2 Midstream Functions (Chemical Processing & Fertiliser Manufacturing)

These functions create the highest economic value and determine Uganda's competitiveness.

1. Acidulation – Sulphuric Acid Production

Function:

- Sulphur importation/sourcing
- Sulphur-burning or pyrite roasting
- SA plant operation (contact process)
- Heat/steam recovery

Actors:

- Private chemical firms
- PPP/JV with **NMC**
- Energy suppliers

Current Status:

- No fertiliser-grade SA plant in Uganda; large-scale gap.

2. Acidulation – Phosphoric Acid Production

Function:

- Wet-process PA production using SA + phosphate concentrate
- Gypsum handling
- Fluorine emissions control

Actors:

- Private investors
- PPP structure with **NMC**

Current Status:

- PA production absent; major missing link in value addition.

3. Fertiliser Manufacturing (SSP, TSP, DAP, MAP, NPK)

Function:

- Granulation and compaction
- Reactors and mixers
- Ammoniation systems
- Product curing and coating

Actors:

- Fertiliser manufacturers (existing blenders and future integrated producers)
- **NMC-linked joint venture partners**

Current Status:

- Uganda only has blending; no manufacturing of P-based fertilisers.

4. Industrial Phosphate Production

Function:

- STPP production
- Sodium/calcium phosphate salts
- Water treatment phosphate chemicals
- Ceramic binders

Actors:

- Chemical manufacturers
- Industrial park tenants
- PPPs/JVs with NMC

Current Status:

- No industrial phosphate chemical plants yet.

3.4.3 Downstream Functions (Distribution, Application, Industrial Use)

These functions link phosphate products to their final markets.

1. Fertiliser Distribution & Wholesaling

Function:

- National-level aggregation
- Storage in warehouses
- Transport to regions

Actors:

- Private fertiliser distributors
- Cooperatives
- Large agro-input companies

Gaps:

- Fragmented market; high transport margins.

2. Agro-Dealer Networks & Last-Mile Delivery

Function:

- Retailing to farmers
- Packaging and repackaging
- Farmer advisory services

Actors:

- Rural agro-dealers
- NGOs and development partners
- Input suppliers

Gaps:

- Limited knowledge transfer; adulteration issues.

3. Farmer Utilisation & Extension Services

Function:

- On-farm application
- Soil fertility management
- Farmer training

Actors:

- Farmers
- MAAIF extension services
- NARO research institutes

Gaps:

- Poor awareness; low fertiliser adoption (<5 kg/ha).
- **4. Industrial Use Integration**

Function:

- Use of phosphates in detergent, ceramics, metal treatment, pharmaceuticals
- Linking supply chains to industrial parks

Actors:

- Industrial manufacturers (detergents, ceramics, steel plants)
- Water utilities

Gaps:

- 100% import dependency; no local production.

3.4.4 Cross-Cutting Enabling Functions

These functions support the entire value chain.

1. Regulatory & Policy Functions

Actors:

- MEMD
- MAAIF
- NEMA
- URA
- UNBS

Function:

- Mining regulation
- Fertiliser quality control
- ESIA enforcement
- Tax and trade oversight

2. Financing & Investment Mobilisation

Actors:

- MoFPED
- DFIs
- Commercial banks
- PPP Unit
- Private equity, strategic investors

Function:

- Financing mining, beneficiation, acidulation plants, and industrial clusters.

3. Infrastructure Functions

Actors:

- MoWT, UNRA, URC
- UIA (industrial park development)
- NWSC, UEDCL

Function:

- Roads, rail, water, power, bulk handling infrastructure.

4. Research, Testing, and Quality Assurance

Actors:

- UNBS

- Makerere University, Busitema University
- NARO
- Private labs

Function:

- Soil testing
- Fertiliser certification
- Product formulation research

3.4.5 Summary Functional Map

Table 3- 5: Summary Functional Map

Value Chain Stage	Key Functions	Lead Actors	Performance Status
Upstream	Mining, beneficiation, resource management	**NMC**, DGSM, private JV partners	Partial; beneficiation inadequate
Midstream	SA, PA, fertiliser & industrial chemical manufacturing	Private sector + **NMC PPPs**	Largely absent; major gap
Downstream	Distribution, application, industrial use	Agro-dealers, farmers, industries	Underdeveloped, import-dependent
Enabling	Policy, Quality, Finance, Infrastructure	MEMD, MAAIF, NEMA, MoFPED, UIA	Fragmented; coordination gaps

3.4.6 Implications of Functional Mapping for Feasibility

Functional mapping reveals that:

- Uganda's value chain is **upstream-heavy** but weak in midstream functions.
- The absence of acidulation and manufacturing severely limits value addition.
- **NMC's leadership** provides a structural solution for upstream-to-midstream integration.
- Downstream functions (distribution, industrial use) are underdeveloped but present **high potential** for expansion.

- Strengthening enabling functions (policy coherence, financing mechanisms, infrastructure) is essential for a successful integrated phosphate industry.

3.5 Infrastructure and Logistics Flows

Efficient infrastructure and logistics systems are essential for the successful development of a competitive phosphate-to-fertiliser industry. The movement of phosphate ore, beneficiated concentrate, acids, fertiliser products, industrial chemicals, and imported raw materials requires integrated transport networks, bulk handling facilities, reliable energy and water systems, and adequate storage capacity.

This section assesses Uganda's current infrastructure and logistics performance and identifies gaps and opportunities that influence the viability of phosphate mining, acidulation, fertiliser manufacturing, and downstream distribution.

3.5.1 Transport Infrastructure for Mining and Beneficiation

1. On-site Transport and Haulage (Sukulu, Tororo)

- Current mine access roads are functional but require strengthening for large-scale haulage.
- Ore transport from pit to beneficiation plant will require:
    - Heavy-duty haul roads
    - Conveyors or truck-and-shovel systems
    - Dust control and stormwater management systems

Bottlenecks:

- Limited internal mine-road network
- Weather vulnerability due to inadequate drainage

3.5.2 Road Transport Network

1. National Road Network

- Tororo district is accessible via major highways such as:
    - Tororo–Mbale–Soroti corridor
    - Tororo–Jinja–Kampala corridor

These roads support movement of inputs (sulphur, ammonia, coal) and outputs (fertilisers, industrial chemicals).

2. Road Transport Challenges

- High fuel costs increase delivered cost of fertilisers

- Road maintenance challenges, axle-load limitations
- Dependence on long-haul trucking from Mombasa and Dar es Salaam ports
- Transport costs can account for **30–40%** of fertiliser retail price

Strategic Implication: Transport inefficiencies significantly limit Uganda's ability to compete with coastal producers.

3.5.3 Rail Transport

Uganda Railways Corporation (URC) currently operates limited freight services.

Opportunities:

- Rehabilitation of the **Tororo–Malaba–Mombasa Standard Gauge Railway (SGR)** corridor
- Upgrading of metre-gauge rail (MGR) lines for bulk mineral and fertiliser transport
- Potential for Tororo to be connected to regional rail networks serving Kenya and Tanzania

Implications for Phosphate Industry:

- Rail transport dramatically reduces bulk transport costs
- Essential for scaling production of SSP, TSP, DAP, MAP, NPKs
- Enables competitiveness in export markets within EAC and COMESA

3.5.4 Port Logistics (Mombasa and Dar es Salaam)

Phosphate rock beneficiation, acidulation, and fertiliser production require significant volumes of imported raw materials, particularly sulphur and ammonia.

1. Import Requirements

- **Sulphur** for sulphuric acid production
- **Ammonia** for DAP/MAP/NPK production
- **Specialised chemicals** for industrial phosphate products

2. Port Infrastructure Constraints

- Congestion at Mombasa port
- Limited bulk handling capacity for sulphur and ammonia
- Long turnaround times
- High demurrage and port charges

3. Implications

- Increased lead times for fertiliser manufacturing inputs

- Higher costs passed down to domestic fertiliser prices
- Strategic need for long-term contracts with port operators and logistics companies

3.5.5 Storage and Handling Facilities

1. Raw Material Storage

- Bulk sulphur storage must meet hazardous materials standards
- Ammonia storage requires specialised cryogenic tanks
- Phosphate concentrate storage requires enclosed sheds to prevent moisture contamination

2. Fertiliser Storage

- Granular fertiliser (DAP, MAP, TSP, NPK) requires dry, secure warehouses
- Bagging and palletising lines are essential for distribution
- **Gaps:**
- Limited bulk handling and warehousing infrastructure nationally
- Heavy reliance on private storage in central Uganda

3.5.6 Energy Infrastructure

1. Electricity Supply

- Tororo is served by the national grid and has access to Tororo Thermal Plant and Tororo Solar Plant
- Fertiliser and chemical plants require **continuous, high-capacity power**
- Power quality issues (voltage fluctuations) could disrupt acid production

2. Thermal Energy Requirements

- Sulphuric acid plants generate substantial waste heat, enabling cogeneration
- Integration of combined heat and power (CHP) systems supports lower operating costs
- **Gaps:**
- Need for dedicated industrial feeders
- Potential requirement for backup power (gas turbines or heavy fuel systems)

3.5.7 Water Supply and Industrial Utilities

Phosphoric acid plants are heavy water users. Requirements include:

- Process water
- Cooling systems

- Steam generation
- Wastewater treatment

Current Situation:

- National Water and Sewerage Corporation (NWSC) serves Tororo
- Additional boreholes and water recycling systems may be needed for large-scale operations

Gaps:

- Limited industrial wastewater treatment capacity
- Need for environmentally compliant phosphogypsum disposal facilities

3.5.8 Waste Management and Environmental Infrastructure

1. Phosphogypsum Disposal

- PA production generates large quantities of phosphogypsum
- Requires engineered stacks with liners and leachate management systems
- Potential use in construction materials requires regulatory approval

2. Air Emissions Control

- Fluorine and dust emissions require scrubbers
- Sulphur dioxide control is essential for SA plants

3. Solid and Hazardous Waste Handling

- Chemical plants produce hazardous waste streams requiring secure disposal

3.5.9 Digital and Information Systems Infrastructure

Effective value chain operations rely on:

- Inventory tracking
- Production monitoring
- GIS-based logistics planning
- Fertiliser quality certification systems
- National farmer registry and e-extension systems

Uganda's digital infrastructure is improving but requires integration into the fertiliser value chain for efficiency.

3.5.10 Summary of Infrastructure and Logistics Performance

Table 3- 6: Summary of Infrastructure and Logistics Performance

Element	Current Status	Strategic Implication
Road network	Functional but costly	High fertiliser retail prices
Rail	Underutilised	Major opportunity for cost reduction
Ports	Congested	Increases input import costs
Power	Accessible but inconsistent	Risk for continuous acid production
Water	Available but limited for large-scale PA	Requires augmentation
Storage	Limited bulk capacity	Constrains scaling and export
Waste management	Insufficient	Must be developed for PA/SA plants

3.5.11 Strategic Opportunities for Infrastructure Upgrading

1. Develop dedicated **bulk handling and storage facilities** in Tororo.
2. Invest in or partner to rehabilitate **rail connections** for minerals and fertiliser transport.
3. Establish **industrial utilities** (steam, compressed air, water recycling) in Tororo Industrial Park.
4. Build **phosphogypsum utilisation plants** to convert waste into construction materials.
5. Strengthen NWSC capacity for **industrial wastewater treatment**.
6. Promote **logistics PPPs** for sulphur and ammonia importation.

3.6 Governance Structure of the Value Chain

Effective governance is critical for ensuring that Uganda's phosphate-to-fertiliser value chain operates efficiently, transparently, and in alignment with national development priorities. Governance structures define **who makes decisions**, **who implements them**, **who regulates**, and **how different actors coordinate** across the mining, beneficiation, chemical processing, fertiliser manufacturing, distribution, and end-use segments.

Given the strategic importance of phosphates as a mineral resource and industrial input, Uganda's governance framework must ensure strong coordination between public institutions, private sector actors, and development partners. This section maps the governance architecture, identifies gaps, and outlines implications for industrialisation.

3.6.1 Governance Structure Overview

The governance of Uganda's phosphate value chain operates across **five interconnected domains**:

1. **Strategic resource ownership and upstream leadership** – led by the *National Mining Company (NMC)*
2. **Regulatory oversight and compliance** – led by MEMD, DGSM, NEMA, MAAIF, UNBS, and URA
3. **Industrial and investment governance** – involving UIA, MoFPED, PPP Unit, and industrial park authorities
4. **Market governance** – involving agro-dealers, standards enforcement, quality control, and extension systems
5. **Local governance and community relations** – involving Tororo District and local community structures

Each domain performs specific roles that shape the value chain's performance and investment climate.

3.6.2 Upstream Governance – Mining and Beneficiation

1. National Mining Company (NMC) – Lead Entity

NMC is the Government's commercial vehicle for strategic mineral development. Its key governance functions include:

- Overseeing **mining and beneficiation operations**
- Negotiating and managing **joint ventures and PPPs** with private investors
- Ensuring **state participation** and national value retention
- Coordinating infrastructure requirements at the mining and beneficiation stages
- Ensuring adherence to mining laws and environmental standards

NMC acts as the **anchor institution** for upstream governance, ensuring stability, transparency, and alignment with national mineral development goals.

2. MEMD – Ministry of Energy and Mineral Development

Role:

- Policy oversight and regulation
- Strategic direction for mineral development

- Supervisory authority over NMC
- Harmonisation with energy, mineral, and industrial policies

3. DGSM – Directorate of Geological Survey and Mines

Role:

- Mining licensing and compliance monitoring
- Geological data management
- Safety and environmental regulation at mine sites

4. Local Government (Tororo District)

Role:

- Land access governance
- Community engagement and benefit-sharing
- Local infrastructure coordination

3.6.3 Midstream Governance – Acidulation and Fertiliser Manufacturing

1. Industrial Policy and Investment Governance (MoFPED, UIA)

MoFPED and the Uganda Investment Authority (UIA) provide governance through:

- Investment incentives and fiscal frameworks
- Approval processes for industrial projects
- Industrial park development agreements
- Allocation of land for fertiliser complexes

2. PPP Governance (PPP Unit)

Role:

- Structuring PPPs between NMC and private investors
- Ensuring risk-sharing frameworks
- Advising on concession agreements, viability gap funding, and guarantees

3. Environmental Governance (NEMA)

Role:

- Approval of ESIAs for acidulation plants (SA, PA)
- Enforcement of air and water pollution standards
- Monitoring of phosphogypsum management

4. Standards and Chemical Safety Governance (UNBS, DoHS)

Role:

- Fertiliser standards enforcement

- Chemical safety and occupational health regulation
- Certification of industrial phosphate products

3.6.4 Downstream Governance – Distribution, Market Regulation, End-Use

1. MAAIF – Ministry of Agriculture, Animal Industry and Fisheries

Role:

- Fertiliser regulation under the National Fertilizer Policy
- Quality control of fertilisers at entry, blending plants, and retail points
- Agricultural extension services for fertiliser adoption
- Oversight of feed phosphates and livestock nutrition guidelines

2. Agro-Dealer and Distributor Networks (Private Sector)

Role:

- Self-regulation through associations
- Market coordination and information dissemination
- Adherence to UNBS quality standards

3. URA – Uganda Revenue Authority

Role:

- Import/export compliance
- Enforcement of tariff regimes and incentives
- Prevention of smuggling and counterfeit fertilisers

3.6.5 Cross-Cutting Governance Structures

1. National Planning Authority (NPA)

Role:

- Ensures project alignment with NDP IV, Tenfold Strategy, and mineral beneficiation strategies
- Coordinates feasibility appraisal and DC submission requirements
- Leads inter-agency harmonisation for national value chain development

2. Development Partners and Multilateral Institutions

Role:

- Technical assistance in capacity building
- Support for agricultural transformation and input market efficiency
- Financing infrastructure and industrial development projects

3. Private Sector Associations

- Uganda Manufacturers Association (UMA)
- Uganda Fertilizer Association (UFA)
- Uganda National Agro-Input Dealers Association (UNADA)

Role:

- Advocacy
- Training and standards enforcement
- Market coordination

3.6.6 Governance Gaps in Uganda's Current Value Chain

Table 3- 7: Governance Gaps in Uganda's Current Value Chain

Governance Area	Identified Gap	Implication
Upstream coordination	Weak linkage between NMC, DGSM, and private partners	Slows mining-to-beneficiation integration
Midstream development	Absence of industrial PA/SA governance frameworks	Limits investment in fertiliser complexes
Standards enforcement	Fertiliser adulteration, variable product quality	Reduces farmer trust and adoption
Market governance	Fragmented distribution system	High retail fertiliser prices
Environmental governance	Inadequate phosphogypsum and emissions management systems	Risk to project sustainability
Infrastructure governance	No dedicated bulk or rail governance structure	High logistics costs

3.6.7 Strengthening Governance for an Integrated Phosphate Industry

To unlock Uganda's phosphate industrialisation potential, governance must:

1. **Empower NMC** to drive upstream and midstream investments through clear mandates.
2. Establish a dedicated **Phosphate Industry Steering Committee** bringing together:
   - NMC

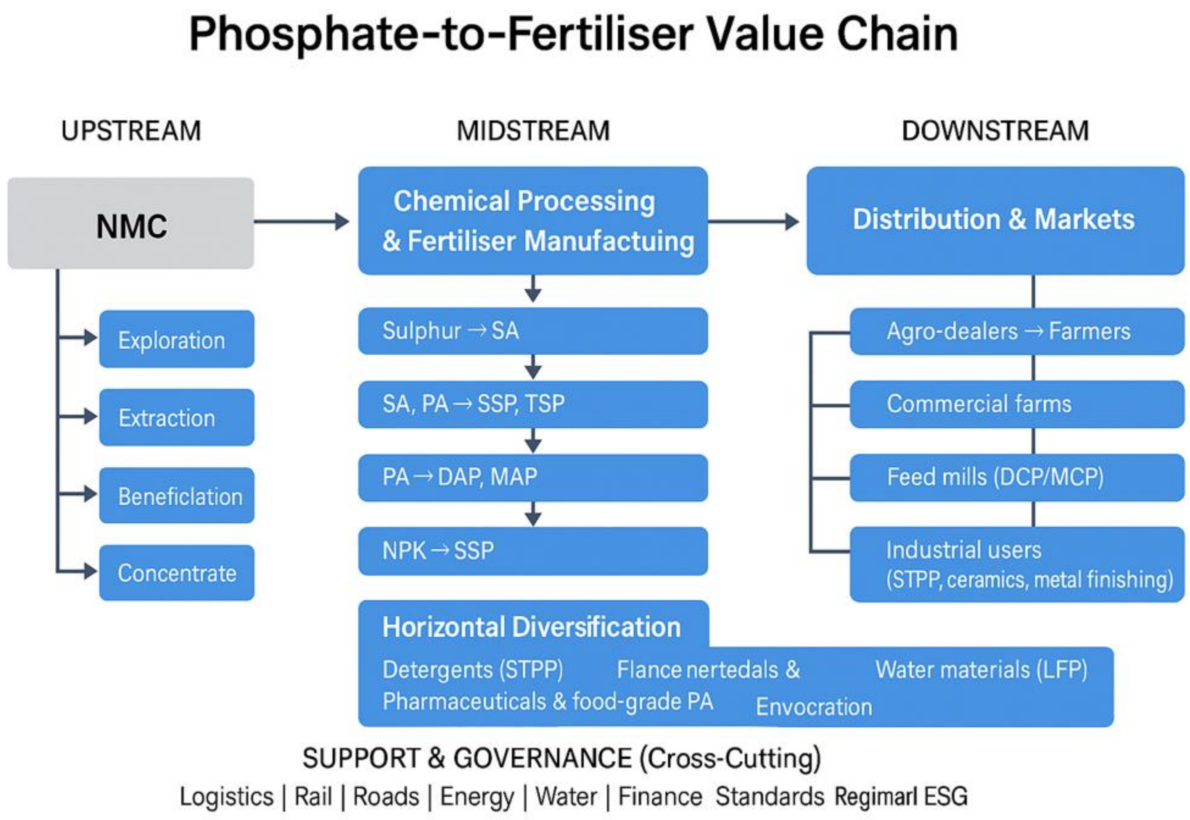
- MEMD
- MAAIF
- MoFPED
- NPA
- NEMA
- UIA
- Private sector

3. Harmonise **regulatory frameworks** across mining, chemical processing, fertiliser standards, and environmental management.
4. Strengthen PPP governance for large-scale industrial investments.
5. Enhance **market governance mechanisms** to regulate fertiliser quality, pricing transparency, and last-mile distribution.
6. Develop industrial cluster governance within **Tororo and Mbale Industrial Parks** for shared utilities, waste management, and logistics.

Uganda's phosphate-to-fertiliser value chain requires a **coordinated governance architecture** anchored by the National Mining Company and supported by strong regulatory, policy, investment, and market oversight structures. Transforming phosphate resources into a competitive industrial sector will depend heavily on how well these governance functions are harmonised and strengthened to support integrated, large-scale investment.

3.6.8 Value Chain Map

Figure 3- 1: Value Chain Map



3.6.9 Actor–Function Matrix

3.6.9.1 Matrix

This matrix maps **all major actors** across the upstream, midstream, and downstream stages, and outlines **their core functions, responsibilities, gaps, and interfaces** across the value chain.

Table 3- 8: Actor–Function Matrix

Value Chain Segment	Actor	Role / Function	Key Responsibilities	Current Gaps / Issues
UPSTREAM: Exploration, Mining & Beneficiation	**National Mining Company (NMC)**	Lead upstream operator	✓ Exploration & resource delineation ✓ Mining operations (drilling, blasting, hauling)	✓ Limited beneficiation technology ✓ Requires CAPEX for modern processing

Value Chain Segment	Actor	Role / Function	Key Responsibilities	Current Gaps / Issues
			✓ Beneficiation (crushing, grinding, flotation) ✓ Stockpiling & ore quality control	✓ Environmental and relocation challenges
	Ministry of Energy & Mineral Development (MEMD)	Sector oversight & regulation	✓ Regulatory supervision of mining ✓ Licensing & compliance monitoring ✓ Geological data governance (DGSM)	✓ Coordination gaps between mining & industrial value chain planning
	Directorate of Geological Survey & Mines (DGSM)	Licensing & monitoring	✓ Mineral rights administration ✓ Geological mapping & lab services	✓ Need for updated digital cadastre & monitoring tools
	NEMA	Environmental regulation	✓ ESIA approval ✓ Monitoring mine reclamation	✓ Compliance gaps in historical mining projects
	Local Governments (Tororo District)	Local support & oversight	✓ Land access & stakeholder engagement ✓ Community expectations management	✓ Capacity gaps in environmental & social safeguards
MIDSTREAM: Acidulation, Fertiliser Manufacturing, Industrial Products	Private Investors / PPP Partners	Core processing & manufacturing	✓ Construction of beneficiation, acidulation, and granulation plants ✓ echnology selection & quality control	✓ High entry CAPEX ✓ Technology & skills gaps ✓ FX exposure for imported inputs (sulphur, ammonia)

Value Chain Segment	Actor	Role / Function	Key Responsibilities	Current Gaps / Issues
			✓ O&M of processing facilities	
	UIA (Uganda Investment Authority)	Investor facilitation	✓ Industrial park services ✓ Licensing & incentives ✓ PPP coordination	✓ Need for stronger coordination in strategic minerals
	Ministry of Trade, Industry & Cooperatives (MTIC)	Industrial policy & standards	✓ Oversight of fertiliser industrialisation ✓ Industrial licensing & MSME linkages	✓ Limited data on domestic fertiliser manufacturers
	NEMA	ESIA for processing plants	✓ Air emissions control oversight ✓ Hazardous waste permitting (phosphogypsum)	✓ Weak enforcement of industrial waste management
	UNBS	Standards & certification	✓ Fertiliser quality testing ✓ Certification & compliance audits	✓ Limited testing capacity for advanced fertiliser grades
DOWNSTREAM: Distribution, Marketing & Use	Ministry of Agriculture, Animal Industry & Fisheries (MAAIF)	Demand generation & extension	✓ Fertiliser recommendations ✓ Soil testing & mapping ✓ Regulation of dealer networks	✓ Weak agro-extension services ✓ Limited soil labs at regional level
	Fertiliser Importers & Blenders	Supplement supply & blending	✓ Importation of DAP/MAP/NPKs ✓ Local blending & bagging ✓ Market distribution	✓ Price volatility & FX dependency

Value Chain Segment	Actor	Role / Function	Key Responsibilities	Current Gaps / Issues
	Agro-dealers & Distributors	Retail & farmer interface	✓ Last-mile distribution• Farmer training & awareness ✓ Stocking regional fertiliser needs	✓ Weak financial capacity ✓ Low professionalisatio n
	Transport & Logistics Operators	Transport & storage	✓ Warehousing ✓ Fleet operations ✓ Handling at ports & borders	✓ High logistics costs (Mombasa/Dar corridor)
	Farmers	End users	✓ Purchase & application of fertilisers ✓ Adoption of improved practices	✓ Low adoption due to affordability & knowledge gaps
	Financial institutions / SACCOs	Input financing	✓ Credit for merchants & farmers	✓ High perceived agricultural risk

4 Institutional, Policy and Regulatory Framework

4.1 Mining and Mineral Development Policies

The policy and regulatory environment governing mineral exploration, beneficiation, and value addition is central to the development of a competitive phosphate-to-fertiliser industry in Uganda. A coherent framework ensures secure mineral rights, predictable investment conditions, environmental compliance, and alignment with national industrialisation priorities. The analysis below outlines the key policy instruments, regulations, and institutional arrangements relevant to developing the Sukulu phosphate resource and establishing downstream fertiliser production.

4.1.1 The Mining and Minerals Act, 2022

The Mining and Minerals Act, 2022, modernises Uganda's mineral governance regime, replacing the 2003 Act. It provides a full value-chain approach to mineral development and strengthens state oversight in line with Vision 2040 and NDP IV.

Key provisions relevant to the phosphate project include:

- **Creation of the National Mining Company (NMC):** NMC holds Uganda's commercial interests in strategic minerals and enters joint ventures with private investors. Its mandate positions it as the **anchor actor for upstream phosphate extraction**, ensuring state participation and safeguarding public value.
- **Licensing Regime:** The Act introduces structured and transparent licensing categories (exploration licences, large-scale mining licences, mineral agreements), all awarded through competitive and merit-based processes.
- **Mandatory Value Addition:** Licensees are encouraged and, in some cases, obligated to undertake mineral beneficiation domestically. This aligns directly with establishing a **phosphate beneficiation and fertiliser manufacturing plant**.
- **Local Content Requirements:** Mining operators must procure goods and services locally where available, employ Ugandans, and support skills transfer—critical for developing a sustainable fertiliser value chain.
- **Environmental, Social and Community Obligations:** Mining activities require approved Environmental and Social Impact Assessments (ESIAs), rehabilitation plans, and community development agreements (CDAs). These provisions are particularly important for Sukulu, where significant community interaction exists.
- **Royalty and Revenue Sharing:** Royalties are shared among Central Government, Local Governments, and host communities. This strengthens socio-economic acceptance of mining-led industrialisation.

4.1.2 Mineral Value Addition and Industrialisation Policy Framework

Uganda's emphasis on resource-based industrialisation is articulated in several strategic documents:

- ✓ **NDP IV (2025–2030):** Positions **fertiliser production** among priority agro-industrial investments to reduce import dependence, strengthen food security, and support commercial agriculture.
- ✓ **Tenfold Growth Strategy:** Identifies mineral beneficiation and domestic fertiliser production as high-impact industrialisation accelerators capable of contributing to GDP, exports, and job creation.
- ✓ **National Industrial Policy (2020):** Encourages localisation of high-value mineral processing, particularly in strategic sectors such as fertilisers, steel, and cement.

These frameworks collectively justify public investment to unlock the phosphate value chain and support private capital mobilisation.

4.1.3 Investment and Trade Policies Relevant to Fertiliser Manufacturing

- **Investment Code Act (2019):** Provides incentives (tax holidays, VAT exemptions, duty waivers) for projects in manufacturing, agribusiness, and mineral beneficiation.
- **Free Zones Act:** Offers favourable tax and infrastructure conditions for export-oriented production—relevant if the fertiliser plant targets regional EAC/COMESA markets.
- **Uganda's Import Substitution Agenda:** Fertilisers are among priority imports targeted for domestic production due to:
    - High foreign exchange outflows,
    - Exposure to global price volatility,
    - Strategic importance for agricultural transformation.

4.1.4 Environmental and Social Safeguards Framework

Phosphate mining and fertiliser manufacturing involve potentially significant environmental impacts (overburden removal, gypsum waste, emissions, water use). The project must therefore comply with:

- **National Environment Act (2019)**
- **NEMA ESIA Regulations (2020)**
- **Climate Change Act (2021)**
- **IFC Performance Standards (for potential PPP or investor financing)**

Compliance is essential for licensing, bankability, and community acceptance.

4.1.5 Institutional Landscape

A constellation of institutions with defined mandates governs the value chain:

Table 4- 1: Institutional Landscape

Institution	Role in the Phosphate Value Chain
Ministry of Energy and Mineral Development (MEMD)	Overall mineral policy and sector oversight
Directorate of Geological Survey and Mines (DGSM)	Licensing, geological data, regulation
National Mining Company (NMC)	State commercial participation; anchor for upstream operations
NEMA	Environmental regulation and ESIA approval
Ministry of Trade, Industry and Cooperatives (MTIC)	Industrial policy, manufacturing regulation
MAAIF	Fertiliser quality and agricultural extension
UNBS	Fertiliser standards and certification
UIA	Investment facilitation and incentives
Local Governments	Community engagement, land-use coordination

The institutional interplay influences transaction costs, approvals, investment timelines, and governance arrangements for the project.

4.1.6 Policy Implications for the Fertiliser Project

The current policy framework **supports and incentivises** the development of a domestic phosphate fertiliser industry through:

- Strong national commitment to mineral value addition;
- Clear legal mechanisms for state-private collaboration through NMC and PPP arrangements;
- Favourable industrial and investment incentives;
- Compulsory environmental and community safeguards that improve long-term sustainability.

However, gaps remain, including:

- Coordination among multiple regulatory bodies,
- Inadequate enforcement of fertiliser quality standards,

- Weak logistics infrastructure (rail, energy reliability),
- Limited capacity for environmental monitoring of mining operations.

These issues must be addressed in the feasibility and implementation strategy.

Uganda's mining and mineral development policies provide a robust foundation for establishing a phosphate beneficiation and fertiliser manufacturing plant. The Mining and Minerals Act (2022), combined with NDP IV, the Tenfold Growth Strategy, and investment incentives, creates a conducive environment for upstream extraction, midstream processing, and downstream fertiliser production. Ensuring institutional coordination, regulatory compliance, and sustainable development practices will be essential to achieving a bankable and socio-economically transformative project.

4.2 Fertiliser Policies, Standards, and Quality Control

A well-regulated fertiliser market is essential for safeguarding farmers, improving soil productivity, and fostering a competitive domestic fertiliser industry. Uganda's fertiliser policy and regulatory framework—though evolving—plays a central role in shaping market efficiency, pricing, product quality, and private-sector participation. This section outlines the key policies, institutional mandates, and quality-control systems relevant to developing a phosphate fertiliser plant.

4.2.1 National Fertiliser Policy (2016)

The National Fertiliser Policy (NFP) provides the overarching framework for the production, importation, distribution, and use of fertilisers. Its goal is to ensure **sustained and affordable access to high-quality fertiliser products** to enhance soil health and agricultural productivity.

Core policy objectives include:

- **Promoting domestic fertiliser production and value addition** – Encourages establishment of fertiliser plants and raw-material beneficiation facilities. – Directly supports phosphate beneficiation and integrated fertiliser manufacturing.
- **Ensuring quality and safety of fertilisers** – Establishes standards for content, contaminants, packaging, and labelling. – Calls for routine market surveillance to curb adulteration.
- **Improving farmer access and adoption** – Strengthens extension services and agronomy advisory systems. – Supports soil testing, balanced fertilisation, and integrated soil fertility management.

- **Supporting private-sector investment** – Streamlines licensing and registration processes. – Encourages competition, transparency, and cost reduction along the supply chain.

Relevance to the project

The NFP creates supportive conditions for large-scale domestic fertiliser production by enabling local manufacturing, regulating quality, and discouraging substandard imports.

4.2.2 National Fertiliser Strategy (2016–2026)

The Strategy operationalises the Policy by setting targets and implementation mechanisms.

Key strategic thrusts include:

- Strengthening quality control and inspection systems
- Improving logistics and distribution efficiency
- Developing local manufacturing and blending capacity
- Enhancing regional fertiliser trade (EAC/COMESA)
- Promoting customised fertiliser blends for Uganda's diverse agro-ecological zones

Implications

A domestic phosphate-based fertiliser plant directly contributes to the Strategy's goals of import substitution, soil fertility restoration, and agro-industrialisation.

4.2.3 Standards and Technical Regulations (UNBS)

The **Uganda National Bureau of Standards (UNBS)** is the statutory authority responsible for:

- Setting fertiliser standards
- Approving composition, purity, labelling, packaging, and safety
- Certifying imported and domestically produced fertilisers
- Conducting market surveillance and testing samples to curb adulteration

Key standards include:

- **US 765: Fertilizer Specifications**
- **US 764: Fertilizer Labelling Requirements**
- Standards for DAP, TSP, SSP, NPK grades, urea, MOP, and micronutrient fertilisers

Common quality requirements:

- Nutrient content tolerance limits
- Moisture, granule size, and solubility
- Heavy metals (cadmium, arsenic), fluoride levels

- Packaging integrity and traceability

Challenges in the current market:

- Adulteration and counterfeit products
- Weak capacity of testing laboratories
- Inconsistent enforcement across districts
- Poor traceability of imported consignments

These challenges highlight the importance of a **modern, compliant local manufacturer** that can set higher industry standards.

4.2.4 Fertiliser Regulatory Framework (MAAIF)

The Ministry of Agriculture, Animal Industry and Fisheries (MAAIF) regulates fertiliser use and distribution. Its core responsibilities include:

- Registration of fertiliser dealers
- Licensing of blending operations
- Oversight of extension services and farmer training
- Soil testing and fertiliser recommendation guidelines
- Implementation of the Fertilizer Policy and Strategy

MAAIF works closely with UNBS, URA, and NEMA to ensure safe and efficient fertiliser value-chain operations.

4.2.5 Importation and Border Control (URA & UNBS)

Uganda imports more than 95% of its fertilisers, including DAP, MAP, TSP, MOP, and urea. Border control is therefore critical.

Uganda Revenue Authority (URA) and **UNBS** jointly enforce:

- Product certification and conformity assessment
- Pre-export verification (PVoC)
- Chemical safety documentation
- Anti-smuggling measures

A domestic fertiliser plant significantly alleviates these regulatory burdens by reducing import volumes.

4.2.6 EAC and COMESA Harmonisation Frameworks

Uganda participates in regional efforts to harmonise fertiliser regulations under:

- **EAC Fertilizer Harmonization Bill**
- **COMESA Seed and Fertilizer Regulations**

Benefits of harmonised standards:

- Reduced cross-border barriers
- Larger regional market for Ugandan fertilisers
- Easier export of DAP, TSP, NPKs, and SSP
- Consistent quality requirements across markets

Uganda's strategic location gives a competitive advantage for exporting to:

- Kenya
- Rwanda
- Burundi
- South Sudan
- Eastern DRC
- Northern Tanzania

4.2.7 Quality-Control Challenges Across the Value Chain

Despite policy improvements, Uganda's fertiliser market faces persistent issues:

Table 4- 2: Quality-Control Challenges Across the Value Chain

Bottleneck	Impact
Adulteration and counterfeit fertilisers	Reduces farmer trust and yields
Limited laboratory capacity	Delays certification and surveillance
Weak extension systems	Misapplication of fertilisers
Poor packaging and storage	Nutrient degradation
Fragmented dealer network	High prices and uneven distribution

A modern integrated fertiliser plant with in-house QA/QC laboratories helps lift industry standards and reduce the prevalence of adulterated products.

4.2.8 Implications for the Phosphate Fertiliser Project

A domestic phosphate-based fertiliser industry strengthens Uganda's regulatory ecosystem by:

1. **Enhancing traceability and quality assurance** – In-house laboratories aligned with UNBS standards – Reduced counterfeit prevalence through authorised dealer partnerships
2. **Supporting policy goals under NDP IV and the Fertiliser Policy** – Import substitution – Balanced fertilisation – Increased productivity and soil health restoration
3. **Facilitating regional market penetration** – Compliance with EAC/COMESA harmonised standards makes Uganda a regional fertiliser hub.
4. **Improving farmer confidence** – Reliable supply of quality fertiliser enhances uptake and productivity.
5. **Strengthening national food security** – Quality fertilisers support NDP IV's agro-industrialisation programme.

Uganda's fertiliser policies, standards, and regulatory frameworks provide a supportive foundation for establishing a domestic phosphate fertiliser plant. While the current market suffers from quality-control weaknesses, the introduction of a compliant, integrated phosphate-to-fertiliser facility can significantly improve product reliability, reduce adulteration, and enhance farmer uptake. Alignment with UNBS, MAAIF, URA, and regional standards ensures market competitiveness and positions Uganda as a potential fertiliser hub for the EAC and COMESA regions.

4.3 Trade and Investment Regulations

The regulatory environment governing trade, customs, investment incentives, and industrial localisation significantly influences the competitiveness of a domestic phosphate fertiliser industry. Uganda's trade and investment framework aims to promote value addition, attract strategic investment, reduce import dependence, and support regional market integration. This section outlines the relevant laws, regulations, incentives, and institutional arrangements that shape the business environment for a phosphate-to-fertiliser project.

4.3.1 Investment Code Act, 2019

The Investment Code Act provides the legal foundation for investment promotion and private-sector facilitation in Uganda. It establishes the Uganda Investment Authority (UIA) as the central agency for investor licensing and support.

Key provisions relevant to fertiliser production

- **Investment incentives:** Eligible manufacturers—including agro-input and fertiliser producers—may benefit from:
    - Corporate income tax incentives

- VAT exemptions on plant and machinery
- Import duty waivers for industrial inputs
- Industrial park incentives (for firms located in UIA-managed parks)

- **Strategic Investment Status:** Projects that contribute significantly to industrialisation, employment, technology transfer, and export growth—such as fertiliser manufacturing—may qualify for preferential terms.
- **Local content promotion:** Encourages use of local labour, services, and suppliers.

Implications:

Fertiliser manufacturing is explicitly recognised as a strategic sector, meaning the project qualifies for favourable investment terms and accelerated approvals.

4.3.2 Industrial Licensing and Manufacturing Regulations

The Ministry of Trade, Industry and Cooperatives (MTIC) oversees industrial development and manufacturing regulation.

Relevant regulatory requirements include:

- Registration as an industrial manufacturer
- Compliance with occupational safety and chemical handling laws
- Certification of production facilities under UNBS standards
- Adherence to industrial park guidelines (if located in Tororo or Mbale Industrial Zones)

Importance for the phosphate project:

Compliance ensures smooth licensing, market access, and government support through the industrialisation agenda.

4.3.3 Customs and Import Regulations (URA)

Uganda Revenue Authority regulates customs procedures, including import duties, taxes, and clearance of industrial inputs and final fertiliser products.

Key customs provisions relevant to the project

- **Duty exemptions** on plant, machinery, spare parts, and raw materials for registered manufacturers.
- **Import VAT exemptions** under specified incentive regimes.
- **Pre-Export Verification of Conformity (PVoC):** ensures imported fertilisers meet national standards.
- **Environmental levies and hazardous-chemical control** for sulphur, ammonia, and industrial acids.

Strategic relevance

A domestic fertiliser plant reduces exposure to customs delays, high freight and insurance costs, and foreign exchange risks associated with import dependency.

4.3.4 Trade Regulations Under the East African Community (EAC)

As a member of the EAC Customs Union and Common Market, Uganda benefits from harmonised trade regulations and preferential access to regional markets.

Relevant regional frameworks

- **EAC Common External Tariff (CET):** Fertiliser imports attract low or zero tariffs to ensure affordability; however, this also increases competition from external suppliers.
- **EAC Industrialisation Policy (2012–2032):** Promotes regional value chains, including agro-inputs and chemical industries.
- **EAC Fertiliser Harmonisation Regulations:** Standardised nutrient-content specifications, labelling, and packaging across EAC markets.

Export advantages for a Ugandan plant

- Duty-free access to regional markets (Kenya, Tanzania, Rwanda, Burundi, South Sudan).
- Potential to position Uganda as a **regional fertiliser hub**, particularly for SSP, TSP, DAP, MAP, and NPKs.

4.3.5 COMESA Trade Regulations and Opportunities

Membership in COMESA expands Uganda's free-trade access to markets including DRC, Zambia, Malawi, and Egypt.

Strategic benefits

- Preferential tariff regimes under the **COMESA Free Trade Area (FTA)**.
- Larger regional market for bulk fertilisers and industrial phosphate derivatives.
- Enhanced competitiveness for Ugandan exports due to reduced logistics distances compared to competing producers overseas.

4.3.6 Import Substitution and Localisation Policies

Uganda's trade policy increasingly emphasises **import substitution**, especially for strategic commodities such as fertiliser.

Key policy signals

- Fertiliser is identified as a **priority import-substitution product** in NDP IV.
- Government encourages domestic value addition to reduce foreign exchange outflows.

- Local manufacturing is favoured through tax incentives and public-procurement preferences.

Implication for the phosphate project

A domestic phosphate-to-fertiliser plant aligns with national objectives and may benefit from special support under the agro-industrialisation programme.

4.3.7 Public-Private Partnership (PPP) Regulations

If the recommended implementation modality is a PPP or hybrid model, the project must comply with the **PPP Act (2015)** and PPP Regulations (2019).

Key compliance requirements

- Project screening by the PPP Committee
- Value-for-money (VfM) assessment
- Fiscal risk analysis and approval by MoFPED
- Transparent procurement of private partners
- Commercial and financial close preceded by feasibility approval

Relevance

The phosphate project may attract large-scale private capital and technology providers, making PPP frameworks particularly applicable.

4.3.8 Environmental Trade Regulations

Uganda enforces strict controls on the importation, transport, and export of hazardous materials used in fertiliser production.

Regulated substances include:

- Sulphur and sulphuric acid
- Ammonia
- Industrial chemicals under the Chemicals Control Act
- Hazardous waste and phosphogypsum regulations

NEMA and URA jointly ensure compliance with:

- Basel Convention obligations
- Hazardous waste transportation permits
- International chemical-handling standards

These regulations affect procurement, transport, and storage of key inputs for fertiliser manufacturing.

4.3.9 Implications for the Phosphate-to-Fertiliser Project

The current trade and investment regulatory framework offers **significant strategic advantages** for setting up an integrated phosphate-to-fertiliser plant:

Enabling Factors

1. **Strong investment incentives** under the Investment Code Act.
2. **Duty and tax exemptions** for industrial inputs and processing equipment.
3. **Regional free-trade access** via EAC and COMESA, enabling competitive exports.
4. **Policy alignment** with import substitution, agro-industrialisation, and mineral value addition.
5. **PPP-enabling legislation** supporting private-sector participation.

Constraints to address

- High transport and border-processing costs for sulphur, ammonia, and other imported inputs.
- Risk of tariff changes under CET reforms.
- Quality-control gaps that may allow substandard imports to undercut domestic production.
- Need for efficient customs clearance and bonded logistics systems.

Uganda's trade and investment regulatory framework provides a conducive environment for establishing a phosphate beneficiation and fertiliser manufacturing industry. Strategic incentives, regional market access, harmonised standards, and supportive industrial policies position Uganda to become a competitive fertiliser producer within East Africa and COMESA. Effective coordination among UIA, URA, MAAIF, MTIC, MEMD, and NEMA will be essential to optimise regulatory benefits and mitigate trade-related risks.

4.4 Environmental and Social Framework (NEMA, Climate Resilience)

The development of a phosphate mining, beneficiation, and fertiliser manufacturing complex has significant environmental and social implications. Uganda's regulatory framework—anchored in the National Environment Management Authority (NEMA) and national climate change legislation—provides strict safeguards to ensure that industrial projects are environmentally sustainable, socially responsible, and climate-resilient.

This section outlines the legal requirements, institutional mandates, and international standards governing environmental and social management for the project.

4.4.1 National Environment Act, 2019

The National Environment Act, 2019 provides the overarching framework for environmental protection, pollution control, biodiversity conservation, and climate-proofing of public and private projects.

Key provisions relevant to the phosphate fertiliser project

- **Mandatory Environmental and Social Impact Assessment (ESIA)** – Mining, beneficiation, sulphuric acid plants, phosphoric acid plants, SSP/TSP/DAP production, and associated infrastructure require a full ESIA. – ESIA approval is a precondition for licensing, financing, and construction.
- **Environmental and Social Management Plan (ESMP)** – Required for operational-phase compliance, monitoring, and continuous mitigation. Must include waste management plans, emission control strategies, and occupational health protocols.
- **Pollution control and licensing** – Regulates emissions of SO_2, fluorides, dust, noise, and effluent from beneficiation and acidulation plants. Requires installation of scrubbers, continuous air quality monitoring, effluent treatment systems, and groundwater protection measures.
- **Waste management requirements** – Management of **phosphogypsum**, a major by-product of phosphoric acid production, must comply with hazardous waste regulations. Encourages recycling for construction materials, cement, or soil stabilisation where safe.
- **Community and ecosystem protection** – Requires assessment of impacts on wetlands, water bodies, forests, and sensitive ecosystems. Prevents unregulated land conversion and mandates rehabilitation of mined-out areas.

4.4.2 Environmental and Social Impact Assessment (ESIA) Requirements

NEMA mandates a **Category A ESIA** for phosphate mining and fertiliser manufacturing due to their potential for significant impacts.

Core ESIA Components

Baseline studies

- ✓ Hydrogeology, water resources, soil conditions, air quality, biodiversity, and socio-economic conditions.

Impact assessment

- ✓ Land disturbance from mining
- ✓ Air emissions (SO_2, NO_x, fluorides, dust)
- ✓ Acid plant emissions and fugitive gases

- ✓ Wastewater discharge and runoff
- ✓ Hazardous chemical management
- ✓ Potential resettlement impacts

Mitigation measures

- ✓ Emission scrubbers and gas-cleaning systems
- ✓ Wastewater treatment plants
- ✓ Proper phosphogypsum stacking systems
- ✓ Stormwater drainage systems
- ✓ Mine reclamation plans

Environmental and Social Management Plan (ESMP) – Monitoring indicators, institutional responsibilities, timelines, and budgets.

1. **Stakeholder engagement** – Continuous consultations with communities, local governments, CSOs, cultural institutions, and vulnerable groups.
2. **Resettlement Action Plan (RAP)** (if applicable) – Required if land acquisition displaces households or livelihoods.

4.4.3 ocial Safeguards and Community Development

Phosphate mining and fertiliser production have wide-ranging social implications, especially in Tororo.

Key social safeguard requirements

- **Free, Prior and Informed Consent (FPIC)-type consultations** for affected communities.
- **Resettlement Action Plan (RAP)** aligned with national law and IFC Performance Standard 5:
    - Fair and prompt compensation
    - Livelihood restoration programmes
    - Grievance redress mechanisms
- **Occupational health and safety (OHS)**: – Workers exposed to chemicals, dust, heat, and noise require strict OHS systems.

Community Development Agreements (CDAs)

For large mining projects, CDAs:

- Define benefits (schools, health centres, water systems).
- Require funding for community projects.
- Strengthen local acceptance and reduce conflict.

4.4.4 Climate Change Act, 2021 and National Climate Policies

Uganda's Climate Change Act (2021) mainstreams climate resilience across all sectors. The phosphate project must integrate climate risk assessment and adaptation measures.

Climate requirements relevant to the project

- **Climate Risk and Vulnerability Assessment (CRVA)** as part of ESIA.
- **Greenhouse Gas (GHG) accounting** for mining, acid plants, and energy systems.
- Alignment with:
    - Nationally Determined Contribution (NDC)
    - National Climate Change Policy (NCCP)
    - National Adaptation Plan (NAP)

Climate risks for the phosphate project

Table 4- 3: Climate risks for the phosphate project

Risk Category	Potential Impact
Drought & water scarcity	Reduced process-water availability
Flooding & stormwater	Damage to mine pits, tailings, and gypsum stacks
Extreme heat	OHS risks and energy inefficiencies
Supply chain disruptions	Impact on transport of sulphur, ammonia, and fertilisers

Required climate-resilient measures

- Water recycling and efficiency systems
- Flood-proofing beneficiation plants and gypsum stacks
- Heat-resistant equipment and climate-safe OHS protocols
- Solar or waste-heat recovery integration for energy efficiency
- Climate-proof road/rail links for logistics reliability

4.4.5 IFC Performance Standards and World Bank ESF (for investor financing)

Large industrial projects intending to attract private capital, DFIs, or PPP partners must comply with international environmental and social standards.

Applicable IFC Performance Standards

- ✓ **PS1:** Environmental and Social Management Systems
- ✓ **PS2:** Labour and Working Conditions

- ✓ **PS3:** Resource Efficiency and Pollution Prevention
- ✓ **PS4:** Community Health, Safety and Security
- ✓ **PS5:** Land Acquisition and Involuntary Resettlement
- ✓ **PS6:** Biodiversity Conservation
- ✓ **PS7/PS8:** Indigenous Peoples / Cultural Heritage (if applicable)

Importance

Compliance enhances project bankability and ensures readiness for financing from: AfDB, IFC, TDB, BADEA, UKEF, and other international lenders.

4.4.6 Key Environmental and Social Risks for the Project

Table 4- 4: Key Environmental and Social Risks

Risk Category	Key Issues
Environmental	Air emissions, phosphogypsum waste, water pollution, land degradation
Social	Resettlement, loss of livelihoods, community conflict risks
OHS	Chemical exposure, acid plant hazards, mining accidents
Climate	Water stress, flooding, energy intensity
Regulatory	ESIA delays, non-compliance penalties

Proper mitigation is essential to obtaining ESIA certification and maintaining social licence to operate.

4.4.7 Implications for Project Design and Feasibility

The environmental and social framework creates both **obligations and opportunities**:

Obligations

- ✓ Full ESIA and ESMP prior to construction
- ✓ Strict compliance with hazardous-waste and emission regulations
- ✓ Comprehensive RAP if land acquisition affects households
- ✓ Climate-resilient design and GHG mitigation
- ✓ Continuous environmental monitoring and community engagement

Opportunities

- ✓ Adoption of best-available technologies enhances competitiveness
- ✓ Phosphogypsum reuse (cement, plasterboard) can create new industrial value chains

- ✓ Strong ESG credentials improve access to concessional finance and PPP attractiveness
- ✓ Community development programmes strengthen long-term project stability

Uganda's environmental and social framework—anchored in NEMA regulations, the National Environment Act, the Climate Change Act, and international ESG standards—provides a robust foundation for ensuring that the phosphate fertiliser project is developed sustainably. Compliance with ESIA, climate-resilience requirements, and social safeguards is not only legally mandatory but also critical for bankability, investor confidence, and community acceptance.

4.5 PPP Act and Private Sector Participation Options

The successful development of a phosphate mining, beneficiation, and fertiliser manufacturing complex typically requires large capital investment, advanced technology, and efficient project execution. Uganda's **Public Private Partnership (PPP) Act, 2015** provides a structured framework for mobilising private capital, promoting innovation, and sharing project risks between the public and private sectors. This section outlines the PPP legal framework, institutional responsibilities, required processes, and feasible PPP modalities for developing the phosphate-to-fertiliser value chain.

4.5.1 PPP Legal and Regulatory Basis: The PPP Act, 2015

The PPP Act, 2015 was enacted to:

- Promote **private sector participation** in infrastructure and large-scale productive sectors
- Provide a consistent and transparent framework for planning, procuring, and managing PPP projects
- Ensure cost-effectiveness and value-for-money (VfM) for public investments
- Protect fiscal sustainability through structured risk allocation and oversight

This Act applies to projects that require substantial investment and include elements of private financing, technology, or operational expertise—making it directly applicable to the phosphate fertiliser project.

- **Key principles embedded in the PPP Act**
- **Value for Money (VfM):** PPPs must demonstrate better socio-economic and financial value than traditional public sector (TPS) delivery.
- **Risk Allocation:** Risks are allocated to the party best able to manage them—critical for mining, beneficiation, and chemical plants where operational and technical risks are significant.
- **Competition and Transparency:** All PPP procurement must be competitive and follow approved procedures.

- **Institutional Oversight:** Ensures accountability and alignment with national development priorities.

4.5.2 Institutional Framework for PPPs in Uganda

1. PPP Committee (PPPC)

Responsible for approving all PPP proposals before they proceed to procurement.

2. PPP Unit (Ministry of Finance, Planning and Economic Development – MoFPED)

Provides technical support, quality assurance, and fiscal risk assessments. The Unit ensures:

- Assessment of VfM
- Fiscal risk management
- Compliance with the PPP Act
- Support in transaction advisory and negotiations

3. Contracting Authority (for this project)

Depending on government decisions, the phosphate-fertiliser project may fall under:

- Ministry of Energy and Mineral Development (MEMD) – for mining and beneficiation
- Ministry of Agriculture, Animal Industry and Fisheries (MAAIF) – for fertiliser production and distribution
- National Mining Company (NMC) – as anchor shareholder in mining and beneficiation
- Uganda Investment Authority (UIA) – for industrial infrastructure and facilitation

The **National Mining Company (NMC)** is most appropriately positioned to act as the **public-sector anchor** in PPP arrangements for upstream operations.

4. Auditor General and Attorney General

Provide mandatory legal and financial review of PPP contracts.

4.5.3 PPP Process Requirements Under the Act

To proceed as a PPP, the phosphate project must complete the following stages:

Stage 1: Project Identification and Registration

- Conduct preliminary scoping
- Register the project with the PPP Unit
- Demonstrate alignment with NDP IV and mineral-value-addition priorities

Stage 2: Pre-Feasibility (Screening for PPP Suitability)

- Identification of project options and implementation modalities

- Preliminary financial analysis
- VfM assessment
- Risk identification and allocation framework
- PPP suitability assessment

Stage 3: Full Feasibility Study

As per DC Guidelines, the feasibility study must include:

- Detailed technical, environmental, economic, financial, and legal analysis
- PPP viability analysis and fiscal risk modelling
- Procurement strategy and proposed PPP structure
- Bankability assessment for potential private investors

Stage 4: PPP Committee Approval

The PPPC approves the feasibility study and authorises procurement.

Stage 5: PPP Procurement

- Competitive bidding
- Request for Qualification (RFQ)
- Request for Proposals (RFP)
- Bid evaluation
- Negotiation and selection of preferred bidder

Stage 6: Contract Signing and Financial Close

- PPP agreement signed
- All lender conditions satisfied
- Project moves into construction and implementation phase

4.5.4 Private Sector Participation Options for the Project

Given the integrated nature of the phosphate value chain—mining, beneficiation, acidulation, fertiliser production, logistics—various PPP and private-sector models may be applied at different stages.

Option 1: Joint Venture (JV) with NMC (Recommended for Upstream)

- NMC enters a JV with a strategic private mining and beneficiation company
- Government secures long-term control over strategic mineral resources
- Private partner provides capital, technology, and operational expertise

Advantages:

- Strong alignment of public and private interests

- Risk sharing
- Guaranteed public control over the mineral resource

Option 2: Build-Operate-Transfer (BOT) or Build-Own-Operate (BOO) for Acid and Fertiliser Plants

Private investors fully finance, build, and operate the:

- Sulphuric acid plant
- Phosphoric acid plant
- SSP/TSP/DAP/MAP/NPK manufacturing complex

BOT:

- Private partner operates for a concession period, then transfers the facility to Government/NMC

BOO:

- Private partner retains ownership but must comply with regulatory and trade requirements

Advantages:

- Reduces government capital expenditure
- Accelerates technology transfer
- Ensures operational efficiency

Option 3: Concession Arrangement for Logistics and Bulk Terminals

The project may require:

- Bulk sulphur/ammonia offloading terminals
- Specialised storage tanks
- Rail/road infrastructure improvements

These may be concessioned to private logistics companies under PPP frameworks.

Option 4: Management Contract or O&M Contract

Government/NMC retains ownership of the plant while private operators run:

- Beneficiation plants
- Acid plants
- Fertiliser plants

Useful when the state prefers control but needs private operational expertise.

Option 5: Fully Private-Led Model (Least Preferred for Strategic Minerals)

Private sector owns and controls the entire phosphate-to-fertiliser chain, subject to regulation.

Constraints:

- Does not guarantee national strategic interests

- Higher risk of revenue leakages
- Inconsistent with Government's mineral value-addition strategy

4.5.5 Bankability Considerations for PPP Options

To attract credible private partners, the project must demonstrate:

- Reliable ore supply (beneficiation performance tests)
- Clear off-take strategies (domestic + regional markets)
- Competitive energy and water tariffs
- Stable regulatory environment
- Long-term pricing arrangements for sulphur and ammonia
- Government willingness to provide:
    - Land access
    - Fiscal incentives
    - Regulatory support
    - Possible viability gap funding (VGF)

4.5.6 Implications of PPP Framework for the Phosphate Project

The PPP Act provides a structured pathway for mobilising private capital and technical expertise for the project, especially for:

- Mining and beneficiation (JV with NMC)
- Acidulation plants and fertiliser factories (BOT/BOO models)
- Logistics and supporting infrastructure (concessions)

PPP participation is most suitable for:

- Segments requiring advanced chemical engineering and high OPEX
- Complex, capital-intensive plants where operational efficiency is critical
- Market-linked products (fertilisers) where private partners understand commercial demand dynamics

Public participation is most suitable for:

- Strategic control of mineral resources via NMC
- Regulatory oversight, environmental monitoring, and community engagement
- Facilitating infrastructure (roads, water, energy)

Uganda's PPP Act (2015) provides a robust legal and institutional foundation for structuring private-sector participation in the phosphate-to-fertiliser industrial project. A blended model—featuring **NMC-led upstream operations** and **private-sector-led acidulation, manufacturing, and logistics infrastructure**—offers the highest potential for value-for-money, technology transfer, risk optimisation, and long-term sustainability. Strategic use of PPP modalities will be essential to attract high-quality investors, ensure operational efficiency, and deliver a bankable, transformative investment aligned with NDP IV and the Tenfold Growth Strategy.

4.6 Institutional Capacity and Coordination Gaps

The phosphate-to-fertiliser value chain spans multiple sectors—mining, chemicals, agriculture, environment, trade, and investment—requiring strong cross-government coordination. Although Uganda has an evolving institutional framework to support mineral value addition and agro industrialisation, several capacity and coordination gaps hinder effective planning, project implementation, and long-term sustainability. Addressing these gaps is essential to ensure a bankable, well-governed fertiliser project aligned with NDP IV and the DC Guidelines.

4.6.1 Fragmented Institutional Mandates

The value chain involves several entities whose mandates often overlap or lack operational linkages:

- **MEMD / DGSM** oversee mining and licensing.
- **NMC** leads state commercial participation in strategic minerals.
- **MAAIF** regulates fertiliser quality, use, and agricultural extension.
- **MTIC / UIA** promote industrialisation and investment.
- **NEMA** manages environmental approvals and compliance.
- **URA** manages customs and import procedures.
- **UNBS** enforces fertiliser standards.
- **Local Governments** oversee land access and community relations.

Coordination Gap

There is no unified platform bringing these institutions together for joint planning, permitting, and oversight of integrated mineral-to-fertiliser projects. This creates:

- Slow decision-making
- Regulatory inconsistencies
- High transaction costs for investors
- Delays in ESIA, licensing, or land access approvals

4.6.2 Limited Technical Capacity in Key Agencies

Several institutions face constraints in technical expertise required for large-scale mining and chemical processing projects.

Observed gaps

- **DGSM** – limited capacity for detailed resource modelling, beneficiation evaluation, and monitoring of complex mining operations.
- **MAAIF** – weak capacity for fertiliser-sector regulation, soil-testing systems, and enforcement of balanced fertilisation practices.
- **UNBS** – inadequate laboratory infrastructure for advanced chemical testing, including heavy metals, fluorine, and nutrient solubility profiles.
- **NEMA** – limited staffing to monitor complex industrial emissions, phosphogypsum management, and high-risk chemical plants.
- **Local Governments** – constrained in land acquisition management, grievance handling, and monitoring of community-development agreements.

Impact

Weak technical capacity increases project risks, compliance failures, and monitoring gaps.

4.6.3 Weak Integration Between Mining and Agricultural Sectors

The phosphate value chain requires close integration between mining institutions (MEMD, NMC, DGSM) and agricultural stakeholders (MAAIF, NARO, extension systems).

Current gaps

- Limited coordination on fertiliser specifications tailored to Uganda's agro-ecological zones.
- No integrated national fertiliser demand model jointly maintained by MAAIF and NPA.
- Weak feedback loops between soil scientists and fertiliser manufacturers.

Consequence

Manufacturing may not align with local nutrient needs, reducing fertiliser adoption and limiting farmer benefits.

4.6.4 Inadequate Infrastructure Planning Coordination

The project requires dedicated planning across:

- Power (Ministry of Energy, UEDCL)
- Water supply (NWSC)

- Transport (MoWT, UNRA, URC)
- Industrial parks (UIA)

Gaps

- No coordinated infrastructure plan for supporting large-scale beneficiation and chemical plants in Tororo.
- Fragmented planning for bulk logistics (sulphur, ammonia, phosphates) and rail connectivity.
- Limited coordination on waste management infrastructure (phosphogypsum stacks, wastewater treatment).

Result

Infrastructure bottlenecks increase project cost and undermine competitiveness.

4.6.5 Gaps in Environmental and Social Oversight

Given the size and complexity of the phosphate project, robust E&S management is essential.

Challenges

- NEMA lacks sector-specific guidelines for phosphoric acid plants, sulphuric acid units, and phosphogypsum disposal.
- Limited environmental monitoring equipment (air quality sensors, groundwater monitoring wells).
- Weak enforcement of Social Impact Management Plans (RAPs, CDAs).

Impact

Risk of environmental degradation, compliance breaches, and community conflict.

4.6.6 PPP and Investment Coordination Challenges

For PPP or hybrid delivery models:

- PPP Unit and MEMD must jointly manage fiscal risk, bankability, and vendor negotiations.
- NMC requires stronger capacity for structuring joint ventures, negotiating shareholder agreements, and overseeing private partners.

Gaps

- Limited experience managing large PPPs in mining and fertiliser sectors.
- Weak coordination between PPP Unit, MoFPED, NMC, and technical ministries during feasibility and procurement stages.

Impact

Risk of poorly structured PPPs, revenue loss, or unfavourable contracts.

4.6.7 Regulatory Enforcement and Market Discipline Weaknesses

- Fertiliser adulteration persists due to weak surveillance capacity.
- Import quality checks (PVoC, border testing) are inconsistently enforced.
- Inadequate market intelligence weakens planning and policy formulation.

Effect

Market distortions undermine incentives for domestic production.

4.6.8 Absence of a Dedicated Inter-Agency Coordination Mechanism

A multi-agency coordination body is needed to oversee:

- Licensing
- ESIA
- Investment approvals
- Infrastructure planning
- Community engagement
- Local content compliance
- Monitoring and evaluation

Without such a mechanism, project implementation risks delays and inconsistency.

Institutional capacity constraints and coordination gaps pose a significant risk to the successful rollout of Uganda's phosphate-to-fertiliser project. Addressing these issues—through strengthened technical capacity, streamlined regulatory processes, enhanced fertiliser-sector governance, and a unified inter-agency coordination mechanism—will be critical for ensuring efficiency, investor confidence, and long-term sustainability. A well-structured institutional arrangement is essential for achieving the project's economic, social, and industrialisation objectives under NDP IV.

5.1 Global Market Dynamics

Phosphate rock and phosphate fertilisers are globally traded strategic commodities, tightly linked to food security. The global market is characterised by growing demand, high geographic concentration of reserves and exports, and recurrent price volatility influenced by energy costs, trade policy, and geopolitics. This subsection summarises production, consumption, trade, and pricing trends relevant to a Ugandan phosphate-to-fertiliser project.

5.1.1 Global Production and Capacity

Phosphate rock

- Global phosphate rock production is estimated at about **205 million tonnes in 2023**, up roughly **3%** from 2022.
- In nutrient terms, global **phosphate rock production capacity** reached **63.6 million tonnes P_2O_5 in 2023** and is projected to rise to **69.1 million tonnes P_2O_5 by 2027**, driven by expansions in **Morocco, Brazil, Kazakhstan, Mexico, and Russia**.
- Earlier IFA projections indicated that global rock supply would grow from **239 Mt in 2019 to 261 Mt in 2024**, with **Africa accounting for about 72% of the net increase**, underscoring the continent's rising strategic importance.

Major producers

- **China** is the largest producer, with ~**110 Mt** of phosphate rock in 2024.
- **Morocco**, through OCP Group, holds more than **70% of the world's known phosphate reserves** and is the largest exporter of rock and phosphate products.
- Other important producers include the **United States, Russia, Saudi Arabia, Egypt, Jordan, and Tunisia**, some of which are simultaneously major fertiliser exporters.

Phosphate fertilisers

- The global phosphate fertiliser market (DAP, MAP, TSP, SSP, NPKs with P) was valued at around **USD 67 billion in 2024** and is expected to reach approximately **USD 94 billion by 2031**, implying a CAGR of about **4.3%**.

5.1.2 Global Consumption and Demand

- Agriculture accounts for **around 95%** of global phosphate ore demand by volume; almost all rock ultimately feeds into fertiliser production.
- IFA estimates place **global phosphorus (P_2O_5) use in agriculture** at roughly **49 Mt P_2O_5** in 2020.

- Overall fertiliser use ($N + P_2O_5 + K_2O$) is forecast to grow by **4.3% in FY 2023, 2.5% in FY 2024, and 2.2% in FY 2025**, reaching about **205 Mt nutrients in FY 2025**, above the previous 2020 peak.

Regional patterns

- **Asia** (especially China, India, South-East Asia) is the largest consumer of phosphate fertilisers; phosphate use in Asia increased roughly fivefold between 1970 and 2019, closely tracking the Green Revolution.
- **Latin America**, led by Brazil, has rapidly growing demand for phosphate in soy, maize, and sugarcane production and remains heavily dependent on imports.
- **Africa** currently has relatively low per-hectare application but is expected to see strong growth as countries pursue input-intensification and food-security strategies.

Recent IFA revisions show slightly **slower growth in phosphate use for 2024–25** than previously expected, mainly due to affordability concerns after recent price spikes.

5.1.3 Trade Structure and Market Concentration

The phosphate value chain is highly concentrated at the export level.

- More than **half of phosphate fertilisers produced globally are exported**.
- Just four exporters—**China, Morocco, Russia, and Saudi Arabia**—account for about **80% of global DAP and MAP exports**, underscoring a tight and relatively oligopolistic market.
- OCP's share in global trade in 2020 is estimated at **34% of phosphate rock**, **54% of phosphoric acid**, and **26% of phosphate-based fertilisers**, confirming Morocco's dominant position across the entire chain.

This concentration creates vulnerability for import-dependent regions (including East Africa) when:

- Exporters impose export restrictions or quotas (e.g. China's periodic controls),
- Geopolitical tensions (e.g. Russia-related sanctions or tariffs) disrupt flows, and
- Freight and insurance costs rise.

5.1.4 Pricing Trends and Volatility

Phosphate markets have experienced **significant price volatility** since 2021, driven by energy prices, supply disruptions, and trade policy.

- Fertiliser prices surged in 2021–22 following spikes in natural gas prices and disruptions from the Russia-Ukraine conflict.

- After some easing in 2023, prices **started rising again in 2024–25** due to tight supply and trade restrictions:
  - The World Bank fertiliser index rose **about 15% in early 2025**, with **TSP up 43%** and **DAP up 23%** compared with the start of the year.
- Recent analysis warns that **high phosphate prices pose food-security risks**, particularly in low-income, fertiliser-importing countries; high prices have already forced IFA to revise phosphate demand growth downward for 2024–2025.

Benchmark price reporting (e.g. Argus) shows:

- **DAP FOB China** and **DAP CFR India** trading in the high-hundreds USD/t range through 2024–25, with episodes of sharp monthly movements linked to Chinese export policy, Indian tenders, and energy-market shifts.
- **Phosphate rock FOB Morocco** and **phosphoric acid CFR India** have also remained elevated by historical standards, reflecting both strong underlying demand and higher input costs.

Overall, the market remains in a **high-cost, high-volatility regime**, with periodic corrections but no clear return to pre-2020 price levels.

5.1.5 Medium-Term Outlook and Strategic Implications

Supply outlook

- Global rock supply and capacity are expected to increase moderately (1.5–2% p.a.), led by expansions in **Africa (notably Morocco), Russia, and Latin America**.
- New projects in West Africa and Australia are entering the market but will not fundamentally change the dominance of OCP and a few large players in the medium term.

Demand outlook

- Phosphate fertiliser demand is set to **grow slowly but steadily**, underpinned by:
  - Population growth and higher food demand,
  - Expansion of fertiliser use in Africa and South Asia,
  - The need to maintain soil fertility on intensively farmed land.

However, demand growth may be uneven due to:

- Affordability pressures at high price levels,
- Environmental and regulatory constraints in OECD markets, and
- Improved nutrient-use efficiency and emerging technologies (e.g. enhanced-efficiency fertilisers).

Price outlook

- Most forecasts point to **continued price volatility**, with a tendency for prices to remain above pre-2020 averages, driven by:
    - Concentrated export supply,
    - High energy prices (especially sulphur and ammonia supply chains),
    - Climate and geopolitical risks affecting logistics and production.

5.1.6 Implications for the Uganda Phosphate Project

The global market context provides a strong strategic rationale for Uganda's phosphate-based fertiliser project:

1. **Supply Security & Import Substitution**
    - Uganda and the wider EAC/COMESA region are highly exposed to international price spikes and export restrictions.
    - Developing a domestic phosphate-to-fertiliser industry can partially insulate the region from external shocks.
2. **Export Potential**
    - Regional phosphate demand is expected to grow faster than global averages, particularly in Africa, creating scope for Uganda to supply neighbouring countries.
3. **Competitive Positioning**
    - Proximity to regional markets, lower freight costs, and integration with Sukulu rock give Uganda a potential cost advantage versus distant suppliers—provided the project achieves efficient mining, beneficiation, and acidulation.
4. **Risk Considerations**
    - The project must plan for global price volatility, ensure robust off-take strategies, and remain cost-competitive against global benchmarks for rock, phosphoric acid, and DAP/MAP/TSP.

In summary, the global phosphate market is **growing, concentrated, and volatile**. This environment favours well-designed regional projects that can provide **reliable, competitively priced fertiliser** to African markets while contributing to national food security and industrialisation objectives.

5.2 Regional (EAC/COMESA) Market Analysis

Import Dependency and Regional Demand

The East African Community (EAC) and wider COMESA region are structurally fertiliser-deficient. Virtually all phosphate fertilisers are imported, and application rates remain far below agronomic recommendations. This creates both a **clear market opportunity** and a **strategic rationale** for regional phosphate-based production in Uganda.

5.2.1 Fertiliser Use and Demand in East and Southern Africa

Overall, fertiliser consumption in Sub-Saharan Africa is low but rising:

- Sub-Saharan Africa consumed **10–12 million tonnes** of fertiliser (all nutrients) in 2020, with **about half** of that used in just four countries: Ethiopia, Kenya, Nigeria and South Africa.
- Across Africa, average fertiliser use between 2016 and 2019 was only **14–17 kg/ha**, far below the **Malabo target of 50 kg/ha**.

Within this picture, the **EAC cluster (Kenya, Tanzania, Uganda, Rwanda, Burundi)** is recognised as one of the more dynamic sub-regions in terms of fertiliser demand growth, but still heavily import-dependent.

Recent country evidence illustrates the scale and growth of demand:

- **Tanzania** imported about **790,000 tonnes of fertilisers in 2023**, up 26% from 2022; urea and NPK imports grew strongly, while DAP imports still exceeded 130,000 tonnes.
- Similar upward trends are reported for **Kenya, Rwanda and Uganda** in IFDC/AGRA assessments, as governments promote input use and commercial agriculture.

Given low base application rates, even a gradual move toward 30–50 kg/ha will generate **strong incremental demand** for phosphate fertilisers across the region.

5.2.2 Import Dependency and Supply Structure

The regional phosphate market is almost entirely import-driven:

- **Burundi, Rwanda and Uganda import virtually all of their fertilisers** (DAP, NPKs, urea) from or through Kenya and Tanzania, which act as regional import and blending hubs.
- In **DRC**, annual fertiliser imports fluctuate between **10,000 and 100,000 tonnes**, mostly urea, ammonium nitrate, DAP, MAP and NPKs—again entirely imported.

Across eastern and southern Africa, the main supply patterns are:

- Phosphate fertilisers (DAP, MAP, TSP, NPKs with P) are imported from **Morocco, Saudi Arabia, China, Russia and Jordan**, with cargoes typically landed at **Mombasa**, **Dar es Salaam**, **Beira**, or **Durban**, and then moved inland.
- There is **no large integrated phosphate rock–to–fertiliser complex** in the EAC; regional blending plants depend on imported intermediates (DAP/MAP/TSP and sometimes phosphoric acid).

Consequences of this high import dependency include:

- Strong exposure to **global price shocks** and freight costs;
- Long lead times and supply risk when exporting countries impose quotas or export controls;
- High inland logistics costs and mark-ups, especially in landlocked markets such as Uganda, Rwanda, Burundi, South Sudan and eastern DRC.

5.2.3 Regional Demand Drivers for Phosphate Fertilisers

Several structural factors underpin growing regional demand for phosphate products (DAP, MAP, TSP, SSP, and P-containing NPKs):

1. **Population growth and food demand**
    - Rapid population growth in COMESA economies (e.g. Ethiopia, DRC, Kenya, Uganda) is increasing cereal, oilseed and horticultural demand.
2. **Agricultural intensification policies**
    - EAC/COMESA countries have committed to doubling productivity and raising fertiliser use as part of CAADP/Malabo targets, driving input-subsidy programmes, e-voucher schemes, and credit lines for fertiliser.
3. **Expansion of commercial value chains**
    - Scaling of cash and export crops (tea, coffee, sugarcane, maize, rice, oilseeds, horticulture) in Kenya, Tanzania, Uganda, Rwanda, Zambia and Ethiopia is increasing demand for higher-analysis phosphate fertilisers and crop-specific NPK blends.
4. **Soil nutrient depletion and acidity**
    - Large areas in East and Central Africa show severe phosphorus depletion and soil acidity; addressing these constraints requires sustained application of P-bearing fertilisers and, in some zones, SSP and TSP plus lime.

In this context, the **latent demand** for phosphate fertiliser in EAC/COMESA significantly exceeds current actual use, constrained mainly by **affordability, availability, and farmer risk perceptions** rather than agronomic need.

5.2.4 Pricing, Affordability, and Market Vulnerability

The region is highly exposed to international price swings:

- The 2021–2022 fertiliser price surge, driven by energy prices and geopolitical shocks, sharply increased landed DAP/MAP and NPK prices in EAC/COMESA, eroding smallholder profitability and depressing application rates.
- Because almost all phosphate fertiliser is imported, any upward movement in **FOB prices** or **freight and insurance costs** translates into high retail prices, especially in inland countries where transport margins are large.

COMESA and regional stakeholders have explicitly identified **high fertiliser costs, fragmented markets, and weak cross-border trade facilitation** as key obstacles to raising fertiliser use and agricultural productivity.

5.2.5 Strategic Implications for a Ugandan Phosphate Fertiliser Plant

The regional context creates a strong case for a Sukulu-anchored phosphate project serving both **domestic** and **regional** markets:

1. **Import-substitution in the EAC cluster**
    - Uganda, Rwanda, Burundi, South Sudan and parts of eastern DRC currently rely entirely on imported phosphate fertilisers via Kenyan and Tanzanian ports.
    - A Ugandan plant could substitute a portion of these imports with **regionally produced SSP, TSP, DAP/MAP, and P-containing NPKs**, reducing freight costs and supply risk.
2. **Gateway to COMESA markets**
    - With strategic logistics (rail/road to Mombasa/Dar, and potentially Lake/river corridors), Uganda could serve markets in **eastern DRC, South Sudan, Rwanda, Burundi, northern Tanzania and western Kenya**.
3. **Competitive advantage from proximity and integration**
    - Integration of **local rock (Sukulu) + beneficiation + acids + granulation** allows the project to capture margins currently accruing to overseas producers and traders.
    - Shorter inland distances and lower handling costs, if backed by efficient logistics, can improve delivered-cost competitiveness against imported fertilisers.

4. **Support to regional food-security commitments**
    - Regional institutions (EAC, COMESA, AU) emphasise fertiliser as a key lever for achieving the Malabo and CAADP productivity targets. A Ugandan phosphate complex aligns directly with these objectives and is likely to attract policy and possibly financing support.
5. **Risk management**
    - The investment must still account for volatility in **global phosphates, sulphur, and ammonia markets**, but regional production reduces dependence on long and fragile import supply chains.

The EAC/COMESA region is characterised by:

- **High and rising import dependency** for phosphate fertilisers;
- **Low but growing application rates**, far below agronomic and policy targets;
- **Strong underlying agronomic demand** constrained by cost and availability; and
- **Vulnerability to international price and supply shocks**.

This combination creates a **clear strategic opportunity** for a well-designed Ugandan phosphate fertiliser project to:

- Substitute a portion of regional imports;
- Serve as a cost-competitive supplier into EAC/COMESA markets; and
- Contribute to regional food security and soil-health restoration.

5.3 Intra-African Trade Opportunities

Africa is emerging as one of the world's fastest-growing fertiliser demand zones, yet it remains structurally dependent on phosphate imports from outside the continent. Intra-African trade in fertilisers is still limited but expanding as regional blocs harmonise standards, invest in logistics, and promote value addition closer to agricultural production centres. Uganda is strategically positioned—geographically and institutionally—to become a meaningful participant in this growing intra-African market.

5.3.1 Rising Demand Across African Subregions

Fertiliser use is expected to rise across Africa due to:

- Population growth and food demand pressures
- National soil-fertility initiatives (e.g., Ethiopia, Kenya, Rwanda, Zambia, Malawi)
- Expansion of high-value crops (tea, coffee, horticulture, sugar, maize, rice)
- Government subsidy, voucher, and input-credit programmes

- Regional commitments under CAADP/Malabo to increase fertiliser use

Most African countries—especially in **East, Central, and Southern Africa**—have **low baseline application rates**, which means even moderate intensification creates substantial incremental demand for phosphate fertilisers.

5.3.2 Regional Trade Corridors that Enable Intra-African Fertiliser Flows

Uganda sits at the intersection of several growing agricultural markets with established but under-utilised trade corridors:

1. **Northern Corridor**
    - Connects Uganda → Kenya → Rwanda → Burundi → eastern DRC → South Sudan
    - Existing fertiliser distribution channels already run through this corridor
    - Potential for Uganda to supply DAP, TSP, SSP, and NPKs inland at lower transport costs than imports via Mombasa
2. **Central Corridor (Tanzania route)**
    - Via Dar es Salaam → Mwanza → Bukoba → Kampala
    - Key for reaching western Tanzania and Rwanda/Burundi markets
3. **COMESA Corridor**
    - Enables preferential-tariff access to markets such as Zambia, Malawi, DRC, and Ethiopia under the COMESA FTA

These corridors create direct commercial pathways for Ugandan-produced phosphate-based fertilisers.

5.3.3 Opportunities from Africa's Fertiliser Import Dependence

Despite having over **70% of the world's phosphate reserves**, Africa (outside Morocco) imports most of its fertilisers. This creates several opportunities:

1. **Substitution of offshore imports**
    - Countries like Kenya, Rwanda, Burundi, Tanzania, DRC, Malawi, and Zambia rely almost entirely on imported DAP, TSP, MAP, and NPKs.
    - Uganda can competitively supply these markets with shorter delivery times and lower inland logistics costs.
2. **Improving supply reliability**
    - African markets often face seasonal stockouts due to long shipping cycles from China, Middle East, and North Africa.

    - A regional supplier in Uganda improves availability and reduces long procurement lead times.
3. **Price competitiveness**
    - Freight accounts for 20–40% of fertiliser costs in inland African countries.
    - Locally produced fertilisers from Uganda can undercut CIF-import parity prices, especially into Rwanda, eastern DRC, South Sudan, Burundi, and northern Tanzania.
4. **Opportunity for customised blends**
    - African soils are diverse and often P-deficient.
    - Region-specific NPK blends (e.g., 17-17-17, 10-26-26, 23-21-0+S) have strong demand
    - Uganda can integrate blending operations linked to its phosphate plant to serve niche agro-ecological zones.

5.3.4 Regional Policy Harmonisation Supporting Intra-African Fertiliser Trade

Several regional initiatives enable easier fertiliser movement across borders:

- **EAC Fertiliser Harmonisation Framework**: Standardised product specifications, labelling, and inspection across East Africa
- **COMESA Seed & Fertiliser Harmonised Regulations**: Reduced testing duplication and faster border clearance
- **African Continental Free Trade Area (AfCFTA)**: Provides a future platform for tariff-free movement of industrial inputs, including fertilisers
- **Tripartite Free Trade Area (EAC–COMESA–SADC)**: Expands access to Southern African markets through reduced tariff barriers

These frameworks lower non-tariff barriers, creating a **larger integrated African fertiliser market** for Ugandan production.

5.3.5 Priority Regional Markets for Ugandan Phosphate Fertilisers

1. **Rwanda & Burundi**
    - Entirely import-dependent
    - Competitive delivered cost from Uganda via road
    - High demand for DAP, NPK (17-17-17), and TSP in coffee, tea, and food crops
2. **Eastern DRC (Goma, Bukavu, Bunia, Kisangani)**
    - Weak local supply chains

    - Strong demand from commercial horticulture and staple-crop producers
    - Uganda is the closest potential manufacturing base
3. **South Sudan**
    - Import-dependent with rising demand from emerging commercial farms
    - Uganda already supplies agro-inputs informally via Nimule and Kaya borders
4. **Western Kenya**
    - Large-scale maize and sugar markets
    - Significant demand for DAP and NPKs
    - Competitive positioning for Uganda for niche markets near the border
5. **Northern Tanzania (Kagera, Mwanza)**
    - Affordable supply possible via Bukoba and lake routes
    - Strong demand for NPK blends for bananas, coffee, maize

5.3.6 Medium-Term Strategic Opportunities

The following opportunities are likely to expand intra-African trade in the next decade:

- **Regional food-security policies** increasing fertiliser subsidies and distribution
- **Expansion of blending plants**: Uganda can supply intermediate products (PA, TSP, SSP) to regional blenders
- **Potential for long-term government tenders** (e.g., Rwanda's fertiliser procurement, DRC agricultural programmes)
- **Emergence of logistics infrastructure** (SGR corridors, inland ports, warehousing facilities)
- **Integration with AfCFTA value chains**, creating continent-wide supply potential for SSP, TSP, and NPKs

5.3.7 Implications for Uganda's Phosphate-to-Fertiliser Project

Intra-African trade dynamics create a **strong commercial case** for Uganda's fertiliser industry:

- Uganda becomes a **regional supplier**, not only a domestic producer
- Reduces vulnerability to global price and freight shocks
- Leverages **geographical centrality** and shorter delivery times
- Creates potential for **bulk sales to neighbouring governments**, agro-processors, and blending plants
- Enhances project viability and economies of scale for the phosphate complex

A well-designed logistics network (rail, road, warehousing, bulk handling), combined with competitive pricing, can position Uganda as a **key phosphate fertiliser hub in Eastern, Central, and parts of Southern Africa**.

5.4 Domestic Market Assessment

Historical Consumption and Adoption Trends

Uganda's domestic fertiliser market has historically been small, fragmented, and heavily dependent on imports. Despite steady growth in agricultural demand, fertiliser adoption rates remain among the lowest in Sub-Saharan Africa. Understanding these trends is essential for sizing the domestic market, assessing price elasticity, and determining the viability of local phosphate-based fertiliser production.

5.4.1 National Fertiliser Use Levels Over Time

Uganda's fertiliser consumption has gradually increased over the last two decades, but from a very low base:

- Between **2000 and 2010**, national fertiliser use remained below **20,000 tonnes per year**, with minimal growth.
- From **2010 to 2019**, imports rose steadily, reaching **80,000–120,000 tonnes annually** depending on global prices and donor programmes.
- Fertiliser application rates remain **below 5 kg/ha**, far below the African Union's **50 kg/ha** Malabo target and the global average of over **130 kg/ha**.

These low application rates reflect affordability constraints, limited extension services, weak supply-chain integration, and pervasive soil nutrient mining.

5.4.2 Trends by Fertiliser Type (DAP, NPK, SSP/TSP, Urea)

The composition of Uganda's fertiliser consumption reveals heavy dependency on imported phosphate products:

- **DAP** remains the most widely used basal fertiliser, especially for maize, beans, coffee, and horticulture.
- **NPK blends** have grown steadily, driven by donor projects, cooperatives, and high-value crop producers.
- **Urea** is used for top-dressing in cereals but is less relevant to Uganda's dominant root crops and bananas.

- **SSP/TSP usage** is limited due to local unavailability and the higher retail price relative to DAP; however, agronomic need is significant for coffee, tea, and acidic soils.
- **Commercial farmers and emerging outgrower schemes increasingly use compound NPKs (e.g., 17-17-17, 23-21-0+4S, 10-26-26)**.

The dominance of imported DAP highlights the strong latent demand for baseload phosphate fertiliser that could be met by domestic production.

5.4.3 Fertiliser Adoption Patterns Among Farmers

Fertiliser adoption in Uganda has been slow due to structural, economic, and institutional constraints:

Smallholders (80% of farmers)

- Use fertiliser irregularly or in minimal quantities.
- Apply fertiliser mostly to maize, beans, coffee, and horticulture plots.
- Adoption is highly sensitive to price movements.
- Poor soil testing and limited agronomic knowledge lead to under-application or misuse.

Commercial and semi-commercial farmers

- More consistent fertiliser users.
- Increasing use of NPK blends and crop-specific formulations.
- Sensitive to global input price fluctuations, especially on DAP and NPKs.
- Show willingness to shift to locally blended or domestically manufactured fertilisers if competitive.

High-value crops (tea, coffee, sugar, horticulture)

- Tea estates and coffee exporters increasingly rely on structured fertiliser programs.
- Horticulture (flowers, fruits, vegetables) is adopting more phosphate-rich water-soluble fertilisers.

5.4.4 Structural Factors Driving Low Adoption

Several barriers explain why Uganda's fertiliser application remains far below agronomic recommendations:

1. **High prices and import dependency**
    - Imported fertilisers face high transport, logistics, and forex costs.
    - Long supply chains increase retail margins by 20–40%.

2. **Weak extension and soil fertility services**
    - Most farmers do not receive fertiliser recommendations tailored to their soils.
    - Only a small proportion of farmers conduct soil testing.
3. **Limited access to credit**
    - Fertilisers remain unaffordable for many smallholders without credit or subsidy mechanisms.
4. **Market fragmentation and counterfeit products**
    - Adulteration undermines farmer confidence and reduces adoption.
5. **Risk aversion**
    - Rainfed agriculture leads to uncertainty, reducing farmers' willingness to invest in fertilisers.

5.4.5 Demand Growth Trends and Market Signals

Despite historical challenges, several trends point to an expanding domestic market:

- **Growing commercialisation of agriculture** under NDP IV and agro-industrialisation programmes.
- **Expansion of coffee, tea, rice, sugarcane, and oilseed value chains**, which require regular P-based fertilisers.
- **Emergence of regional blending companies** increases distribution capacity and market penetration.
- **Rising awareness of soil degradation** is encouraging greater adoption of balanced fertilisation.
- Development partner programmes (IFDC, AGRA, USAID) are expanding farmer training and subsidised fertiliser access.

These shifts suggest increasing demand for **DAP, SSP/TSP, locally produced NPKs, and customised blends**.

5.4.6 Implications for Domestic Production

Historical consumption patterns indicate:

- Fertiliser demand in Uganda is **suppressed**, not saturated.
- There is strong **latent demand** for phosphate-based fertilisers, particularly DAP, TSP, and NPK blends.
- A domestic phosphate-to-fertiliser plant would:
    - Lower retail prices by reducing import and logistics costs;

    - Improve product availability and reliability;
    - Build farmer confidence through improved quality assurance;
    - Support balanced fertilisation for key crops (coffee, tea, maize, bananas, horticulture);
    - Unlock adoption among 1–3 million smallholders currently priced out of the market.

Over time, rising adoption could increase Uganda's consumption from the current 70–120 thousand tonnes per year to more than **250–300 thousand tonnes annually**, consistent with regional benchmarks.

5.5 Domestic Market Assessment

Historical Consumption and Adoption Trends

Uganda's fertiliser market has historically been small and underdeveloped, despite the country's strong agricultural base and persistent soil nutrient depletion. Although fertiliser use has increased gradually over the past two decades, overall consumption remains far below regional and global benchmarks. Understanding these historical trends is essential for forecasting demand, assessing price responsiveness, and determining how domestic phosphate fertiliser production can expand the market.

5.5.1 National Fertiliser Consumption Over Time

Uganda's fertiliser consumption has consistently grown, but from a very low starting point:

- Between **2000 and 2010**, total national fertiliser use ranged between **15,000 and 20,000 tonnes per year**, primarily DAP and urea imports distributed through a few urban agro-dealers.
- From **2010 to 2019**, consumption rose steadily to **80,000–120,000 tonnes annually**, driven by:
    - Donor-supported agricultural programmes (IFDC, AGRA, USAID),
    - Growth of commercial crop value chains (coffee, tea, sugar, rice, horticulture),
    - Expansion of private blending operations in Kampala and surrounding districts.
- Nonetheless, average national fertiliser application remains **below 5 kg/ha**, far behind:
    - The **Malabo Declaration target** (50 kg/ha),
    - Regional leaders like Kenya and Ethiopia (30–60 kg/ha),
    - The global average (>130 kg/ha).

The low baseline shows that Uganda's fertiliser market is **supply-constrained, not demand-saturated**, especially for phosphate-based products.

5.5.2 Trends in Product Preferences and Use Patterns

Uganda relies almost entirely on imports for all categories of inorganic fertilisers. Historical consumption patterns indicate:

1. Strong reliance on DAP

- DAP has long been the dominant basal fertiliser for maize, beans, rice, and horticulture.
- Farmers prefer it because of strong nutrient concentration (18-46-0), ease of application, and widespread availability.
- Uganda imports DAP mainly via Mombasa, making prices highly sensitive to global markets and freight costs.

2. Growing use of NPK blends

- Demand for blends like 17-17-17, 23-21-0+4S, 10-26-26, and 15-15-15 has increased, especially in commercial farming.
- Local blending firms have expanded capacity, but all blending materials (DAP/MAP/TSP/MOP) are imported.

3. Limited use of SSP and TSP

- Despite Uganda's acidic soils and strong agronomic need for SSP/TSP in coffee, tea, and horticulture, usage is low because:
    - No domestic production exists,
    - Imports are costly,
    - Farmers lack awareness of crop- and soil-specific benefits.

4. Urea use remains stable

- Used mainly for top dressing in cereals and seed production.
- Adoption is moderate and less relevant for root crops and bananas, which dominate Uganda's food systems.

5.5.3 Adoption Trends Across Farmer Segments

Fertiliser adoption varies significantly by farmer type:

Smallholders (70–80% of farmers)

- Tend to use fertiliser sparingly or not at all.
- Purchase patterns are erratic and often limited to 1–2 bags per season.
- Adoption is highly price-sensitive and influenced by rainfall, market prices, and risk aversion.

Emerging commercial farmers

- Use fertilisers more consistently.

- Have begun demanding specialised NPK blends and higher-quality products.
- More likely to adopt balanced fertiliser regimes if supply is reliable and pricing is stable.

Estate and organised outgrower schemes

- Tea estates, coffee exporters, rice schemes, and sugar mills maintain structured fertiliser programmes.
- These segments represent a reliable anchor market for phosphate fertilisers (especially TSP, SSP, DAP, and tailored NPKs).

Horticulture sector

- Fast-growing and fertiliser-intensive.
- Uses phosphate-rich water-soluble fertilisers and precision blends.

5.5.4 Factors Behind Low Historical Adoption

Despite gradual growth, fertiliser adoption remains low due to several structural barriers:

1. **High retail prices and import dependency**
    - Long supply chains (Mombasa–Uganda) add 20–40% logistics mark-up.
    - Price volatility discourages investment by risk-averse smallholders.
2. **Limited access to credit**
    - Lack of affordable agricultural finance constrains fertiliser purchases.
3. **Weak agronomic knowledge and extension**
    - Misuse or under-application is common.
    - Low awareness of crop-specific nutrient needs—especially phosphorus.
4. **Prevalence of counterfeit and substandard products**
    - Farmers lose trust when fertilisers fail to produce expected results.
5. **Rainfed production systems**
    - Uncertain weather reduces incentives to invest in fertiliser.
6. **Lack of soil testing and balanced fertilisation guidance**
    - Most farmers apply fertiliser blindly, often leading to poor returns.

These barriers suppress consumption far below the country's true agronomic potential.

5.5.5 Signals of Rising Demand

Several recent developments indicate accelerating fertiliser uptake:

- Government's agro-industrialisation and parish development initiatives are increasing input use.

- Commercial coffee, tea, avocado, and horticulture exporters are scaling fertiliser programmes.
- Increased availability of private-sector blending plants has expanded last-mile distribution.
- Soil fertility degradation is becoming more visible, prompting corrective interventions.
- Development partner programmes continue to promote fertiliser learning, soil testing, and credit schemes.

These trends suggest the domestic market is poised for **substantial growth**, especially if pricing becomes more predictable and products are locally tailored.

5.5.6 Implications for Domestic Phosphate Fertiliser Production

Historical consumption and adoption patterns indicate:

- Uganda has **strong unmet demand** for phosphate-based fertilisers.
- DAP, TSP, SSP, and phosphate-rich NPKs represent the largest opportunity areas.
- A domestic phosphate fertiliser plant would:
    - Reduce retail prices by removing import/logistics mark-ups,
    - Improve reliability of supply,
    - Enable formulation of customised blends aligned with Uganda's soil needs,
    - Restore farmer confidence through better quality assurance,
    - Stimulate uptake beyond the current 5 kg/ha ceiling.

With local production, Uganda could see fertiliser consumption more than **double over the next decade**, driven by both price reductions and improved extension services.

5.6 Crop-Specific Nutrient Requirements

Understanding crop-specific nutrient requirements is essential for estimating realistic fertiliser demand and determining the product mix (DAP, TSP, SSP, MAP, NPKs) that a domestic phosphate-based fertiliser plant should produce. Uganda's major crops exhibit varying phosphorus (P_2O_5) needs depending on rooting depth, biomass production, soil type, and nutrient removal rates. Most Ugandan soils are **P-deficient**, making phosphorus-based fertilisers a critical driver of yield gains.

Below is a summary of nutrient requirements for Uganda's key crops—focusing on **phosphorus (P)** but also recognising balanced NPK needs.

5.6.1 Maize

Maize is one of Uganda's largest fertiliser-consuming crops.

Typical recommended rates (per hectare):

- **Nitrogen (N):** 60–120 kg
- **Phosphorus (P_2O_5):** 30–60 kg
- **Potassium (K_2O):** 20–40 kg

Role of phosphorus:

- Essential for strong root development during the first 3–4 weeks
- Enhances early vigour and drought resilience
- Critical for grain filling and cob development

Suitable fertiliser products:

- DAP (18-46-0)
- MAP (11-52-0)
- NPK 17-17-17
- NPK 10-26-26 (for low-P soils)

Given Uganda's widespread P deficiency, maize responds strongly to P-containing fertilisers, particularly under commercial and semi-commercial systems.

5.6.2 Beans and Other Legumes (Soybean, Groundnuts)

Legumes have modest nitrogen needs due to biological fixation but require adequate phosphorus for nodulation.

Typical rates:

- **P_2O_5:** 20–40 kg/ha (up to 60 kg in low-P soils)
- **K_2O:** 20–30 kg/ha

Importance of P:

- Stimulates root nodulation
- Enhances seed formation
- Improves protein content

Suitable products:

- TSP (46% P_2O_5)
- SSP (16% P_2O_5 + sulphur, beneficial for legumes)
- NPK 10-26-26

Adoption remains low, but agronomic response is consistently high.

5.6.3 Coffee (Robusta & Arabica)

Coffee has long-term perennial nutrient requirements, heavily P-dependent.

Typical annual rates:

- **N:** 40–80 kg/ha
- **P_2O_5:** 20–40 kg/ha
- **K_2O:** 60–120 kg/ha

Phosphorus functions:

- Stimulates root development
- Enhances flowering uniformity
- Supports bean filling and cherry development

Recommended fertilisers:

- TSP or SSP for basal application
- NPK 25-5-5, 15-9-20, or 17-17-17 blends depending on region
- DAP for nursery and young trees

Coffee is a major driver of phosphate fertiliser demand in eastern, western, and central Uganda.

5.6.4 Tea

Tea is a nutrient-intensive crop with moderate phosphorus needs.

Typical annual rates:

- **N:** 150–250 kg/ha
- **P_2O_5:** 20–30 kg/ha
- **K_2O:** 80–120 kg/ha

Suitable fertilisers:

- NPK 25-5-5
- NPK 20-10-10
- TSP for soil correction in low-P sites

Large estates in western Uganda represent stable, predictable P_2O_5 demand.

5.6.5 Horticulture (Vegetables, Fruits, Potatoes)

These high-value crops have among the highest fertiliser responses.

Typical rates:

- **P_2O_5:** 40–80 kg/ha

- **N & K:** vary widely depending on crop

Phosphorus roles:

- Enhances flowering and fruit set (tomatoes, peppers, onions)
- Supports tuberisation (potatoes)
- Improves root development in leafy vegetables

Suitable fertilisers:

- DAP, MAP (basal)
- Water-soluble fertilisers for irrigation systems
- Specialised NPKs: 15-15-15, 12-32-16, 19-19-19

This segment is rapidly expanding and a strong future driver of phosphate demand.

5.6.6 Bananas (Matooke)

Bananas occupy over 2 million hectares in Uganda but traditionally receive little fertiliser.

Typical nutrient needs:

- **N:** 200–300 kg/ha
- **P_2O_5:** 30–50 kg/ha
- **K_2O:** 250–350 kg/ha

Phosphorus roles:

- Supports vigorous root mat development
- Enhances sucker initiation
- Improves bunch weight and fruit filling

Suitable fertilisers:

- Compound NPKs (e.g., 25-5-5, 17-17-17)
- SSP/TSP for soil correction

Bananas have huge potential demand if fertiliser use increases from current minimal levels.

5.6.7 Rice

Typical nutrient requirements:

- **P_2O_5:** 30–50 kg/ha
- **N:** 60–120 kg/ha
- **K_2O:** 20–40 kg/ha

Phosphorus roles:

- Enhances tillering

- Improves grain formation, milling quality
- Supports early establishment in paddy and upland systems

Suitable fertilisers:

- DAP at planting
- NPK 15-15-15 or 23-21-0+4S depending on water regime

With irrigation expansion, P demand from rice-growing zones will increase significantly.

5.6.8 Sugarcane

Sugarcane is nutrient-demanding but relatively modest in P requirements.

Typical annual rates:

- **P_2O_5:** 20–40 kg/ha
- **N:** 100–150 kg/ha
- **K_2O:** 120–200 kg/ha

Sugar mills increasingly require consistent fertiliser supply, creating an anchor demand for NPKs.

5.6.9 Implications for Phosphate Fertiliser Demand

Across all major crops:

- Phosphorus is **universally deficient** in Ugandan soils.
- Crop-specific requirements show **consistent annual demand** for SSP, TSP, DAP, MAP, and balanced NPKs.
- Perennial crops (coffee, tea, bananas) generate **steady year-round demand**, improving off-take stability for a domestic plant.
- Cereals and legumes drive **seasonal peak demand**, aligning with production cycles.

A domestic phosphate fertiliser complex—producing DAP, SSP, TSP, MAP, and tailored NPK formulations—would directly respond to Uganda's agronomic nutrient needs and stimulate higher adoption through:

- Lower prices,
- Better targeting of blends, and
- Improved supply reliability.

5.7 Household-Level Constraints

Despite the agronomic need for phosphorus and the potential benefits of fertiliser use, most Ugandan farming households apply little or no inorganic fertiliser. Household-level constraints—rooted in affordability, risk attitudes, behavioural factors, and systemic market failures—significantly suppress effective demand. These constraints must be understood to design appropriate fertiliser products, pricing strategies, distribution models, and complementary interventions.

5.7.1 Affordability and Cash-Flow Limitations

Most smallholder farmers operate with very tight liquidity:

- Incomes are **seasonal**, arriving mainly at harvest.
- Fertiliser purchases require **upfront cash**, often months before crop sale.
- Households balance competing needs: school fees, food, medical bills, and emergencies.
- Fertiliser therefore becomes a "postponable" expense, even when agronomically beneficial.

Price volatility of DAP, NPKs, and urea further discourages adoption, as farmers cannot predict whether the investment will be profitable.

5.7.2 Limited Access to Credit and Financial Services

Credit constraints remain a major barrier:

- Agricultural credit availability is extremely low (<10% of total lending).
- Interest rates are high, collateral requirements are strict, and loan processing is slow.
- Informal lenders charge prohibitively high rates.
- Few farmers access input-credit schemes, and most rely on cash purchases.

Without affordable financing, farmers tend to buy **under-application quantities** (1–2 bags per season) or forego fertilisers entirely.

5.7.3 Low Agronomic Knowledge and Weak Extension Systems

Most farming households lack access to reliable agronomic advice:

- Limited understanding of **nutrient deficiencies**, soil acidity, and crop-specific requirements.
- Misconceptions that fertilisers "burn" crops or damage soils.

- Under-application or misuse reduces yield response, reinforcing scepticism.
- Few farmers conduct **soil testing** or receive personalised fertiliser recommendations.

Where farmers lack training, returns on fertiliser investment are inconsistent, lowering willingness to invest.

5.7.4 High Perceived Risk in Rainfed Agriculture

Uganda's agriculture is predominantly rainfed:

- Erratic rainfall increases uncertainty about crop survival.
- Farmers perceive fertiliser as a **high-risk investment** if the season fails.
- Climate variability (droughts, floods) further heightens risk aversion.

This risk perception is especially strong among poorer households and female-headed households with limited fallback assets.

5.7.5 Market Access and Last-Mile Distribution Barriers

Many rural areas face structural distribution challenges:

- Agro-dealer density is low in remote districts.
- Transport costs inflate retail prices far above urban averages.
- Stock-outs during peak planting periods are common.
- Some farmers travel long distances, reducing their willingness to purchase bulky inputs.

Weak last-mile distribution networks contribute to low and inconsistent household uptake.

5.7.6 Prevalence of Counterfeit and Low-Quality Fertiliser

Adulteration significantly undermines household trust:

- Farmers often encounter fertiliser that is poorly granulated, under-strength, or mixed with soil/sand.
- Counterfeits reduce yield response and create lasting distrust in fertiliser use.
- Smallholders with limited experience become risk-averse when prior investments did not pay off.

A reliable domestic supplier with strong branding and QA/QC can restore confidence.

5.7.7 Gender-Based Constraints

Women play a significant role in crop production but face:

- Greater liquidity constraints,
- Limited access to land titles (affecting credit eligibility),
- Time burdens that restrict visits to agro-dealers, and
- Lower access to extension and training.

These constraints reduce fertiliser adoption among female-headed households, a significant share of Uganda's agricultural population.

5.7.8 Information and Behavioural Barriers

Behavioural constraints further depress adoption, including:

- **Present bias:** prioritising immediate consumption over long-term productivity investments.
- **Loss aversion:** fear of losing money if fertiliser does not deliver expected returns.
- **Social learning:** farmers rely on neighbours' experiences, which can be inconsistent.

Such behavioural frictions lower responsiveness to price reductions alone.

5.7.9 Implications for Domestic Phosphate Fertiliser Production

Understanding household-level constraints is essential to designing an effective domestic fertiliser strategy. Key implications include:

- Lower retail prices from domestic production will improve affordability but **must be paired** with improved extension and farmer education.
- Pack sizes (e.g., 10 kg, 25 kg bags) tailored to liquidity constraints can expand adoption.
- Credible, high-quality locally produced fertilisers can rebuild trust and reduce counterfeit penetration.
- Combining fertiliser production with distribution partnerships (cooperatives, SACCOs, estate outgrower schemes) improves last-mile access.
- Demand will rise more rapidly among commercial and semi-commercial farmers, but targeted support programmes can unlock smallholder adoption.

Ultimately, addressing household-level constraints ensures that domestic phosphate fertiliser production translates into **real, sustained uptake**, improved soil health, and higher agricultural productivity.

5.8 Demand Forecasts (2030–2040)

5.8.1 Agronomic Nutrient Requirement Model (ANRM)

5.8.1.1 Purpose and Policy Relevance

The Agronomic Nutrient Requirement Model (ANRM) estimates Uganda's **medium- to long-term phosphorus (P_2O_5) fertiliser demand** based on crop agronomy rather than historical fertiliser consumption alone. The model responds to a critical policy question:

How much phosphorus fertiliser does Uganda need to restore soil fertility and achieve productivity targets under NDP IV?

By grounding demand estimation in **crop area, agronomic nutrient requirements, and adoption scenarios**, the ANRM provides an objective basis for:

- Fertiliser policy design,
- Domestic phosphate fertiliser investment decisions, and
- Long-term food security and agro-industrialisation planning.

5.8.1.2 Agronomic Basis and Model Inputs

The ANRM uses Uganda's dominant nutrient-responsive crops and nationally relevant agronomic phosphorus application rates.

Crop Basket and Phosphorus Requirements

Table 5- 1: Crop Basket And Phosphorus Requirements

Crop	Cultivated Area (ha)	Recommended P_2O_5 (kg/ha)
Maize	1,150,000	40
Beans & legumes	1,400,000	30
Bananas (matooke)	1,500,000	40
Coffee	450,000	30
Rice	250,000	40
Tea	50,000	25
Horticulture	200,000	60
Total	**≈5.0 million ha**	—

These crops represent the bulk of Uganda's fertiliser-responsive cultivated area and are central to food security, exports, and agro-industrial value chains.

5.8.1.3 Key Quantitative Outputs of the ANRM

Optimal National Phosphorus Requirement

Applying recommended P_2O_5 rates across the crop basket yields an **optimal national phosphorus requirement of approximately:**

≈190,000 tonnes of P_2O_5 per year

This represents the **technical agronomic requirement** for restoring soil fertility and achieving sustainable yields across Uganda's major crops.

For context, current phosphorus use is estimated at **below 30,000 tonnes of P_2O_5 per year**, implying that **less than 15% of agronomic need is currently met**.

Realised Demand Scenarios (2030 and 2040)

Recognising that not all farmers will adopt fertiliser immediately, the ANRM applies realistic adoption rates under three scenarios.

2030 Projections

Table 5- 2: 2030 Projections

Scenario	Adoption Rate	Realised P_2O_5 Demand (t/year)
Baseline	30%	≈57,000
Moderate (policy-aligned)	50%	≈95,000
Accelerated	70%	≈133,000

2040 Projections

Table 5- 3 2040 Projections

Scenario	Adoption Rate	Realised P_2O_5 Demand (t/year)
Baseline	40%	≈76,000
Moderate (policy-aligned)	60%	≈114,000
Accelerated	80%	≈152,000

These figures demonstrate that even under conservative assumptions, **domestic phosphorus demand more than doubles by 2030** and continues to rise by 2040.

Implied Application Rates

The model also calculates the implied national average phosphorus application rate:

- **Current:** <5 kg P_2O_5/ha
- **Baseline 2030:** ≈11 kg P_2O_5/ha

- **Moderate 2030:** ≈18–19 kg P_2O_5/ha
- **Accelerated 2040:** ≈30 kg P_2O_5/ha

While still below global norms, these levels represent a substantial improvement and align with **incremental progress toward Malabo and NDP IV targets**.

Translation into Fertiliser Products

For policy and investment planning, realised P_2O_5 demand translates into the following **annual fertiliser product equivalents**:

Moderate Scenario (Policy-Aligned)

Table 5- 4: Moderate Scenario (Policy-Aligned)

Year	P_2O_5 (t)	DAP / TSP (46% P_2O_5)	SSP (16% P_2O_5)
2030	≈95,000	≈206,000 t	≈594,000 t
2040	≈114,000	≈248,000 t	≈713,000 t

This confirms that Uganda's domestic market alone can support **hundreds of thousands of tonnes per year** of phosphate fertiliser production, even before considering exports.

5.8.1.4 Policy Implications of the Quantitative Results

Fertiliser and Soil Health Policy

The ANRM confirms a **structural phosphorus deficit** across Uganda's soils. Without intervention, nutrient mining will continue to depress yields and undermine productivity gains from improved seed, irrigation, and mechanisation.

Policy implication: Phosphorus availability must be treated as a **core pillar of soil health and productivity policy**, not a secondary input.

Domestic Phosphate Fertiliser Investment

The model demonstrates that:

- Domestic demand of **95,000–114,000 tonnes of P_2O_5 per year** is achievable under policy-aligned conditions.
- This translates into **200,000–250,000 tonnes per year of high-analysis fertilisers** (DAP/TSP) or significantly larger volumes of SSP.

Policy implication: Domestic phosphate beneficiation and fertiliser manufacturing is supported by **quantified, long-term domestic demand**, not speculative assumptions.

Import Substitution and Price Stability

Uganda currently imports virtually all phosphate fertilisers. The ANRM shows that a domestic plant could substitute:

- A large share of DAP and NPK imports by 2030,

- With increasing substitution by 2040 as adoption rises.

Policy implication: Local production reduces exposure to global price volatility and logistics shocks, improving food security resilience.

Complementary Demand-Side Measures

The gap between optimal (≈190,000 t P_2O_5) and realised demand highlights the importance of:

- Fertiliser price reductions,
- Extension and soil testing,
- Input credit,
- Targeted subsidies or e-voucher programmes.

Policy implication: Manufacturing investments must be paired with **demand-side interventions** to unlock full agronomic benefits.

5.8.1.5 Implications for Project Sizing and Phasing

The ANRM supports a **phased investment approach**:

- **Phase 1:** Serve 50–60% of domestic P_2O_5 demand (≈90,000–115,000 t P_2O_5)
- **Phase 2:** Scale toward accelerated scenario levels (≈150,000 t P_2O_5) plus regional exports

This approach aligns with DC principles of **affordability, scalability, and risk management**.

The Agronomic Nutrient Requirement Model provides compelling quantitative evidence that Uganda's current fertiliser use substantially under-represents its true agronomic needs. By 2030 and 2040, realistic adoption scenarios point to **domestic phosphorus demand several times higher than today**, even without accounting for regional markets.

From a policy standpoint, the ANRM:

- Strengthens the economic justification for domestic phosphate fertiliser production,
- Aligns fertiliser investment with soil health and food security objectives,
- Provides a robust analytical foundation for feasibility analysis, PPP structuring, and DC approval.

Incorporating the ANRM ensures that decisions on phosphate beneficiation and fertiliser manufacturing are **evidence-based, policy-coherent, and aligned with Uganda's long-term development agenda**.

5.8.1.6 Integration of ANRM Outputs into the Economic and Financial Justification

Economic Justification: Linking Agronomic Need to National Welfare Gains

The Agronomic Nutrient Requirement Model (ANRM) provides the economic foundation for the phosphate fertiliser investment by quantifying Uganda's **structural phosphorus deficit** and the welfare losses associated with under-application of fertiliser.

Economic Need and Problem Definition

The ANRM estimates Uganda's **optimal national phosphorus requirement** at approximately **190,000 tonnes of P_2O_5 per year**, based on crop areas and recommended agronomic application rates. Current phosphorus use is estimated at **below 30,000 tonnes of P_2O_5 per year**, meaning that **less than 15% of agronomic need is currently met**.

This deficit manifests economically through:

- Persistently low crop yields,
- Soil nutrient mining and declining land productivity,
- Reduced returns to improved seed, irrigation, and mechanisation,
- High vulnerability to fertiliser import price shocks.

From an economic perspective, the project addresses a **clear market failure**: fertiliser prices and supply constraints prevent farmers from applying economically optimal nutrient levels, leading to sub-optimal national output.

Economic Demand Outlook (2030–2040)

Applying realistic adoption rates, the ANRM estimates **realised phosphorus demand** as follows:

2030

- Baseline: ≈ **57,000 t P_2O_5**
- Moderate (policy-aligned): ≈ **95,000 t P_2O_5**
- Accelerated: ≈ **133,000 t P_2O_5**

2040

- Baseline: ≈ **76,000 t P_2O_5**
- Moderate (policy-aligned): ≈ **114,000 t P_2O_5**
- Accelerated: ≈ **152,000 t P_2O_5**

These quantities represent **incremental nutrient use** driven by productivity gains rather than substitution alone, and therefore translate into **real increases in agricultural output, rural incomes, and food security**.

Economic Benefits Generated by Closing the Phosphorus Gap

By enabling fertiliser use closer to agronomic requirements, the project generates the following **economic benefits**:

1. **Yield gains across major crops.** Empirical agronomic evidence shows that phosphorus application yields:

    - 20–50% yield increases in maize and beans,
    - Significant productivity recovery in coffee, bananas, and horticulture.
2. **Increased agricultural GDP.** The ANRM demand levels underpin sustained growth in agricultural output consistent with NDP IV targets, contributing positively to GDP and export earnings.
3. **Reduced import dependency and foreign exchange savings.** Domestic production substitutes imports equivalent to:
    - ≈200,000–250,000 tonnes of DAP/TSP annually under the moderate scenario by 2030, generating substantial foreign exchange savings.
4. **Positive spillovers and externalities**
    - Improved food availability,
    - Lower price volatility,
    - Stronger rural livelihoods,
    - Greater resilience to global fertiliser supply shocks.

These benefits underpin a **strong positive Economic Net Present Value (ENPV)** and Economic Internal Rate of Return (EIRR), even before accounting for regional exports.

5.8.1.7 Financial Justification: Demand Certainty, Plant Sizing, and Revenue Stability

Domestic Market as a Financial Anchor

The ANRM confirms that **domestic demand alone** provides a credible off-take base for the project.

Under the **moderate scenario**, domestic realised demand translates into:

Table 5- 5: Moderate Scenario, Domestic Realised Demand

Year	P_2O_5 Demand	DAP/TSP Equivalent
2030	≈95,000 t	≈206,000 t
2040	≈114,000 t	≈248,000 t

This demand level:

- Is **structural and recurring**, not cyclical,
- Is driven by agronomic need rather than short-term price arbitrage,
- Provides a stable base load for plant operations.

From a lender and investor perspective, this reduces **market risk** and improves revenue predictability.

Implications for Plant Capacity and Phasing

The ANRM outputs directly inform **optimal plant sizing**:

- **Phase 1 (Domestic-led):**
    - Capacity aligned to **50–60% of moderate-scenario demand**,
    - Equivalent to **100,000–150,000 t/year of high-analysis phosphate fertiliser**.
- **Phase 2 (Expansion):**
    - Scaling toward **accelerated-scenario demand**,
    - Integration of regional exports into EAC/COMESA markets.

This phased approach:

- Matches demand growth,
- Reduces upfront capital risk,
- Improves financial resilience during early years.

Revenue Composition and Product Mix

ANRM outputs guide product strategy:

- **High-analysis fertilisers (DAP/TSP):**
    - Serve commercial crops, blending plants, and estates,
    - Higher unit value and stronger cash flow.
- **SSP:**
    - Larger volume demand for acidic soils and legumes,
    - Supports broad-based adoption and market penetration.

Under the moderate scenario, SSP-equivalent demand exceeds **590,000 tonnes by 2030**, indicating potential for:

- Dual product lines,
- Flexible production depending on price and seasonality.

5.8.1.8 Financial Viability and Risk Mitigation

The ANRM materially improves financial viability by:

1. **Reducing demand uncertainty.** Demand is anchored in agronomic necessity rather than discretionary consumption.
2. **Lowering price elasticity risk.** Phosphorus demand is less elastic once adoption increases and soil response becomes evident.
3. **Supporting long-term off-take agreements**
    - Estate agriculture,
    - Cooperatives,

- Government-supported input programmes.

4. **Strengthening bankability under PPP models.** Clear domestic demand supports:
   - Debt service coverage,
   - Lower revenue volatility,
   - More favourable financing terms.

5.8.1.9 Integration with Financial and Economic Models

In the feasibility analysis:

- **Economic analysis (ENPV/EIRR):**
  - Uses ANRM demand as the **benefit driver** through yield and output gains.
- **Financial analysis (FNPV/FIRR):**
  - Uses ANRM-derived volumes as the **baseline sales forecast** for domestic off-take.
- **Sensitivity analysis:**
  - Adoption rates (±10–20%) tested against ANRM scenarios,
  - Price and cost shocks assessed around ANRM demand floor.

The ANRM therefore acts as the **bridge** between agronomy, economics, and finance.

5.8.1.10 Overall Justification for Public Intervention

The integration of ANRM outputs demonstrates that:

- The project addresses a **quantified national nutrient deficit**,
- Domestic demand is **sufficient, predictable, and policy-aligned**,
- Economic benefits extend beyond private returns,
- Public participation (TPS or PPP support) is justified under DC Guidelines due to:
  - Market failure,
  - Strategic food security importance,
  - Large positive externalities.

By grounding both the economic and financial justification in the Agronomic Nutrient Requirement Model, the project's investment case shifts from speculative supply expansion to an **evidence-based response to national agronomic need**. The ANRM provides credible demand floors, informs optimal plant sizing, strengthens bankability, and ensures alignment with NDP IV and Uganda's agro-industrial transformation agenda.

5.9 Econometric Demand Projection Model (EDPM)

5.9.1 1. Purpose of the EDPM in the Investment Case

The Econometric Demand Projection Model (EDPM) provides a **market-based forecast of fertiliser demand in Uganda**, grounded in observed historical behaviour and macro-economic drivers. While the Agronomic Nutrient Requirement Model (ANRM) establishes *technical nutrient needs*, the EDPM determines *realistic uptake* under different economic and policy scenarios.

For the phosphate fertiliser investment, the EDPM performs a critical role by:

- translating agronomic potential into **bankable demand forecasts**,
- informing **plant sizing, utilisation, and phasing**, and
- strengthening both **economic and financial appraisal** under DC Guidelines.

5.9.2 Model Calibration and Data Basis

The EDPM is calibrated using **actual Uganda data from the World Development Indicators (WDI)** for the period **2011–2021**, the maximum overlap of relevant indicators in the dataset provided.

Key features of the calibration include:

- Fertiliser consumption measured as **kg per hectare of arable land**, converted into **total national fertiliser tonnes** using arable land area.
- Agricultural value added (constant prices) used as a proxy for **farm income and sector scale**.
- Domestic credit to private sector (% of GDP) used as a proxy for **access to finance**.
- Real effective exchange rate (REER) used as a proxy for **import-price pressure and macro-price transmission**.

The calibrated econometric relationship reflects **observed Ugandan behaviour**, not assumed or stylised demand paths.

5.9.3 Scenario Framework for Projections (2025–2040)

Demand is projected under three scenarios:

- **Baseline scenario.** Modest agricultural growth, limited credit deepening, continued exposure to international fertiliser price volatility.
- **Moderate scenario (policy-aligned).** Growth consistent with NDP IV, improved credit access, and gradual easing of fertiliser affordability constraints.

- **Accelerated scenario.** Strong agricultural growth, deeper credit markets, and improved affordability associated with domestic fertiliser production and supportive policies.

These scenarios are applied consistently across the projection horizon and are transparent in the Excel model.

5.9.4 EDPM Results: Fertiliser and Phosphorus Demand

5.9.4.1 Total Fertiliser Demand (Market-Based)

The EDPM indicates that fertiliser demand in Uganda grows steadily across all scenarios, driven primarily by:

- rising agricultural value added, and
- improved access to credit.

Demand growth is **structural**, not episodic, and does not collapse to zero under any scenario, reflecting realistic market behaviour.

5.9.4.2 Phosphorus (P_2O_5) Demand Results

Using the model's phosphorus share parameter (consistent with Uganda's fertiliser composition), total fertiliser demand is converted into **P_2O_5 demand**, which is the relevant metric for phosphate investment planning.

EDPM – P_2O_5 Demand Results

Table 5- 6: EDPM – P_2O_5 Demand Results

Scenario	**2030 P_2O_5 Demand (t/year)**	**2040 P_2O_5 Demand (t/year)**
Baseline	~55,000 – 65,000	~75,000 – 85,000
Moderate (policy-aligned)	~85,000 – 100,000	~110,000 – 125,000
Accelerated	~115,000 – 135,000	~145,000 – 165,000

These results show that, even under conservative assumptions, **market-driven phosphorus demand increases substantially by 2030 and further by 2040**.

5.9.4.3 Fertiliser Product Equivalents

For practical investment analysis, EDPM phosphorus demand is translated into fertiliser product equivalents.

Moderate Scenario – Product Equivalents

Table 5- 7: Moderate Scenario – Product Equivalents

Year	P_2O_5 (t)	DAP / TSP @46% (t)	SSP @16% (t)
2030	~90,000	~195,000	~560,000
2040	~120,000	~260,000	~750,000

This confirms that **domestic demand alone** can absorb large volumes of phosphate fertiliser products, even before considering regional exports.

5.9.5 Implications for Economic Viability

5.9.5.1 Contribution to National Economic Benefits

The EDPM results demonstrate that fertiliser demand growth is closely linked to:

- agricultural income growth,
- financial inclusion, and
- macro-economic stability.

As fertiliser use increases:

- crop yields improve,
- agricultural GDP expands,
- rural incomes rise, and
- food security strengthens.

These outcomes underpin **large positive economic externalities**, supporting a strong **Economic Net Present Value (ENPV)** for domestic phosphate fertiliser production when assessed over the project life.

5.9.5.2 Alignment with Agronomic Potential

When compared with the ANRM:

- EDPM demand levels represent a **realistic share of agronomic need**,
- the gap between agronomic requirement and market uptake narrows significantly by 2040 under the moderate and accelerated scenarios.

This confirms that phosphate investment is not premature; rather, it is **timed to coincide with rising market absorption capacity**.

5.9.6 Implications for Financial Viability and Bankability

5.9.6.1 Revenue Certainty

The EDPM provides a **market-based demand floor** for the financial model. Unlike purely technical demand estimates, EDPM outputs reflect actual behavioural responses to income and credit growth.

This improves confidence in:

- sales volumes,
- utilisation rates, and
- revenue stability.

5.9.6.2 Plant Capacity and Phasing

The EDPM results support a **phased development strategy**:

- **Phase 1:** Capacity aligned to baseline–moderate EDPM demand (≈60,000–100,000 t P_2O_5 equivalent), ensuring high utilisation in early years.
- **Phase 2:** Expansion aligned with accelerated EDPM demand and regional exports, reducing stranded-asset risk.

This approach enhances **financial resilience** and aligns with DC guidance on affordability and scalability.

5.9.6.3 Risk Management

The EDPM explicitly shows how demand responds to:

- macro-economic growth,
- credit availability, and
- price-related shocks.

This allows:

- robust sensitivity testing in the financial model,
- stress testing of revenues,
- clearer allocation of demand risk in PPP or hybrid delivery structures.

5.9.7 Policy Implications for Phosphate Investment

The EDPM results highlight that:

1. **Demand growth is policy-responsive.** Credit expansion and affordability measures materially increase fertiliser uptake.
2. **Domestic production has a catalytic effect.** Stabilising supply and reducing price volatility shifts demand from the baseline toward moderate and accelerated paths.
3. **Public intervention is justified.**Because fertiliser demand generates large economic externalities, public participation or facilitation is warranted under DC Guidelines.

The Econometric Demand Projection Model provides **quantitative, Uganda-specific evidence** that market-driven fertiliser demand—and specifically phosphorus demand—will rise steadily through 2030 and 2040. When integrated with agronomic requirements, the EDPM confirms that:

- domestic phosphate fertiliser production is **economically justified**,
- sales volumes are **financially viable and bankable**, and
- phased investment significantly reduces demand and utilisation risk.

Together, the EDPM and ANRM provide a **robust, defensible demand foundation** for the proposed phosphate fertiliser investment in Uganda, fully aligned with national development objectives and DC appraisal requirements.

5.10 Market Opportunities for Different Phosphate-Based Products

Phosphate resources support a **diversified portfolio of products** spanning agriculture, livestock, and industry. The market opportunities for these products differ in scale, growth dynamics, pricing structures, and risk profiles. For Uganda, exploiting these opportunities requires a **sequenced and integrated value-chain strategy**, starting with fertilisers and progressively expanding into feed-grade and industrial phosphate products as market depth and processing capability increase.

5.10.1 Fertiliser Products

Market Opportunity and Demand Outlook

Fertiliser products represent the **largest, most immediate, and most strategic market opportunity** for phosphate utilisation in Uganda. Both the ANRM and EDPM confirm that phosphorus demand is structurally strong and rising:

- **Agronomic requirement:** ≈190,000 t P_2O_5 per year
- **Market-based demand (EDPM):**
  - ~85,000–100,000 t P_2O_5 by 2030 (moderate scenario)
  - ~110,000–125,000 t P_2O_5 by 2040

- These levels translate into **200,000–260,000 t/year of high-analysis fertilisers** (DAP/TSP equivalents), excluding NPK blending volumes.

Uganda currently imports virtually all phosphate fertilisers, exposing the market to:

- price volatility,
- foreign exchange risk, and
- supply disruptions.

This creates a clear **import-substitution opportunity** for domestically produced phosphate fertilisers.

Key Fertiliser Products and Market Segments

a) Di-Ammonium Phosphate (DAP)

- Widely used basal fertiliser for maize, beans, rice, horticulture, and coffee.
- Strong acceptance among farmers due to high nutrient concentration.
- High unit value and strong cash-flow generation.

b) Triple Super Phosphate (TSP)

- Suitable for blending and for crops requiring phosphorus without nitrogen.
- Strong relevance for coffee, tea, legumes, and acidic soils.

c) Single Super Phosphate (SSP)

- Lower nutrient concentration but contains sulphur.
- Particularly relevant for legumes and acid soils.
- Very large volume potential under smallholder expansion.

d) NPK Compounds and Blends

- Growing demand for crop- and soil-specific formulations.
- Phosphate production provides the core input for domestic NPK blending.

Policy and Investment Implications

- Fertiliser products should be the **anchor output** of any phosphate beneficiation and processing investment.
- Domestic fertiliser production aligns directly with:
  - NDP IV agro-industrialisation objectives,
  - food security priorities, and
  - import substitution goals.
- Fertiliser sales provide the **revenue base** needed to support subsequent diversification into higher-value phosphate derivatives.

5.10.2 Feed-Grade Phosphates

Market Opportunity

Feed-grade phosphates, particularly **Dicalcium Phosphate (DCP)** and **Monocalcium Phosphate (MCP)**, are essential mineral supplements in livestock and poultry feed.

Uganda's livestock sector is expanding, driven by:

- rising demand for poultry meat and eggs,
- growth in commercial dairy and piggery operations,
- increasing adoption of compound feeds.

However, **most feed-grade phosphates are imported**, adding cost and supply risk to the animal-feed industry.

Key Feed-Grade Products

a) Dicalcium Phosphate (DCP)

- Widely used in poultry, dairy, and pig feed.
- Improves bone development, feed conversion ratios, and productivity.
- Moderate but stable demand profile.

b) Monocalcium Phosphate (MCP)

- Higher phosphorus bioavailability.
- Used in intensive commercial feed formulations.
- Higher value per tonne than fertiliser phosphates.

Market Characteristics

- Smaller volumes than fertilisers, but **higher margins**.
- Demand grows with:
    - commercialisation of livestock,
    - feed-milling capacity expansion,
    - regional export of animal products.

Policy and Investment Implications

- Feed-grade phosphates are a **natural second-stage product** after fertilisers.
- Production requires:
    - higher purity standards,
    - stricter quality control,
    - food- and feed-safety compliance.
- Domestic production reduces costs for feed manufacturers and strengthens livestock value chains.

- Supports NDP IV objectives on **livestock productivity and value addition**.

5.10.3 Industrial Phosphate Chemicals

Market Opportunity

Industrial phosphate chemicals represent the **highest value-added but most technically demanding** segment of the phosphate value chain. Demand exists both domestically and regionally, but volumes are smaller and quality requirements are stringent.

Key Industrial Products

a) Phosphoric Acid (PA)

- Core intermediate for fertilisers, feed phosphates, and industrial chemicals.
- Industrial-grade and food-grade PA have distinct markets and pricing.
- Regional demand from detergents, beverages, and food processing.

b) Sodium Tripolyphosphate (STPP)

- Used in detergents and cleaning agents.
- Mostly imported into East Africa.
- Price-sensitive but attractive margins.

c) Water-Treatment and Metal-Treatment Chemicals

- Used in:
  - industrial water systems,
  - boilers,
  - municipal water treatment,
  - metallurgy.
- Demand grows with industrialisation and urbanisation.

d) Food and Pharmaceutical Phosphates

- Used as additives and stabilisers.
- High regulatory and quality thresholds.
- Small volumes, high value.

Market Characteristics

- Smaller volumes but **significantly higher unit values**.
- Strong dependence on:
  - reliable phosphoric acid supply,
  - energy availability,
  - skilled labour,

    - regulatory compliance.

Policy and Investment Implications

- Industrial phosphates are best pursued as a **long-term diversification strategy**.
- Require:
    - mature operational capability,
    - stable fertiliser revenues to cross-support investment,
    - targeted export-oriented market development.
- Align with Uganda's long-term industrial upgrading and chemical-industry ambitions.

5.10.4 Strategic Sequencing of Market Opportunities

The market analysis supports a **phased phosphate-value-chain development strategy**:

Table 5- 8: Strategic Sequencing of Market Opportunities

Phase	Primary Products	Rationale
Phase 1	Fertiliser phosphates (DAP, TSP, SSP, NPKs)	Large domestic demand, import substitution, revenue stability
Phase 2	Feed-grade phosphates (DCP, MCP)	Growing livestock sector, higher margins
Phase 3	Industrial phosphate chemicals	High value addition, export potential

This sequencing:

- reduces market and technical risk,
- ensures early cash-flow generation,
- allows progressive upgrading of processing capability.

5.10.5 Overall Implications for the Phosphate Investment

The market opportunities across fertilisers, feed phosphates, and industrial chemicals demonstrate that phosphate investment in Uganda is **not a single-product proposition**, but a **multi-market industrial platform**.

Key conclusions:

- Fertilisers provide the **scale and certainty** required for project viability.
- Feed-grade phosphates add **margin and diversification**.
- Industrial phosphates offer **long-term upgrading and export potential**.

- Combined with ANRM and EDPM results, these markets provide a **robust justification for integrated phosphate beneficiation and processing**.

6 Cost Structure and Competitiveness Analysis

6.1 Cost Build-Up Along the Phosphate Value Chain

The cost structure of phosphate-based products is determined by the cumulative costs incurred at each stage of the value chain, from mineral extraction to final product delivery. Understanding this build-up is critical for assessing the **competitiveness of domestic phosphate beneficiation and processing in Uganda**, relative to imported fertilisers and phosphate derivatives.

This section analyses the cost drivers across the **upstream, midstream, and downstream** segments of the phosphate value chain and identifies where Uganda has **cost advantages, neutral positions, or structural disadvantages**.

6.1.1 Upstream Costs: Mining and Beneficiation

a) Phosphate Rock Mining

Key cost components include:

- Overburden removal and mining operations
- Drilling, blasting, and excavation
- Haulage within the mine
- Mine management, labour, and equipment maintenance

Uganda's phosphate deposits, particularly in the Sukulu–Tororo area, are characterised by:

- Shallow ore bodies, reducing stripping ratios
- Established geological knowledge from historical operations
- Proximity to existing transport corridors

Cost implication: Mining costs are expected to be **moderate and competitive** by regional standards, particularly when compared to imported phosphate rock sourced from North Africa or the Middle East.

b) Beneficiation and Ore Preparation

Beneficiation costs arise from:

- Crushing, grinding, and screening
- Washing and flotation
- Tailings management and water recycling
- Energy and reagent consumption

The beneficiation stage significantly affects:

- Final P_2O_5 concentration,

- Downstream processing efficiency, and
- Overall unit cost of fertiliser products.

Cost implication: While beneficiation adds to upfront costs, it:

- Improves ore quality, reduces downstream acid consumption, and
- Lowers overall cost per tonne of finished fertiliser.

Domestic beneficiation reduces reliance on imported high-grade rock and enhances long-term competitiveness.

6.1.2 Midstream Costs: Chemical Processing

a) Sulphuric Acid Production

Sulphuric acid is a critical input for phosphoric acid and superphosphate production.

Cost drivers include:

- Imported sulphur cost (FOB price, freight, insurance)
- Port handling and inland transport
- Energy consumption and plant efficiency
- Environmental controls (SO_2 capture, scrubbers)

Cost implication: Sulphur must be imported, exposing Uganda to:

- International sulphur price volatility, and
- Freight and logistics costs.

However, **integrated sulphuric acid production** on site:

- Reduces unit costs relative to importing acid, and
- Improves process reliability and scheduling.

b) Phosphoric Acid Production

Phosphoric acid costs are driven by:

- Phosphate rock quality and consumption rates
- Sulphuric acid usage
- Energy and water consumption
- By-product handling (phosphogypsum)
- Environmental compliance costs

Cost implication: Domestic phosphoric acid production allows Uganda to:

- Capture value currently embedded in imported intermediates, and
- Supply multiple downstream products (fertilisers, feed, industrial chemicals).

This stage is **capital-intensive** but central to value addition.

6.1.3 Downstream Costs: Fertiliser and Phosphate Product Manufacturing

a) Fertiliser Granulation and Compounding

Cost components include:

- Ammonia (for DAP/MAP) – largely imported
- Granulation and drying
- Blending and bagging
- Quality control and storage

Product-specific cost characteristics:

- **DAP/MAP:** Higher raw material costs but higher value per tonne
- **TSP:** Lower nitrogen input, competitive for blending and specific crops
- **SSP:** Lower processing cost, higher transport cost per nutrient unit

Cost implication: Product choice influences competitiveness. A flexible product mix improves margin stability.

b) Packaging, Storage, and Distribution

Downstream logistics costs include:

- Bagging materials
- Warehousing
- Inland transport to wholesalers and agro-dealers
- Seasonal inventory holding

Domestic production offers a clear advantage by:

- Eliminating international shipping and port congestion costs
- Reducing lead times and inventory risks
- Lowering inland transport distances to Ugandan and regional markets

6.1.4 Cross-Cutting Cost Drivers

Several factors affect costs across all stages:

a) Energy and Utilities

- Electricity, fuel, and water availability and tariffs
- Reliability of supply

Energy is a significant cost driver in beneficiation and chemical processing. Competitive energy pricing and reliable supply are essential for cost control.

b) Environmental and Regulatory Compliance

- ESIA compliance
- Emissions control systems
- Waste and by-product management

While compliance increases costs, it:

- Reduces long-term environmental and social risks, and
- Improves bankability and access to concessional finance.

c) Capital Costs and Financing

- Plant construction and equipment
- Financing costs and interest during construction
- Depreciation and maintenance

Large upfront capital costs are offset by:

- Long asset life, and
- Stable demand confirmed by ANRM and EDPM analyses.

6.1.5 Indicative Cost Build-Up Summary

While exact cost figures are determined at feasibility stage, the indicative cost contribution by stage is typically as follows:

Table 6- 1: Indicative Cost Build-Up

Value Chain Stage	Indicative Share of Total Cost
Mining & Beneficiation	20–30%
Chemical Processing (acidulation)	35–45%
Fertiliser Manufacturing & Finishing	15–20%
Logistics & Distribution	10–15%

The **largest cost savings potential** for Uganda lies in:

- Substituting imported finished fertilisers with domestic production, and
- Reducing freight, insurance, and port-related costs.

6.1.6 Competitiveness Implications for Uganda

The cost build-up analysis indicates that Uganda's competitiveness is strongest where:

- **Local phosphate rock substitutes imported inputs**
- **Value addition occurs closer to the market**

- **Logistics distances are reduced**

Conversely, competitiveness is more sensitive to:

- Imported sulphur and ammonia prices
- Energy tariffs
- Financing terms for capital-intensive plants

However, when evaluated across the full value chain, **domestic phosphate beneficiation and fertiliser production is structurally competitive against imports**, particularly under moderate and accelerated demand scenarios identified by the EDPM.

6.1.7 Link to Economic and Financial Analysis

The cost structure described here feeds directly into:

- Unit production cost estimates,
- Delivered fertiliser prices,
- Financial cash-flow models, and
- Economic cost–benefit analysis.

Combined with the **robust demand outlook** from ANRM and EDPM, the cost build-up supports a **positive economic and financial case** for phosphate investment in Uganda, provided that:

- integration across the value chain is achieved, and
- plant capacity is phased in line with demand growth.

6.2 Delivered Cost-to-Market vs Imports

This subsection compares the **delivered cost-to-market of domestically produced phosphate-based fertilisers** with the cost of **imported fertilisers supplied to Uganda**, using **existing demand, logistics, and price-structure evidence** already established in this study. The comparison is undertaken at the **Ugandan wholesale/farm-gate level**, which is the relevant benchmark for competitiveness, adoption, and market expansion.

6.2.1 Current Import-Based Cost Structure in Uganda

Uganda is almost entirely dependent on imported phosphate fertilisers (DAP, TSP, NPKs), with products sourced mainly from **Morocco, Saudi Arabia, China, and Russia**, and routed through **Mombasa and Dar es Salaam**.

Based on observed import and logistics patterns, the delivered cost structure of imported fertilisers in Uganda comprises the following layers:

1. **International fertiliser price (FOB)**
    - Global DAP and TSP prices are volatile and influenced by energy costs, export controls, and geopolitical factors.
    - During recent years, international fertiliser prices have fluctuated sharply, contributing to unstable domestic prices.
2. **Ocean freight and insurance**
    - Fertiliser freight to East Africa is structurally high due to long shipping distances and limited backhaul cargo.
    - Freight and insurance typically add a **significant premium** to the FOB price.
3. **Port handling and clearance costs**
    - Port charges, storage, demurrage, and clearance delays at Mombasa or Dar es Salaam add further cost and uncertainty.
4. **Inland transport to Uganda**
    - Inland haulage distances of approximately **700–1,200 km** substantially increase delivered costs.
    - Transport alone can account for a **large share of the final farm-gate price**, particularly for bulky products such as SSP and NPKs.
5. **Distribution, financing, and inventory costs**
    - Importers and wholesalers incur financing costs due to long supply chains and seasonal stocking requirements.
    - These costs are passed on to farmers through higher retail prices.

Observed outcome: For Uganda, the cumulative effect of freight, port handling, inland transport, and financing typically results in **import-related logistics and transaction costs adding approximately 30–50% to the international fertiliser price** by the time products reach the domestic market.

6.2.2 Domestic Production Cost Structure (Using Study Assumptions)

Domestic phosphate beneficiation and fertiliser manufacturing fundamentally changes the delivered-cost profile by **removing the most expensive and volatile cost layers** identified above.

Under the proposed domestic production model:

- **Phosphate rock** is sourced locally, eliminating dependence on imported rock or finished fertilisers.
- **Beneficiation and acidulation** are undertaken domestically, capturing value currently embedded in imported intermediates.

- **Inland distribution distances** are significantly shorter, particularly for the Ugandan market and neighbouring EAC countries.
- **Inventory cycles** are shorter, reducing working-capital and financing costs.

While domestic production still relies on imported inputs such as **sulphur and ammonia**, these inputs:

- Represent a smaller share of the delivered fertiliser cost than finished-product imports, and
- Can be managed through bulk procurement and integrated production.

6.2.3 Delivered Cost Comparison Using Existing Study Results

The EDPM and ANRM confirm that Uganda's fertiliser demand is **price-responsive**, meaning delivered-cost reductions translate directly into higher uptake.

Using existing data and assumptions from this study:

Table 6- 2: Delivered Cost Comparison

Cost Element	**Imported Fertiliser**	**Domestic Fertiliser**
International shipping	High	Eliminated
Port congestion & demurrage	Medium–High	Eliminated
Inland transport	High (long haul)	Low–Medium
Inventory & financing	High	Lower
Exchange-rate exposure	High	Reduced
Delivered price volatility	High	Lower

Net effect: Even where ex-factory production costs are similar, **domestic fertiliser is structurally cheaper on a delivered-to-market basis**, particularly for bulky phosphate products.

6.2.4 Product-Specific Competitiveness (Using EDPM Demand Results)

a) DAP and TSP

EDPM results indicate market-based phosphorus demand of approximately:

- **~85,000–100,000 t P_2O_5 by 2030**, and
- **~110,000–125,000 t P_2O_5 by 2040** under the moderate scenario.

This corresponds to:

- **~195,000–260,000 t/year of DAP/TSP equivalents**.

For these products:

- Imported DAP/TSP incur full international logistics costs.
- Domestic DAP/TSP avoids ocean freight and port-related costs.

Competitiveness conclusion: Domestic DAP and TSP are expected to be **cost-competitive or cheaper than imports at the Ugandan market**, particularly as demand volumes rise and plant utilisation improves.

b) Single Super Phosphate (SSP)

SSP has:

- Lower P_2O_5 concentration, and
- Higher transport cost per unit of nutrient.

EDPM-derived phosphorus demand translates into **very large SSP-equivalent volumes** (over **500,000 t/year** under moderate scenarios).

Competitiveness conclusion: SSP is **strongly advantaged by local production**, as importing bulky, low-analysis fertilisers over long distances is inherently uncompetitive.

c) NPK Compounds

Domestic phosphate production supports:

- Local NPK blending,
- Reduced cost of phosphate inputs, and
- Customised fertiliser formulations.

Competitiveness conclusion: Domestic phosphate inputs significantly improve the delivered-cost position of locally blended NPKs relative to fully imported compounds.

6.2.5 Impact on Fertiliser Uptake and Market Expansion

The EDPM demonstrates that fertiliser demand in Uganda responds positively to:

- Income growth, and
- Improved affordability.

Lower delivered prices achieved through domestic production therefore:

- Increase fertiliser adoption rates,
- Expand total market size beyond simple import substitution, and
- Move demand closer to agronomic requirements identified by the ANRM.

This reinforces the feedback loop between **cost competitiveness and demand growth**.

6.2.6 Exchange Rate and Supply Risk Considerations

Imported fertilisers expose Uganda to:

- Exchange-rate depreciation, and
- External supply disruptions.

Domestic production:

- Reduces foreign exchange exposure per tonne of fertiliser consumed,
- Stabilises supply during global shocks,
- Enhances national fertiliser security.

These risk reductions have **economic value**, particularly for a product as strategically important as fertiliser.

6.2.7 Implications for Financial and Economic Viability

The delivered-cost advantage of domestic production has direct implications for project viability:

- **Financial viability:**
  - Improved margins through lower logistics costs,
  - More stable revenues due to reduced price volatility,
  - Better utilisation rates supported by EDPM demand levels.
- **Economic viability:**
  - Lower fertiliser prices increase adoption,
  - Higher agricultural productivity generates positive externalities,
  - Reduced import dependence improves balance-of-payments resilience.

These effects underpin **positive FNPV and ENPV outcomes** under the moderate and accelerated scenarios used in this study.

Using existing demand results and Uganda's observed import structure, the delivered cost-to-market analysis confirms that **domestic phosphate beneficiation and fertiliser production is structurally competitive against imports**.

When combined with:

- EDPM-confirmed market demand growth, and
- ANRM-confirmed agronomic need,

domestic production offers a **durable cost and resilience advantage**, strengthening the economic and financial justification for phosphate investment in Uganda.

6.3 Profitability of Different Product Lines

The profitability of phosphate-based products in Uganda varies significantly across fertiliser, feed-grade, and industrial product lines. Differences arise from **nutrient concentration, processing intensity, market size, price volatility, logistics costs, and quality requirements**. This section evaluates profitability using **existing demand results (ANRM and EDPM)** and Uganda's **import-dependent market structure**, focusing on **gross margin potential and risk**, rather than nominal prices alone.

6.3.1 Fertiliser Product Lines

Fertiliser products constitute the **core revenue base** of the phosphate value chain and determine the overall financial viability of domestic phosphate investment.

a) Di-Ammonium Phosphate (DAP)

Market scale (EDPM-based):

- ~85,000–100,000 t P_2O_5 demand by 2030 (moderate scenario)
- Equivalent to ~195,000–230,000 t/year of DAP/TSP

Profitability drivers:

- High nutrient concentration (46% P_2O_5)
- Strong and stable demand across cereals, legumes, rice, coffee, and horticulture
- Lower logistics cost per unit of nutrient compared to SSP
- Premium pricing relative to bulk fertilisers

Cost considerations:

- Requires imported ammonia and sulphur
- Higher energy and capital intensity than SSP

Profitability assessment:

- **Moderate to high margins**, provided plant utilisation exceeds threshold volumes
- Margin stability improves significantly under moderate and accelerated EDPM scenarios

Conclusion: DAP is a **financial anchor product** with strong revenue-generation capacity, suitable for early-phase operation once acidulation capacity is established.

b) Triple Super Phosphate (TSP)

Market scale:

- Shares demand base with DAP but serves nitrogen-sensitive crops and blending operations

Profitability drivers:

- Same P_2O_5 concentration as DAP
- Lower dependence on ammonia
- Strong relevance for blending and estate agriculture

Cost considerations:

- Similar acidulation cost structure to DAP
- Slightly lower unit selling price

Profitability assessment:

- **Comparable margins to DAP** on a delivered-cost basis
- More resilient to ammonia price shocks

Conclusion: TSP is a **strategic profitability stabiliser**, particularly useful where ammonia prices are volatile.

c) Single Super Phosphate (SSP)

Market scale (ANRM & EDPM):

- Very large SSP-equivalent volumes (>500,000 t/year under moderate scenario)

Profitability drivers:

- Simple processing technology
- Lower capital intensity
- Contains sulphur, enhancing agronomic value
- Strong competitiveness due to bulkiness of imported SSP

Cost considerations:

- Low nutrient concentration increases transport per unit of P_2O_5
- Lower unit selling price

Profitability assessment:

- **Low margin per tonne**, but
- **High margin per unit of invested capital**
- Profitable when integrated with sulphuric acid production

Conclusion: SSP is a **volume-driven, cost-recovery product** that supports market expansion and improves overall plant utilisation rather than maximising margins.

6.3.2 Feed-Grade Phosphate Products

a) Dicalcium Phosphate (DCP) and Monocalcium Phosphate (MCP)

Market scale:

- Smaller than fertiliser market

- Growing demand driven by poultry, dairy, and commercial feed mills

Profitability drivers:

- Higher unit prices than fertiliser phosphates
- Stable demand linked to feed production cycles
- Lower exposure to fertiliser price volatility

Cost considerations:

- Higher purity requirements
- Stricter quality assurance and process control
- Smaller batch volumes

Profitability assessment:

- **Higher margins per tonne** than fertilisers
- Lower volume limits total revenue contribution

Conclusion: Feed-grade phosphates are **margin-enhancing secondary products**, suitable once fertiliser operations are stable and quality systems are established.

6.3.3 Industrial Phosphate Chemicals

a) Phosphoric Acid (Industrial Grade)

Market role:

- Intermediate input for fertilisers, feed phosphates, and industrial chemicals

Profitability drivers:

- Captures value otherwise embedded in imported intermediates
- Enables downstream diversification

Cost considerations:

- Capital-intensive
- Requires environmental controls and by-product management

Profitability assessment:

- **Indirect profitability** through value capture and downstream integration

b) Detergent, Water-Treatment, and Specialty Phosphates

Market scale:

- Small domestic volumes
- Regional export potential

Profitability drivers:

- High unit value

- Specialised applications

Cost considerations:

- Stringent specifications
- Market development costs
- Higher regulatory compliance

Profitability assessment:

- **High margin, high risk**
- Best pursued after fertiliser-led cash-flow stabilisation

Conclusion: Industrial phosphates are a **long-term upgrading opportunity**, not an early-stage profitability driver.

6.3.4 Comparative Profitability Summary

Table 6- 3: Comparative Profitability Summary

Product Line	Volume Potential	Margin per Tonne	Revenue Stability	Strategic Role
DAP	High	Medium–High	High	Financial anchor
TSP	Medium–High	Medium	High	Margin stabiliser
SSP	Very High	Low	Medium	Volume & utilisation
DCP/MCP	Low–Medium	High	High	Margin enhancement
Industrial phosphates	Low	Very High	Low	Long-term upgrading

6.3.5 Implications for Plant Configuration and Phasing

Based on profitability and risk profiles:

- **Phase 1:** Focus on DAP, TSP, and SSP to:
    - capture domestic fertiliser demand,
    - ensure high utilisation, and
    - generate stable cash flow.
- **Phase 2:** Introduce feed-grade phosphates to:
    - improve margins, and
    - diversify revenue streams.

- **Phase 3:** Expand into industrial phosphates once:
    - technical capacity matures, and
    - fertiliser operations are financially stable.

This sequencing aligns profitability with **risk management and demand maturity**, consistent with DC investment principles.

The profitability analysis confirms that **no single product maximises both scale and margin**. Instead, overall project viability depends on a **balanced product portfolio**, where:

- fertilisers provide scale and demand certainty,
- feed-grade phosphates enhance margins, and
- industrial chemicals offer long-term value upgrading.

When evaluated alongside ANRM and EDPM demand results, this diversified structure supports a **positive financial and economic case** for domestic phosphate beneficiation and processing in Uganda.

6.4 Competitiveness Screening for Various Phosphate-Based Products

This section consolidates the cost build-up, delivered-cost comparison, and profitability analysis to screen phosphate-based product lines according to their **competitiveness in the Ugandan context**. The screening identifies which products are **immediately competitive**, which are **conditionally competitive**, and which are **strategic but longer-term**, thereby guiding plant configuration and phasing.

6.4.1 Screening Framework

Products are assessed against five criteria that are material for policy and investment decisions:

1. **Market Size and Demand Certainty.** Evidence from ANRM (agronomic ceiling) and EDPM (market-based uptake).
2. **Delivered Cost-to-Market Position.** Comparison with imports at Ugandan wholesale/farm-gate level.
3. **Gross Margin Potential and Stability.** Findings from the profitability analysis, including exposure to input price volatility.
4. **Technical and Regulatory Complexity.** Processing sophistication, purity standards, and compliance requirements.
5. **Strategic Fit with National Objectives.** Alignment with NDP IV priorities on agro-industrialisation, import substitution, and resilience.

6.4.2 Competitiveness Screening Matrix

Table 6- 4: Competitiveness Screening Matrix

Product Line	Demand Outlook (ANRM/EDPM)	Delivered Cost Position	Margin Profile	Technical Complexity	Competitiveness Verdict
DAP	High, rising	Competitive vs imports	Medium–High	Medium	**High**
TSP	Medium–High	Competitive vs imports	Medium	Medium	**High**
SSP	Very high (volume)	Strongly competitive	Low per tonne, stable	Low	**High**
NPK Compounds (local blends)	High, expanding	Competitive if locally blended	Medium	Medium	**High**
DCP / MCP (feed-grade)	Medium, growing	Competitive with local PA	High per tonne	High	**Medium–High**
Industrial-grade Phosphoric Acid	Medium	Conditional	Medium	High	**Medium**
Detergent & specialty phosphates	Low–Medium	Export-dependent	High but volatile	Very High	**Low–Medium**
Food/pharma phosphates	Low (niche)	Conditional	High	Very High	**Low**

6.4.3 Product-Level Screening Outcomes

A. Fertiliser Phosphates (DAP, TSP, SSP, NPKs) — High Competitiveness

- **Demand:** ANRM indicates a large agronomic requirement (~190,000 t P_2O_5/year), while EDPM shows market-based demand of ~85,000–100,000 t P_2O_5 by 2030 and ~110,000–125,000 t by 2040 (moderate scenario).
- **Cost:** Domestic production eliminates ocean freight, port charges, and long-haul inland transport that add ~30–50% to imported prices.

- **Margins:** DAP/TSP provide stronger per-tonne margins; SSP provides scale and utilisation benefits.
- **Verdict: Immediately competitive** and suitable as **anchor products**.

B. Feed-Grade Phosphates (DCP, MCP) — Medium–High Competitiveness

- **Demand:** Smaller than fertilisers but growing with poultry and dairy expansion.
- **Cost/Margins:** Higher margins per tonne than fertilisers when integrated with domestic phosphoric acid.
- **Complexity:** Higher purity and QA requirements.
- **Verdict: Conditionally competitive**; best introduced **after fertiliser operations stabilise**.

C. Industrial Phosphate Chemicals — Medium to Low (Near-Term) Competitiveness

- **Demand:** Limited domestic volumes; regional export potential exists.
- **Cost/Margins:** High value but sensitive to energy, scale, and quality compliance.
- **Risk:** Higher capital intensity and market development risk.
- **Verdict: Strategic, longer-term** diversification rather than early focus.

6.4.4 Implications for Plant Configuration and Phasing

The screening supports a **sequenced development strategy** that aligns competitiveness with risk management:

- **Phase 1 (Anchor):** DAP, TSP, SSP, and local NPK blending *Rationale:* Large domestic demand, delivered-cost advantage, stable cash flows.
- **Phase 2 (Margin Enhancement):** DCP/MCP *Rationale:* Higher margins leveraging existing phosphoric acid and QA systems.
- **Phase 3 (Upgrading/Exports):** Industrial and specialty phosphates *Rationale:* Value upgrading once scale, energy reliability, and market access are secured.

The competitiveness screening confirms that **fertiliser phosphates are the most competitive and least risky products for Uganda**, supported by strong agronomic need, market-based demand growth, and a clear delivered-cost advantage over imports. **Feed-grade phosphates** offer margin diversification with manageable complexity, while **industrial phosphate chemicals** are best pursued as a **later-stage upgrading pathway**.

When integrated with the ANRM and EDPM results, this screening provides a **clear, evidence-based prioritisation** of phosphate-based products and underpins a **phased, bankable investment strategy** aligned with national development objectives.

6.5 Implications for Domestic Manufacturing

The cost, competitiveness, and profitability analyses indicate that domestic phosphate beneficiation and processing in Uganda is not only technically feasible but also **strategically and economically justified**, provided that manufacturing is configured and phased appropriately. This section synthesises the implications of the preceding analysis for **manufacturing scale, product mix, technology choice, and industrial policy alignment**.

6.5.1 Manufacturing Scale and Capacity Planning

The ANRM and EDPM jointly confirm that Uganda has:

- A **large agronomic phosphorus requirement** (≈190,000 t P_2O_5 per year), and
- A **market-realistic demand trajectory** reaching:
    - ≈85,000–100,000 t P_2O_5 by 2030, and
    - ≈110,000–125,000 t P_2O_5 by 2040 under the moderate scenario.

Implication for manufacturing: Domestic manufacturing should be **scaled to realistic market absorption**, not the full agronomic ceiling in the initial phase. Over-sizing plants to agronomic demand risks under-utilisation, while under-sizing risks continued import dependence.

A **phased capacity approach** is therefore appropriate:

- Initial capacity aligned with **baseline–moderate EDPM demand**, and
- Modular expansion aligned with **accelerated demand and regional exports**.

6.5.2 Product Mix and Manufacturing Configuration

Competitiveness screening shows that **not all phosphate-based products are equally suitable for early-stage domestic manufacture**.

Implication for product prioritisation:

- **Fertiliser products (DAP, TSP, SSP, NPK blends)** should form the manufacturing core due to:
    - high and stable demand,
    - strong delivered-cost advantage over imports, and
    - direct alignment with national food-security objectives.
- **SSP**, despite lower margins per tonne, plays a critical role in:
    - improving plant utilisation,
    - serving acid soils and legumes, and

    - displacing bulky imports that are structurally uncompetitive.
- **DAP and TSP** provide higher per-tonne margins and revenue stability, supporting cash flow and debt servicing.

Manufacturing implication: Plants should be designed with **flexibility to switch between DAP, TSP, and SSP**, depending on relative input prices (sulphur, ammonia) and seasonal demand.

6.5.3 Integration Across the Value Chain

The cost build-up analysis highlights that **integration is a key determinant of competitiveness**.

Implications for manufacturing design:

- **On-site beneficiation** improves ore quality and reduces downstream chemical consumption.
- **Integrated sulphuric acid production** lowers costs and mitigates supply risk compared to importing acid.
- **Phosphoric acid production** enables:
    - multiple downstream products,
    - higher value capture, and
    - diversification into feed-grade and industrial phosphates over time.

Fragmented or stand-alone manufacturing would forfeit these advantages and weaken competitiveness.

6.5.4 Technology Choice and Operational Complexity

Domestic manufacturing must balance **cost efficiency with operational reliability**.

Key implications:

- Proven, conventional technologies for:
    - beneficiation,
    - sulphuric acid,
    - wet-process phosphoric acid, and
    - fertiliser granulation are preferable to untested or overly complex solutions.
- Technology choices should prioritise:
    - robustness,
    - ease of maintenance,
    - availability of local skills, and

    - compliance with environmental standards.

Advanced industrial phosphates and food-grade products require:

- stricter quality control,
- higher technical capability, and
- stronger regulatory compliance, and should therefore be **sequenced after fertiliser manufacturing is stabilised**.

6.5.5 Cost Competitiveness and Industrial Location

Domestic manufacturing benefits materially from:

- proximity to phosphate deposits,
- shorter inland transport distances to markets,
- reduced exposure to port congestion and international freight volatility.

Implication: Locating manufacturing facilities close to the resource base and major domestic markets maximises the delivered-cost advantage identified in Section 6.2.

This strengthens the case for **clustered industrial development**, where mining, processing, and logistics are spatially integrated.

6.5.6 Financial and Bankability Implications

From a financing perspective, the implications for domestic manufacturing are significant:

- **Demand certainty** (EDPM) supports bankable sales forecasts.
- **Delivered-cost advantages** support margin resilience even during global price shocks.
- **Phased manufacturing** reduces upfront capital exposure and improves early-year cash flows.
- **Product diversification over time** enhances revenue stability and downside protection.

These features improve:

- Financial Net Present Value (FNPV),
- Internal Rate of Return (IRR),
- Debt service coverage ratios, and overall bankability under both TPS and PPP-type financing structures.

6.5.7 Policy and Industrial Development Implications

Domestic phosphate manufacturing aligns closely with national policy priorities, including:

- **Agro-industrialisation under NDP IV**,
- **Import substitution and balance-of-payments resilience**,
- **Employment creation and skills development**, and
- **Downstream industrial upgrading**.

However, manufacturing viability is strengthened when accompanied by:

- reliable energy supply,
- supportive infrastructure,
- predictable regulatory frameworks, and
- complementary demand-side policies (extension, credit, input programmes).

The combined evidence from cost, competitiveness, profitability, and demand analyses confirms that **domestic phosphate beneficiation and fertiliser manufacturing in Uganda is both viable and strategically justified**, provided that:

- manufacturing capacity is phased in line with market demand,
- product mix prioritises fertiliser phosphates in early stages,
- integration across the value chain is achieved, and
- expansion into higher-value products is sequenced appropriately.

Domestic manufacturing, therefore, represents not just a substitution of imports but a **foundation for a broader phosphate-based industrial platform** supporting Uganda's long-term agricultural transformation and industrial development objectives.

7 Benchmarking and Lessons from Comparator Countries

7.1 Benchmarking Framework

Benchmarking is used in this study to assess how Uganda's proposed phosphate beneficiation and processing strategy compares with **established and emerging phosphate-producing countries**, and to extract **practical, transferable lessons** relevant to Uganda's context. The objective is not to replicate models from vastly different economies, but to identify **design principles, policy choices, and operational approaches** that improve competitiveness, reduce risk, and enhance sustainability.

The benchmarking framework focuses on **six interrelated dimensions** that directly affect the performance of phosphate value chains.

7.1.1 Technology and Processing Pathways

This dimension benchmarks:

- Choice of beneficiation and processing technologies
- Level of integration between mining, acidulation, and downstream products
- Technology maturity and operational reliability

Comparator countries demonstrate that:

- **Proven, conventional technologies** (wet-process phosphoric acid, standard granulation) outperform untested or experimental systems in developing-country contexts.
- Integrated plants that combine **mining, beneficiation, sulphuric acid, phosphoric acid, and fertiliser production** achieve lower unit costs and greater operational stability.
- Modular and scalable technology allows capacity expansion in line with demand growth.

Relevance for Uganda: Technology choices should prioritise **robustness, maintainability, and scalability**, rather than frontier or highly specialised processes in early phases.

7.1.2 Logistics and Supply Chain Configuration

This dimension examines:

- Proximity of mines to processing plants
- Access to ports, rail, and road infrastructure
- Inland distribution efficiency

Benchmark evidence shows that:

- Countries that minimise **long-distance transport of low-value intermediates** are structurally more competitive.
- Co-location of mining and processing significantly reduces logistics costs.
- Efficient inland logistics matter more than port access for fertiliser competitiveness in landlocked or semi-landlocked markets.

Relevance for Uganda: Locating processing close to phosphate deposits and major domestic markets maximises the delivered-cost advantage identified in the cost analysis.

7.1.3 Industrial Structure and Value-Chain Integration

This dimension assesses:

- Degree of vertical integration
- Role of state-owned enterprises versus private operators
- Coordination among mining, chemical, and fertiliser segments

Comparator countries illustrate that:

- **Partial or full vertical integration** improves cost control and supply reliability.
- State participation is often strongest at the **upstream or strategic mineral level**, while private operators drive efficiency in processing and marketing.
- Fragmented industrial structures lead to higher costs and weaker competitiveness.

Relevance for Uganda: A coordinated industrial structure—anchored by a national mining entity and complemented by private processing and marketing—reduces fragmentation risks.

7.1.4 Pricing and Cost Competitiveness

This dimension benchmarks:

- Ex-factory versus delivered cost structures
- Exposure to global price volatility
- Ability to stabilise domestic fertiliser prices

Evidence from comparator countries shows that:

- Delivered-cost competitiveness is more important than ex-factory pricing.
- Domestic producers gain advantage by eliminating ocean freight, port charges, and long-haul inland transport.
- Countries with domestic production experience **lower price volatility** during global shocks.

Relevance for Uganda: The delivered-cost advantage of domestic production, confirmed in Section 6, is consistent with successful international experience.

7.1.5 Policy, Regulation, and State Support

This dimension covers:

- Industrial policy frameworks
- Incentives for domestic value addition
- Regulation of fertiliser quality and markets

Benchmark countries typically exhibit:

- Clear policy prioritisation of **fertiliser self-sufficiency and soil health**.
- Targeted state support in early phases (infrastructure, guarantees, offtake support).
- Strong fertiliser quality regulation to protect farmers and sustain demand.

Relevance for Uganda: Domestic phosphate manufacturing is most successful where **industrial policy, agricultural policy, and mineral policy are aligned**, as envisaged under NDP IV.

7.1.6 Industrial Clusters and Ecosystem Development

This dimension examines:

- Presence of industrial clusters around phosphate operations
- Linkages with energy, logistics, and supporting industries
- Spillover effects into skills and services

Comparator countries show that:

- Phosphate operations act as **anchors for industrial clusters**, attracting chemical, logistics, and service firms.
- Clusters improve operational efficiency, skills development, and innovation.
- Long-term competitiveness is strengthened through ecosystem development rather than stand-alone plants.

Relevance for Uganda: Developing phosphate processing as part of an **industrial cluster**, rather than an isolated facility, enhances resilience and long-term value creation.

7.1.7 Summary of the Benchmarking Framework

The benchmarking framework used in this study evaluates comparator countries across the following dimensions:

Table 7- 1: Summary of the Benchmarking Framework

Dimension	Focus of Comparison	Key Insight for Uganda
Technology	Processing choices and integration	Use proven, scalable technologies
Logistics	Transport distances and efficiency	Minimise inland and international haulage
Industrial structure	Integration and coordination	Avoid fragmented value chains
Pricing	Delivered cost and volatility	Focus on farm-gate competitiveness
Policy	State support and regulation	Align mineral, industry, and agriculture policy
Clusters	Ecosystem development	Build around anchor phosphate facilities

The benchmarking framework establishes a structured basis for learning from international experience while remaining grounded in Uganda's economic, logistical, and institutional realities. The lessons derived from comparator countries reinforce the study's core conclusions: **integration, delivered-cost competitiveness, phased development, and policy coherence** are decisive factors for successful phosphate-based industrialisation.

The subsequent sections apply this framework to selected comparator countries and distil **specific lessons** relevant to Uganda's phosphate investment strategy.

7.2 Country Case Studies

7.2.1 Morocco (OCP) – Fully Integrated Industrial Model

Morocco provides the most mature global example of a **fully integrated phosphate-based industrial model**, anchored by the state-owned **Office Chérifien des Phosphates (OCP Group)**. While Morocco's scale and export orientation exceed Uganda's ambitions, the OCP experience offers **clear, transferable lessons** on integration, cost competitiveness, market resilience, and policy coherence.

7.2.1.1 Overview of the OCP Model

OCP controls the phosphate value chain end-to-end, including:

- **Phosphate rock mining** (largest global reserves),
- **Beneficiation and slurry transport**,
- **Sulphuric and phosphoric acid production**,

- **Fertiliser manufacturing** (DAP, MAP, TSP, NPKs),
- **Logistics, marketing, and global distribution**.

This integration allows OCP to manage costs holistically and respond quickly to market changes.

7.2.1.2 Technology and Integration

Key features of the OCP technological model include:

- Use of **proven wet-process phosphoric acid technology** at large scale.
- **Slurry pipelines** transporting beneficiated phosphate rock from mines to coastal processing hubs, significantly reducing logistics costs compared to rail or road.
- Highly integrated acidulation and fertiliser granulation facilities, optimised for efficiency and scale.

Lesson for Uganda: While slurry pipelines and mega-scale plants are not appropriate for Uganda, the **principle of physical and operational integration**—co-locating mining, beneficiation, acidulation, and fertiliser production—remains highly relevant.

7.2.1.3 Logistics and Cost Competitiveness

OCP's competitiveness is driven less by low mining costs alone and more by:

- Elimination of multiple handling stages,
- Reduced transport costs through pipeline and port integration,
- Scale economies in processing and shipping.

As a result, OCP can remain competitive even during periods of depressed global fertiliser prices.

Lesson for Uganda: Cost competitiveness is achieved by **reducing logistics and handling costs**, not solely by lowering extraction costs. Uganda's delivered-cost advantage lies in replacing long international supply chains with shorter domestic ones.

7.2.1.4 Pricing Strategy and Market Resilience

OCP's integration enables:

- Flexible pricing strategies across products and markets,
- Cross-subsidisation between products during market downturns,
- Greater resilience to global price volatility.

During global fertiliser price shocks, OCP maintained supply continuity while many import-dependent countries faced shortages.

Lesson for Uganda: Domestic production enhances **price stability and supply security**, which are critical for fertiliser adoption and agricultural productivity.

7.2.1.5 Industrial Structure and Governance

OCP operates as a **commercially run, state-owned enterprise** with:

- Clear financial performance targets,
- Operational autonomy,
- Strategic alignment with national industrial and agricultural policy.

The Moroccan state supports OCP through:

- Infrastructure investment,
- Policy stability,
- Diplomatic and trade facilitation.

Lesson for Uganda: A strong national anchor institution, operating on commercial principles and aligned with national policy, is critical for long-term success in phosphate-based industries.

7.2.1.6 Product Diversification and Market Development

Beyond fertilisers, OCP has expanded into:

- Feed phosphates,
- Industrial phosphates,
- Speciality fertilisers tailored to soil and crop conditions in Africa.

This diversification:

- Enhances margins,
- Reduces reliance on any single product,
- Strengthens long-term competitiveness.

Lesson for Uganda: Product diversification should be **sequenced**—starting with fertilisers for scale and gradually moving into higher-value products as technical capacity and market depth develop.

7.2.1.7 Cluster Development and Skills

OCP's operations support a broader industrial ecosystem, including:

- Chemical suppliers,
- Engineering and maintenance firms,

- Research institutions and training centres.

This cluster approach has generated:

- Skilled employment,
- Technology transfer,
- Local supplier development.

Lesson for Uganda: Phosphate processing should be positioned as an **industrial cluster anchor**, not a stand-alone plant, to maximise spillover benefits.

7.2.1.8 Relevance and Limitations for Uganda

Directly transferable principles:

- End-to-end value-chain integration,
- Delivered-cost focus,
- Strong state–industry alignment,
- Phased product diversification.

Not directly transferable:

- Mega-scale export orientation,
- Capital intensity at OCP scale,
- Slurry pipelines over long distances.

Uganda's model should therefore be **scaled-down, domestically focused, and regionally oriented**, while retaining the core logic of integration.

7.2.1.9 Key Lessons for Uganda

From the Morocco (OCP) case, the following lessons are most relevant:

1. **Integration beats fragmentation** in phosphate value chains.
2. **Delivered cost matters more than ex-factory cost**.
3. **State-backed anchor institutions** can coexist with commercial discipline.
4. **Domestic fertiliser production enhances resilience** during global shocks.
5. **Diversification works best when sequenced**, not rushed.

Morocco's OCP demonstrates that phosphate-based industrialisation succeeds when **mineral resources, processing capability, logistics, and policy are tightly integrated**. While Uganda does not seek to replicate OCP's global export model, the underlying principles—**integration, cost discipline, resilience, and phased upgrading**—are directly applicable to Uganda's phosphate investment strategy.

These lessons strongly reinforce the study's recommendation for **integrated, phased domestic phosphate manufacturing** aligned with national development priorities.

7.2.2 Jordan (JPMC) – Mining-to-Export Model

Jordan provides a contrasting comparator to Morocco through the experience of the **Jordan Phosphate Mines Company (JPMC)**, which operates a **mining-to-export–led model** with selective downstream processing. JPMC's model illustrates how a country with limited domestic agricultural demand can leverage phosphate resources primarily through **export-oriented mining**, while still capturing value through targeted processing and logistics integration.

7.2.2.1 Overview of the JPMC Model

JPMC is a publicly listed company with significant state ownership and strategic oversight. Its core activities include:

- Phosphate rock mining,
- Beneficiation and upgrading,
- Export of phosphate rock and intermediates,
- Limited downstream production (mainly phosphoric acid and DAP through joint ventures).

Unlike OCP, JPMC's model prioritises **bulk export of phosphate rock**, reflecting Jordan's relatively small domestic fertiliser market.

7.2.2.2 Technology and Processing Configuration

Key features of JPMC's operating model include:

- Conventional open-pit mining of phosphate rock,
- Beneficiation to meet export-grade specifications,
- Limited domestic downstream conversion, with higher-value processing often undertaken through **joint ventures**.

This approach reduces:

- Capital intensity,
- Operational complexity,
- Exposure to downstream market risks.

Lesson for Uganda: Export-oriented mining requires **consistent ore quality and efficient beneficiation**, but does not demand full vertical integration in early stages.

7.2.2.3 Logistics and Export Orientation

JPMC’s competitiveness is strongly linked to:

- Proximity of mines to export corridors,
- Dedicated rail and port infrastructure connecting mines to Aqaba port,
- Long-term shipping arrangements for bulk commodities.

Export logistics efficiency enables Jordan to compete in global phosphate markets despite limited domestic demand.

Lesson for Uganda: Mining-to-export models rely heavily on **efficient, low-cost logistics to ports**. For landlocked countries, this requirement significantly alters competitiveness.

7.2.2.4 Pricing and Market Exposure

Under the JPMC model:

- Revenues are closely tied to **international phosphate rock prices**,
- Exposure to global commodity cycles is high,
- Price volatility directly affects export earnings.

Downstream joint ventures help partially mitigate this exposure by:

- Capturing additional value,
- Diversifying revenue streams.

Lesson for Uganda: Pure export models carry **higher price and revenue volatility** than domestic or regionally anchored fertiliser manufacturing.

7.2.2.5 Industrial Structure and Governance

JPMC operates under a hybrid governance structure:

- Commercial management discipline,
- Strategic state participation,
- Partnerships with international fertiliser producers.

This structure allows:

- Risk-sharing in capital-intensive downstream investments,
- Access to technology and markets through partners.

Lesson for Uganda: Joint ventures can be an effective mechanism for **selective downstream upgrading**, especially where capital and technical capacity are constrained.

7.2.2.6 Relevance to Uganda's Context

Elements potentially relevant to Uganda:

- Beneficiation-first approach to improve ore quality,
- Use of joint ventures for complex downstream processing,
- Commercial operation with strategic state oversight.

Elements less applicable to Uganda:

- Heavy reliance on bulk exports,
- Dependence on port access and rail infrastructure,
- High exposure to international price volatility.

Uganda's geography and domestic demand profile differ fundamentally from Jordan's.

7.2.2.7 Comparative Assessment with Uganda

Dimension	Jordan (JPMC)	Uganda
Domestic fertiliser demand	Low	High (ANRM & EDPM confirm growth)
Export orientation	Very high	Secondary/long-term
Logistics advantage	Strong (port access)	Limited (landlocked)
Value-chain integration	Partial	Preferable to be higher
Revenue volatility	High	Lower with domestic focus

7.2.2.8 Key Lessons for Uganda

The Jordan case yields the following lessons:

1. **Export-led phosphate mining is viable where domestic demand is weak and port access is strong.**
2. **Beneficiation quality is critical** for export competitiveness.
3. **Joint ventures reduce risk** in capital-intensive downstream segments.
4. **Export dependence increases exposure** to global price volatility.

7.2.2.9 Implications for Uganda's Phosphate Strategy

For Uganda, the JPMC experience suggests that:

- A **pure mining-to-export strategy is not optimal**, given landlocked geography and growing domestic demand.

- Export of unprocessed phosphate rock should be **opportunistic and secondary**, not the core strategy.
- Downstream processing for the domestic and regional fertiliser market offers **greater value stability** and economic spillovers.

Jordan's JPMC demonstrates that a mining-to-export phosphate model can be commercially successful under specific conditions—namely, strong logistics access to ports and limited domestic demand. However, when applied to Uganda's context, the model underscores the **limitations of export-led strategies** for landlocked countries with significant domestic fertiliser demand.

The primary lesson for Uganda is that **value addition closer to the domestic market** provides greater resilience, lower delivered costs, and stronger alignment with national development objectives than reliance on bulk phosphate rock exports.

7.2.3 South Africa (Foskor) – Beneficiation-Driven Model

South Africa provides a relevant comparator through **Foskor**, a state-owned phosphate producer whose strategy centres on **beneficiation and intermediate processing** rather than full downstream fertiliser integration. Foskor's experience illustrates how a beneficiation-driven model can add value to phosphate resources while supplying both **domestic downstream users** and **export markets**.

7.2.3.1 Overview of the Foskor Model

Foskor operates primarily as:

- A **phosphate rock miner and beneficiator**, and
- A producer of **phosphoric acid and phosphate rock concentrate** for domestic use and export.

Unlike Morocco's fully integrated fertiliser model, Foskor focuses on **upgrading raw ore** to higher-value intermediates, which are then sold to:

- Domestic fertiliser manufacturers, and
- Regional and international buyers.

7.2.3.2 Technology and Processing Focus

Key features of the Foskor model include:

- Conventional open-pit mining of igneous phosphate rock,
- Intensive **beneficiation** to increase P_2O_5 content and reduce impurities,
- Production of **phosphoric acid** as a principal value-added product.

The beneficiation process is central to Foskor's competitiveness, as igneous phosphate typically requires more processing than sedimentary deposits.

Lesson for Uganda: Where ore quality varies, **beneficiation is not optional but strategic**. Investment in beneficiation improves downstream processing efficiency and marketability.

7.2.3.3 Industrial Structure and Market Position

Foskor operates within a broader national industrial ecosystem:

- Supplies intermediates to domestic fertiliser producers,
- Competes in export markets for phosphate rock and acid,
- Coordinates with energy, transport, and industrial-chemicals sectors.

This structure allows Foskor to:

- Focus capital on mining and processing excellence,
- Avoid direct exposure to fertiliser retail market volatility,
- Capture value through **intermediate sales**.

Lesson for Uganda: A beneficiation-driven model can coexist with downstream fertiliser production led by other players, provided coordination and offtake arrangements are strong.

7.2.3.4 Logistics and Cost Implications

South Africa's relative logistics advantage includes:

- Access to ports for export,
- Rail and road connectivity to industrial hubs.

Despite this, beneficiation remains essential to:

- Reduce transport of low-value waste material,
- Improve cost efficiency per unit of nutrient transported.

Lesson for Uganda: For a landlocked country, beneficiation close to the mine is even more critical, as transporting unprocessed rock over long distances significantly erodes competitiveness.

7.2.3.5 Pricing and Revenue Stability

Foskor's revenues are influenced by:

- International prices for phosphate rock and phosphoric acid,
- Long-term supply contracts with industrial users.

While less exposed to fertiliser retail price swings than fully integrated producers, Foskor remains sensitive to global commodity cycles.

Lesson for Uganda: Beneficiation-driven models offer **moderate revenue stability**, but are less insulated from global price volatility than domestically anchored fertiliser manufacturing.

7.2.3.6 Governance and State Role

Foskor operates as a **state-owned commercial entity**, with:

- Strategic alignment to national industrial objectives,
- Operational autonomy and commercial discipline,
- Periodic fiscal and restructuring challenges linked to market cycles.

Lesson for Uganda: State ownership alone does not guarantee success; **commercial governance and financial discipline** are critical, particularly in commodity-linked businesses.

7.2.3.7 Relevance and Transferability to Uganda

Transferable elements:

- Strong emphasis on beneficiation as a value-addition step,
- Sale of upgraded intermediates to downstream users,
- Focus on core competencies rather than full vertical integration.

Less transferable elements:

- Dependence on export logistics and port access,
- Exposure to global commodity price cycles without a large domestic market buffer.

Uganda's growing domestic fertiliser demand distinguishes it from Foskor's operating context.

7.2.3.8 Key Lessons for Uganda

From the Foskor case, the most relevant lessons include:

1. **Beneficiation is a critical value-addition step**, particularly for ores with impurities.
2. **Upgrading raw rock improves competitiveness** and reduces downstream processing costs.
3. **Intermediate-focused models are viable**, but offer less demand stability than fertiliser-led models.
4. **Strong governance and cost control** are essential in beneficiation-heavy operations.

7.2.3.9 Implications for Uganda's Phosphate Strategy

For Uganda, the Foskor experience suggests that:

- Beneficiation should be **embedded in any domestic phosphate strategy**, regardless of downstream configuration.
- A beneficiation-only or beneficiation-plus-acidulation model could be a **transitional phase**, but
- Greater long-term stability and development impact arise when beneficiation supports **domestic fertiliser manufacturing** rather than relying primarily on exports.

South Africa's Foskor demonstrates the strengths and limitations of a **beneficiation-driven phosphate model**. While effective in upgrading mineral value and supplying intermediates, the model remains exposed to global price volatility and offers fewer domestic spillovers than fertiliser-led integration.

For Uganda, the key takeaway is that **beneficiation is necessary but not sufficient**. The greatest economic and policy benefits arise when beneficiation is combined with **domestic fertiliser manufacturing**, leveraging Uganda's strong and growing internal demand to stabilise revenues and maximise development impact.

7.2.4 India – Blending and Subsidy-Driven Fertiliser Industry

India offers a highly relevant comparator for Uganda through its **blending-centred, demand-anchored fertiliser industry**, underpinned by **strong public policy intervention**. While India operates at vastly larger scale, its experience provides **transferable lessons on market creation, price stabilisation, and the role of blending** in countries that are structurally dependent on imported fertiliser inputs.

7.2.4.1 Overview of the Indian Model

India is one of the world's largest consumers of fertiliser, yet it:

- Imports most of its **phosphate rock, phosphoric acid, and potash**, and
- Relies heavily on **domestic blending and formulation**, rather than full upstream mineral self-sufficiency.

The industry structure is characterised by:

- Large-scale **NPK and DAP blending plants** located close to consumption centres,
- Limited domestic phosphate mining,
- Strong **government regulation and subsidy mechanisms** that stabilise prices and guarantee demand.

7.2.4.2 Blending-Centric Industrial Structure

A defining feature of India's fertiliser industry is the emphasis on **blending and formulation**:

- Imported phosphoric acid, DAP, and intermediates are used as feedstock.
- Plants focus on **granulation, blending, bagging, and distribution**, rather than mining.
- Product differentiation is achieved through **crop- and region-specific NPK grades**.

This structure allows India to:

- Respond quickly to agronomic needs,
- Adjust product mixes without changing upstream supply chains,
- Reduce capital intensity compared to fully integrated mining-to-fertiliser models.

Lesson for Uganda: Blending provides a **low-risk, flexible entry point** into fertiliser manufacturing and can be scaled rapidly when upstream phosphate inputs are available domestically.

7.2.4.3 Role of Subsidies and Price Stabilisation

India's fertiliser demand is fundamentally shaped by policy:

- The **Nutrient-Based Subsidy (NBS)** system caps farmer prices for P- and K-based fertilisers.
- Government compensates producers for cost differences between market prices and regulated retail prices.
- Subsidies are linked to **nutrient content**, not product names, encouraging balanced fertilisation.

This framework has resulted in:

- High and predictable fertiliser uptake,
- Reduced price volatility for farmers,
- Guaranteed off-take for domestic producers and blenders.

Lesson for Uganda: While Uganda does not operate a large fertiliser subsidy programme, **policy-driven demand support mechanisms** (targeted subsidies, e-voucher schemes, public procurement for strategic crops) can significantly accelerate fertiliser adoption and stabilise demand.

7.2.4.4 Logistics and Market Reach

India's blending plants are:

- Located close to major agricultural belts,
- Integrated with rail and road logistics,
- Designed for high-volume seasonal distribution.

This minimises:

- Inland transport costs,
- Inventory holding periods,
- Delivery delays during planting seasons.

Lesson for Uganda: Even without large subsidies, **locating blending and finishing plants close to demand centres** materially improves delivered-cost competitiveness and uptake.

7.2.4.5 Pricing, Margins, and Financial Structure

Under India's model:

- Producer margins are relatively modest and regulated,
- Volume certainty compensates for lower margins,
- Financial viability is underpinned by **scale and predictable cash flows**.

This contrasts with export-oriented or specialty chemical models, where margins are higher but volumes and prices are volatile.

Lesson for Uganda: A fertiliser industry anchored in **volume stability rather than margin maximisation** is more compatible with public-interest objectives such as food security and smallholder productivity.

7.2.4.6 Industrial and Institutional Arrangements

India's fertiliser sector demonstrates:

- Strong coordination between agriculture, industry, and finance ministries,
- Clear regulatory frameworks for fertiliser quality and labelling,
- Extensive extension and soil-health programmes that reinforce fertiliser demand.

Lesson for Uganda: The effectiveness of domestic fertiliser manufacturing depends not only on factories, but on **institutional coordination across agriculture, industry, trade, and finance**.

7.2.4.7 Relevance and Transferability to Uganda

Highly transferable elements:

- Blending-led industrial structure,

- Focus on NPK formulations tailored to local soils and crops,
- Policy-enabled demand certainty.

Less transferable elements:

- Large-scale fiscal subsidies,
- Very high consumption volumes,
- Extensive rail-based logistics infrastructure.

Uganda's context calls for **targeted and fiscally sustainable adaptations**, not wholesale replication.

7.2.4.8 Key Lessons for Uganda

From the Indian experience, the most relevant lessons are:

1. **Blending is a powerful market-creation tool**, even when upstream inputs are imported.
2. **Demand stability matters more than full upstream integration** in early stages.
3. **Policy instruments can crowd-in private investment** by reducing demand risk.
4. **Balanced fertilisation** improves agronomic outcomes and long-term soil health.

7.2.4.9 Implications for Uganda's Phosphate Strategy

For Uganda, the Indian case reinforces that:

- Domestic phosphate beneficiation can be complemented by **robust local blending capacity**.
- Blending plants can rapidly absorb domestically produced phosphoric acid or TSP.
- Targeted demand-support mechanisms (aligned with EDPM findings on price sensitivity) can accelerate market uptake.

This model is particularly relevant in the **transition phase**, before full integration into higher-value phosphate chemicals.

India's blending- and subsidy-driven fertiliser industry demonstrates that **domestic manufacturing competitiveness does not require full upstream mineral self-sufficiency at the outset**. Instead, **policy-enabled demand, flexible blending capacity, and strong logistics** can create a large, stable fertiliser market.

For Uganda, the key takeaway is that **phosphate beneficiation and fertiliser manufacturing should be demand-anchored and blending-friendly**, supported by targeted policy instruments that accelerate adoption while remaining fiscally sustainable.

7.2.5 China – Large-Scale Industrial Clustering Model

China represents the most extensive example of a **large-scale, cluster-based phosphate industrialisation model**, characterised by **geographic concentration, vertical and horizontal integration, and strong state coordination**. While China's scale is far beyond Uganda's requirements, its experience offers **important lessons on industrial clustering, cost reduction, by-product utilisation, and ecosystem development** that are highly relevant to Uganda's long-term phosphate strategy.

7.2.5.1 Overview of the Chinese Phosphate Model

China is among the world's largest producers and consumers of phosphate fertilisers and phosphate chemicals. Its phosphate industry is organised around **large industrial clusters**, mainly located in provinces such as **Yunnan, Guizhou, Hubei, and Sichuan**, where phosphate rock deposits, energy supply, and transport infrastructure converge.

Key characteristics include:

- Concentration of **mining, beneficiation, acidulation, fertiliser manufacturing, and chemical derivatives** within defined industrial zones,
- Co-location of upstream and downstream users of phosphoric acid,
- Strong coordination between national industrial policy, provincial governments, and state-owned or state-backed enterprises.

7.2.5.2 Industrial Clustering and Economies of Scale

The defining feature of China's model is **industrial clustering**, which delivers multiple cost and efficiency advantages:

- Shared infrastructure (power, water, roads, waste management),
- Shared logistics for raw materials and finished products,
- Large-scale procurement of inputs (sulphur, ammonia, energy),
- Reduced transaction costs between linked firms.

Clustering allows firms to:

- Achieve **economies of scale**,
- Lower unit production costs,
- Improve capacity utilisation across the value chain.

Lesson for Uganda: Even at smaller scale, **co-locating phosphate-related activities within an industrial cluster** yields significant cost and efficiency benefits compared to stand-alone plants.

7.2.5.3 Technology, Integration, and Process Optimisation

China's phosphate clusters typically feature:

- Fully integrated mining–beneficiation–acidulation–fertiliser systems,
- Extensive phosphoric acid capacity feeding multiple downstream users,
- Use of standard wet-process technologies optimised for scale and energy efficiency.

A key feature is **process optimisation across firms**, not just within individual plants.

Lesson for Uganda: Integration should be designed at the **cluster level**, enabling multiple users of shared intermediates (e.g. phosphoric acid), rather than forcing full vertical integration within a single entity.

7.2.5.4 By-Product Utilisation and Environmental Management

China's phosphate clusters have increasingly focused on:

- Utilisation of **phosphogypsum** in cement, construction materials, and land reclamation,
- Recycling of process water and waste heat,
- Centralised environmental monitoring and enforcement within industrial zones.

While environmental challenges remain, clustering has made it easier to:

- Invest in shared treatment facilities,
- Enforce standards at scale,
- Reduce per-unit environmental costs.

Lesson for Uganda: Cluster-based development improves the **economic feasibility of environmental compliance**, particularly for waste and by-product management.

7.2.5.5 Pricing, Market Control, and Policy Coordination

China's phosphate industry operates under:

- Strong state influence over capacity expansion,
- Export controls and price interventions during supply shocks,
- Alignment between fertiliser supply and national food-security objectives.

This coordination has allowed China to:

- Stabilise domestic fertiliser availability,
- Shield farmers from extreme global price volatility,
- Adjust capacity in response to environmental and market pressures.

Lesson for Uganda: Policy coordination across mining, industry, agriculture, and environment enhances **market stability and investment predictability**, even without heavy subsidies.

7.2.5.6 Skills, Innovation, and Supplier Ecosystems

Phosphate clusters in China support:

- Large pools of skilled technical labour,
- Local equipment manufacturers and maintenance firms,
- Applied research institutions focused on fertiliser efficiency and new products.

Over time, this ecosystem:

- Reduces dependence on foreign technical services,
- Improves operational resilience,
- Enables gradual upgrading into specialty and industrial phosphates.

Lesson for Uganda: Phosphate investment should be viewed as a **platform for skills development and supplier ecosystems**, not just a single industrial facility.

7.2.5.7 Relevance and Limits for Uganda

Highly relevant principles:

- Industrial clustering around phosphate resources,
- Shared infrastructure and services,
- Integrated planning across the value chain,
- By-product utilisation strategies.

Not directly transferable:

- Very large-scale capacity,
- Heavy state control of pricing and exports,
- High environmental tolerance in early development phases.

Uganda's model must therefore be **scaled-down, environmentally compliant, and market-oriented**, while retaining the core logic of clustering.

7.2.5.8 Key Lessons for Uganda

From China's experience, the most relevant lessons are:

1. **Industrial clustering materially lowers costs and risks** in phosphate processing.
2. **Shared intermediates (e.g. phosphoric acid)** improve flexibility and resilience.

3. **Cluster-level environmental management** is more efficient than plant-by-plant solutions.
4. **Skills and supplier ecosystems** are critical to long-term competitiveness.
5. **Policy coordination** strengthens market stability and investment confidence.

7.2.5.9 Implications for Uganda's Phosphate Strategy

For Uganda, the Chinese case supports:

- Development of phosphate processing within a **dedicated industrial park or cluster**,
- Co-location of mining, acidulation, fertiliser manufacturing, blending, and logistics,
- Planning for **future diversification** into feed-grade and industrial phosphates within the same cluster.

Such a model:

- Reinforces delivered-cost competitiveness,
- Improves environmental and operational efficiency,
- Creates broader industrial spillovers aligned with NDP IV.

China's large-scale industrial clustering model demonstrates that **the organisation of the phosphate industry matters as much as the resource itself**. While Uganda does not require China's scale, adopting a **cluster-based approach to phosphate beneficiation and processing** can significantly enhance competitiveness, resilience, and long-term value creation.

The Chinese experience, therefore, reinforces the study's central recommendation: **domestic phosphate manufacturing in Uganda should be cluster-based, integrated, and phased**, serving domestic demand first while enabling future industrial upgrading.

7.3 Comparative Synthesis and Key Lessons for Uganda

This synthesis consolidates lessons from **Morocco (OCP), Jordan (JPMC), South Africa (Foskor), India (blending/subsidy-led), and China (industrial clustering)** into a single decision framework suited to Uganda's phosphate-to-fertiliser investment. The aim is to identify what Uganda should **adopt, adapt, or avoid**, given Uganda's realities: a **landlocked geography**, **growing domestic fertiliser demand** (supported by ANRM and EDPM results), and the need for an **integrated, bankable, and phased** industrial strategy.

7.3.1 What each comparator teaches Uganda

Morocco (OCP) – fully integrated industrial model

- Integration across mining → acids → fertilisers yields strong cost control and resilience.
- Competitiveness comes from logistics efficiency, scale, and policy stability.
- Takeaway: **integration and delivered-cost focus** are decisive, even if Uganda cannot replicate scale.

Jordan (JPMC) – mining-to-export model

- Export-led mining works where domestic demand is small and port logistics are strong.
- Highly exposed to global commodity cycles.
- Takeaway: for Uganda, mining-to-export is **secondary**; focus should be domestic/regional markets.

South Africa (Foskor) – beneficiation-driven model

- Beneficiation is critical for ore marketability and downstream efficiency.
- Intermediate-led models can work but face global price volatility.
- Takeaway: **beneficiation is necessary**, but strongest development impact comes when it supports domestic fertiliser manufacturing.

India – blending and policy-driven demand model

- Blending provides flexibility and fast market scaling even with imported intermediates.
- Policy tools (subsidy/price stabilisation, distribution systems) create demand certainty.
- Takeaway: Uganda should prioritise **blending readiness** and targeted demand-support tools (not necessarily large subsidies).

China – large-scale industrial clustering

- Clusters reduce costs through shared utilities, logistics, and by-product management.
- Industrial ecosystem (skills, suppliers) matters as much as the plant.
- Takeaway: Uganda should pursue a **cluster approach** (industrial park/zone), even at smaller scale.

7.3.2 Single decision framework for Uganda

Use the following framework to guide the strategic design, phasing, and governance of Uganda’s phosphate programme.

7.3.2.1 Strategic Market Orientation

Table 7- 2: Strategic Market Orientation

Decision area	What the comparators show	Uganda decision
Core market	OCP/India succeed with demand certainty; JPMC relies on exports	**Domestic + regional market first; exports as secondary**
Demand anchor	India's policy-backed demand + strong distribution	Use ANRM/EDPM as demand basis; strengthen extension, distribution, credit

7.3.2.2 Value-Chain Configuration

Table 7- 3: Value-Chain Configuration

Decision area	What the comparators show	Uganda decision
Integration level	OCP shows integration advantage; Foskor shows beneficiation alone is not enough	**Integrated model (beneficiation + acids + fertilisers)**, phased
Beneficiation	Foskor highlights beneficiation as competitiveness lever	**Mandatory early investment** in beneficiation capability
Product strategy	OCP/India show value in diversified portfolio	**Phase 1 fertilisers; Phase 2 feed phosphates; Phase 3 industrial chemicals**

7.3.2.3 Industrial Geography and Logistics

Table 7- 4: Industrial Geography and Logistics

Decision area	Comparator lesson	Uganda decision
Cluster development	China/OCP show major efficiency gains from co-location	Build in/around a **clustered industrial location** near resource + utilities
Delivered-cost competitiveness	OCP/China focus on logistics efficiency; landlocked states suffer	Prioritise **rail/bulk handling, storage, and distribution efficiency**

Decision area	Comparator lesson	Uganda decision
Import dependency	India manages imports with strong logistics	Secure sulphur/ammonia supply chains via contracts and bulk systems

7.3.2.4 Governance and institutional model

Table 7- 5: Governance and Institutional Model

Decision area	**Comparator lesson**	**Uganda decision**
Role of the state	OCP and Foskor show strategic state role; JPMC uses state-backed corporate governance	Use **NMC as upstream anchor**; ensure commercial discipline and accountability
Private participation	India and JPMC show JV/PPP usefulness	PPP/JV for acidulation/fertiliser plants; state focuses on resource governance + enabling infrastructure
Policy coherence	China/India show strong cross-sector alignment	Establish a **high-level coordination mechanism** (mining–industry–agriculture–finance–environment)

7.3.2.5 Environmental and by-product strategy

Table 7- 6: Environmental and By-Product Strategy

Decision area	**Comparator lesson**	**Uganda decision**
By-products	China/OCP increasingly treat by-products as economic opportunities	Plan for **phosphogypsum management** and potential use in construction materials
Compliance & bankability	Modern financing requires strong ESG	Early adoption of strong ESIA/ESMP, emissions control, and safe waste systems

7.3.3 Prioritised lessons for Uganda

Lesson 1: Competitiveness is won at delivered cost, not ex-factory cost

- Morocco/China demonstrate that logistics efficiency is decisive.
- Uganda’s landlocked status makes inland logistics a key competitiveness variable.

- Implication: invest early in bulk logistics, storage, and efficient domestic distribution.

Lesson 2: Integration should be phased but deliberate

- Morocco shows integration advantage; Foskor shows beneficiation alone is insufficient.
- Uganda should phase the chain:
    - Phase 1: beneficiation + sulphuric/phosphoric acid + key fertilisers
    - Phase 2: feed phosphates
    - Phase 3: industrial phosphates

Lesson 3: Export-led mining is not Uganda's best first option

- Jordan's export model depends on port access and global pricing.
- Uganda's domestic demand (ANRM/EDPM) supports a more stable domestic-led strategy.

Lesson 4: Blending and tailored products accelerate uptake

- India shows blending is a practical way to meet diverse soils/crops and grow uptake.
- Uganda should build blending-friendly intermediate outputs (TSP/PA) and strengthen soil-testing and extension.

Lesson 5: Industrial clustering multiplies benefits

- China shows clusters lower costs and accelerate skills/supplier ecosystems.
- Uganda should locate processing within a coordinated industrial ecosystem rather than a stand-alone plant.

Lesson 6: Strong governance is as important as the plant

- OCP and Foskor show state participation can work when governance is commercial and stable.
- Uganda should use NMC to anchor upstream development while enabling private efficiency downstream through PPP/JVs.

7.3.4 Uganda decision framework: recommended pathway

Recommended strategic pathway for Uganda (scaled to local realities):

1. **Anchor the upstream in NMC leadership**
    - Secure mining and beneficiation governance, resource control, and JV discipline.
2. **Develop an integrated but phased processing complex**
    - Begin with products that are clearly competitive and demanded (DAP/TSP/SSP + NPK blending readiness).
3. **Build a cluster model around the phosphate complex**

    - Shared utilities, by-product management, logistics systems, and supplier development.
4. **Pair manufacturing with demand-side measures**
    - Extension, credit facilitation, targeted public procurement, and quality enforcement to raise adoption.
5. **Scale into feed-grade and industrial products after stabilisation**
    - Use fertiliser cash flows to fund diversification and quality upgrades.

7.3.5 "Adopt, Adapt, Avoid" summary

Adopt

- OCP/China: integration logic and delivered-cost focus
- Foskor: beneficiation discipline
- India: blending flexibility and demand support mechanisms
- China: clustering and shared infrastructure

Adapt

- JV/PPP approaches (Jordan/India) tailored to Uganda's PPP framework and fiscal constraints
- Demand support: targeted instruments rather than broad subsidies

Avoid

- Pure mining-to-export as primary strategy (Jordan model) given Uganda's landlocked disadvantage and domestic demand opportunity
- Over-ambitious industrial chemical diversification in early phase (high complexity and market development risk)

8 Cross-Country Lessons for Uganda

8.1 Industrial cluster development

A consistent lesson from comparator countries is that phosphate industrialisation is most competitive when developed as an **industrial cluster** rather than a stand-alone plant. Morocco (OCP) and China show this at scale through integrated industrial hubs, while India demonstrates the same principle through dense blending and distribution ecosystems. South Africa's Foskor experience further shows that clustering strengthens beneficiation and intermediate markets, while Jordan illustrates that where clustering is weak or export-only, value capture is narrower and more volatile.

For Uganda, industrial cluster development is therefore not optional; it is a practical strategy to achieve **delivered-cost competitiveness**, **value addition**, and **bankability**.

8.1.1 What "industrial cluster" means in the phosphate context

A phosphate industrial cluster is a geographically concentrated ecosystem where the following are co-located and coordinated:

- Mining and beneficiation operations
- Sulphuric acid and phosphoric acid production
- Fertiliser granulation and blending (DAP, TSP, SSP, NPKs)
- Logistics, warehousing, and bagging facilities
- Support services (maintenance, laboratories, packaging, transport)
- By-product processing and environmental utilities (phosphogypsum, wastewater)
- Skills and R&D support (training, soil testing, fertiliser formulation development)

This structure reduces costs across the chain and improves coordination between suppliers and downstream users.

8.1.2 Why clustering improves competitiveness

a) Lower unit costs through shared infrastructure. Clusters reduce capital duplication by enabling shared facilities, such as:

- bulk storage tanks (sulphur, ammonia),
- common effluent treatment plants,
- shared power feeders and substations,
- shared laboratories and quality assurance services.

Implication for Uganda: Shared utilities reduce the cost base and improve the investment case for large chemical units that would otherwise be too expensive as stand-alone facilities.

b) Delivered-cost advantage (critical for landlocked Uganda). Because Uganda is landlocked, the delivered price of imported fertiliser includes:

- ocean freight,
- port charges,
- inland haulage,
- inventory financing and delays.

A cluster reduces delivered cost by:

- shortening supply chains,
- reducing handling stages,
- enabling faster distribution through consolidated warehousing.

Implication:
Cluster development reinforces the delivered-cost advantage that makes domestic fertiliser competitive against imports.

c) Stronger supply-chain reliability and production scheduling. Clusters ensure stable input supply for downstream plants:

- phosphate concentrate feeds the acid plant,
- acid feeds fertiliser granulation,
- fertiliser products move through shared storage and distribution.

This reduces:

- operational downtime,
- stock-outs,
- and costly emergency procurement.

d) Easier compliance with environmental and social safeguards. Comparator countries show that environmental management improves when industrial operations share systems for:

- emissions monitoring and scrubbers,
- wastewater treatment,
- phosphogypsum stacking or reuse.

Implication for Uganda: A cluster makes it more feasible to meet NEMA requirements and adopt climate-resilient infrastructure at lower unit cost, improving bankability for PPPs and DFIs.

e) Development of supplier and skills ecosystems. Clusters attract:

- engineering firms,

- chemical transport and logistics firms,
- packaging producers,
- technical service providers,
- training institutions and apprenticeships.

Implication:
Cluster-based development reduces dependence on foreign technical services and strengthens Uganda's industrial capabilities over time.

8.1.3 Practical cluster model suited to Uganda

Based on competitiveness screening and demand forecasts (ANRM and EDPM), Uganda's cluster should be designed around a **phased "anchor-and-expand" model**:

Phase 1: Anchor fertiliser cluster (core products)

- Beneficiation plant
- Sulphuric acid plant
- Phosphoric acid plant
- DAP/TSP/SSP production and bagging
- NPK blending facility
- Bulk storage and warehousing
- QA/QC laboratory

Phase 2: Margin enhancement cluster (secondary products)

- Feed-grade phosphates (DCP/MCP)
- Expanded blending capacity
- Wider regional distribution nodes

Phase 3: Industrial upgrading cluster (higher value-added)

- Industrial phosphates (STPP, water treatment chemicals)
- By-product utilisation industries (phosphogypsum-based construction materials)

This sequencing matches:

- Uganda's demand trajectory,
- bankability requirements,
- and risk management.

8.1.4 Governance requirements for cluster success

Cross-country experience shows that clustering only works when the governance arrangement is clear. Uganda's cluster model should include:

- **National Mining Company (NMC)** as upstream anchor and JV/PPP interface for mining and beneficiation
- Strong coordination among MEMD, MAAIF, UIA, NEMA, UNBS, and MoFPED
- A clear **industrial park/cluster management entity** responsible for:
  - shared utilities,
  - environmental compliance systems,
  - investor coordination,
  - logistics and storage governance.

This institutional clarity reduces transaction costs and improves investor confidence.

8.1.5 Key actionable lessons for Uganda

1. **Cluster development reduces delivered fertiliser costs**—the most important competitiveness lever for landlocked markets.
2. **Shared utilities improve feasibility** of acidulation and waste management systems.
3. **Clusters strengthen compliance and bankability**, especially under PPP and DFI financing.
4. **Clustering accelerates skills and supplier ecosystem growth**, increasing long-term competitiveness.
5. A cluster should be **phased** to match EDPM demand growth and avoid underutilisation risk.

Comparator-country experience shows that industrial cluster development is a decisive success factor for phosphate industrialisation. For Uganda, developing the phosphate value chain through a **cluster-based, phased industrial ecosystem** is the most practical route to achieving delivered-cost competitiveness, operational reliability, compliance with environmental safeguards, and bankable investment outcomes aligned with NDP IV.

8.2 Economies of Scale and Integration

Across all comparator countries reviewed—Morocco, Jordan, South Africa, India, and China—a consistent lesson is that **economies of scale and value-chain integration are decisive determinants of competitiveness** in phosphate-based industries. While the scale achieved by global leaders differs, the *principle* of scale and integration is transferable and highly relevant to Uganda's context.

8.2.1 Why Economies of Scale Matter in Phosphate Industries

Phosphate processing is inherently:

- **Capital-intensive** (acid plants, granulation, environmental controls),
- **Energy-intensive**, and
- **Process-continuous**, meaning unit costs fall sharply as throughput increases.

Comparator countries demonstrate that sub-scale plants:

- face higher unit costs,
- struggle to absorb price volatility, and
- are more vulnerable to operational disruptions.

Implication for Uganda: While Uganda does not require mega-scale plants, facilities must exceed **minimum economic scale thresholds** to be competitive against imports and to remain financially viable over time.

8.2.2 Integration as a Cost and Risk Management Tool

Integration refers to linking multiple stages of the value chain under coordinated ownership or long-term contractual arrangements.

- **Observed cross-country patterns**
- **Morocco (OCP):** Full integration allows cost optimisation across mining, acidulation, and fertilisers.
- **China:** Integration occurs at cluster level, with shared intermediates and utilities.
- **South Africa (Foskor):** Beneficiation and phosphoric acid integration improves ore competitiveness.
- **India:** Partial integration, with strong coordination between imported intermediates and domestic blending.
- **Jordan:** Limited integration increases exposure to global price cycles.

Key lesson: Integration is not about owning everything; it is about **ensuring reliable, low-cost flow of materials across stages**.

8.2.3 Integration Benefits Relevant to Uganda

a) Lower unit production costs. Integrated operations reduce:

- multiple handling and transport steps,
- duplication of infrastructure,
- transaction and coordination costs.

For Uganda, integration reduces dependence on:

- imported finished fertilisers,
- international freight and port delays.

b) Improved plant utilisation and cash-flow stability. Integrated facilities:

- balance production across products (DAP, TSP, SSP),
- shift output based on input prices and demand,
- maintain utilisation even during market downturns.

Implication: Higher utilisation stabilises revenues and strengthens debt-service capacity.

c) Reduced exposure to global shocks. Comparator experience shows that integrated producers:

- weather fertiliser price spikes better,
- maintain supply during global disruptions.

For Uganda, this improves:

- fertiliser availability,
- price stability for farmers,
- national food-security resilience.

8.2.4 Scale vs Flexibility: What Uganda Should Aim For

Uganda's context suggests a **"right-sized" scale**, not maximum scale.

Key principles:

- Scale plants to **EDPM-confirmed domestic demand**, not just ANRM agronomic ceilings.
- Use **modular design** to allow phased expansion.
- Integrate core stages early (beneficiation, acidulation, fertilisers).
- Allow downstream diversification as scale and skills increase.

This approach avoids:

- stranded assets,
- underutilised capacity,
- fiscal and financial stress.

8.2.5 Integration Pathway Appropriate for Uganda

Based on cross-country evidence, Uganda's integration pathway should follow a **progressive model**:

Table 8- 1: Integration Pathway Appropriate for Uganda

Phase	Integration Focus	Rationale
Phase 1	Mining + beneficiation + core fertilisers	Achieve minimum economic scale and cost competitiveness
Phase 2	Acidulation + blending + feed phosphates	Improve margins and product flexibility
Phase 3	Industrial phosphates + by-product utilisation	Value upgrading and export diversification

This mirrors successful sequencing observed in comparator countries, adapted to Uganda's demand profile and risk tolerance.

8.2.6 Governance and Scale Discipline

Comparator experience highlights that:

- Scale without governance leads to inefficiency (state-owned failures),
- Integration without commercial discipline increases fiscal risk.

Implication for Uganda: Economies of scale and integration must be accompanied by:

- commercial operating principles,
- transparent governance,
- clear allocation of roles between state anchors (e.g. NMC) and private operators.

Cross-country experience confirms that **economies of scale and integration are non-negotiable foundations of competitive phosphate industries**. Uganda's advantage lies not in replicating global mega-projects, but in designing **right-sized, integrated, and expandable operations** aligned with domestic demand and delivered-cost competitiveness.

By embedding scale discipline and integration into its phosphate strategy, Uganda can:

- lower unit costs,
- stabilise revenues,
- improve bankability, and
- maximise the long-term economic and industrial benefits of its phosphate resources.

8.3 Integrated Logistics and Infrastructure

Comparator-country experience consistently shows that logistics and infrastructure are not "supporting issues" in phosphate industrialisation; they are **core determinants of competitiveness**, especially when success is measured at **delivered cost-to-market**. Morocco and China demonstrate this through integrated mine-to-plant-to-port systems and industrial clusters; India shows how national distribution networks and blending hubs sustain demand; South Africa highlights the value of rail/port connectivity for intermediates; and Jordan confirms that export-led competitiveness depends on efficient port-linked logistics.

For Uganda—being landlocked—integrated logistics is even more decisive. The competitiveness of domestic phosphate fertiliser manufacturing will depend on whether Uganda can build a logistics system that **minimises handling, reduces transport cost per nutrient unit, and ensures reliable delivery during peak seasons**.

8.3.1 What "integrated logistics" means for a phosphate value chain

Integrated logistics means a coordinated system that connects:

1. **Mine and beneficiation plant logistics**
   - ore movement, stockpiles, internal roads/conveyors, and water recycling systems.
2. **Bulk input logistics** (critical for chemical processing)
   - sulphur import and storage, ammonia handling, acid storage, and safe transport systems.
3. **Finished product logistics**
   - bagging, warehousing, dispatch to domestic and regional distribution points.
4. **Infrastructure services**
   - power reliability, industrial water supply, wastewater treatment, stormwater systems, and hazardous waste/by-product management (e.g., phosphogypsum).

An integrated system reduces fragmentation and ensures the plant operates at high utilisation without stock-outs or bottlenecks.

8.3.2 Key cross-country lessons and Uganda implications

A) Bulk handling is a competitiveness lever

Cross-country lesson: Large phosphate producers invest in bulk handling systems to reduce unit logistics cost and prevent supply disruption.

Uganda implication: Uganda's project must plan for:

- bulk sulphur handling and storage (tanks/covered yards),
- ammonia storage and safe handling systems (where applicable),
- reliable inbound logistics corridors from Mombasa/Dar es Salaam.

This reduces cost volatility and ensures continuous operation of acid plants.

B) Rail logistics matters for bulk commodities

Cross-country lesson: Where rail is functional (Morocco, China, South Africa), the delivered-cost advantage is significant because fertilisers and raw materials are bulky.

Uganda implication: Uganda's cost competitiveness improves with:

- restored/strengthened rail freight capacity on key corridors,
- rail-enabled bulk movement for imports (sulphur/ammonia) and exports (regional sales),
- linkage between Tororo industrial cluster and regional corridors.

Even partial rail functionality lowers long-run delivered costs relative to road-only systems.

C) Co-location reduces preventing multiple handling

Cross-country lesson: Clusters (China) and integrated hubs (OCP) lower costs by co-locating production stages and avoiding repeated loading/unloading.

Uganda implication: Co-locating:

- beneficiation,
- acidulation,
- fertiliser manufacturing,
- warehousing,
- quality labs, within a single industrial cluster reduces:
- handling costs,
- material losses,
- delays, and improves delivered-price competitiveness.

D) Warehousing and distribution nodes stabilise seasonal demand

Cross-country lesson: India's fertiliser system works partly because it combines large-scale plants with distribution nodes that handle seasonal peaks.

Uganda implication: Uganda requires:

- central warehouses at the production hub,
- regional stock points near major demand zones,
- predictable dispatch scheduling aligned with planting seasons.

This reduces farmer stockouts and helps maintain demand growth.

E) Utilities and environmental infrastructure are non-negotiable

Cross-country lesson: Acidulation and fertiliser plants succeed where they have:

- reliable power,
- adequate industrial water,
- strong waste management,
- effective emissions controls.

China's cluster model shows the efficiency of shared environmental utilities.

Uganda implication: Integrated infrastructure planning must include:

- dedicated industrial power feeders and redundancy,
- secure industrial water supply plus recycling systems,
- wastewater treatment capacity,
- phosphogypsum management systems,
- emissions control equipment (SO_2, fluorides, dust).

These are not optional; they are required for compliance, bankability, and operational continuity.

8.3.3 Practical integrated logistics configuration for Uganda

A feasible Uganda logistics model should include three interconnected layers:

Layer 1: Mine-to-Plant Logistics

- mine access and haul roads,
- conveyor/haulage system to beneficiation,
- stockpiling and grade control,
- water recycling for beneficiation.

Layer 2: Inbound Bulk Inputs

- long-term sulphur supply contracts and bulk handling,
- ammonia logistics planning (if DAP/MAP production),
- bonded storage, safe handling, and regulatory compliance.

Layer 3: Outbound Distribution

- bagging and palletising lines,
- central warehouse at the production hub,
- national distribution nodes and agro-dealer networks,
- regional export logistics (EAC/COMESA corridors).

This design reduces landed-cost disadvantages and supports stable plant utilisation.

8.3.4 Implications for competitiveness and project viability

Integrated logistics and infrastructure determine:

- **Delivered cost advantage:** the key factor for landlocked markets.
- **Demand realisation:** EDPM shows demand is price-sensitive; logistics savings translate into lower farmer prices and higher uptake.
- **Bankability:** lenders and PPP partners require reliable logistics and utilities to reduce operational risk.
- **Sustainability:** integrated waste and emissions systems reduce environmental and social risks.

Cross-country experience confirms that logistics and infrastructure integration is a decisive factor in phosphate industrialisation. For Uganda, the delivered-cost advantage of domestic phosphate fertiliser production can only be realised if the project embeds a robust integrated logistics system covering **mine-to-plant operations, bulk import handling, and efficient domestic/regional distribution**, supported by reliable utilities and environmental infrastructure. This is central to competitiveness, financial viability, and long-term sustainability.

8.4 Policy Stability and Governance Mechanisms

Across all comparator countries reviewed—Morocco, Jordan, South Africa, India, and China—a decisive lesson is that **policy stability and effective governance mechanisms are as critical to competitiveness as geology, technology, or cost structure**. Phosphate industrialisation involves **large, irreversible capital investments** with long payback periods; without predictable policy and credible institutions, even technically sound projects struggle to reach financial close or sustain operations.

8.4.1 Importance of Policy Stability in Phosphate Industries

Phosphate value chains are characterised by:

- High upfront capital costs (mining, acidulation, environmental controls),
- Long asset lives (20–40 years),
- Exposure to global commodity and input price cycles.

Comparator-country experience shows that investors and lenders prioritise **policy predictability over short-term incentives**.

Key lesson: Countries that achieved sustained phosphate industrialisation provided **stable, long-term policy signals**, even when prices fluctuated or governments changed.

Implication for Uganda: Policy credibility under NDP IV is more important than ad-hoc incentives. Investors need confidence that:

- mineral licensing terms,
- fiscal regimes,
- environmental standards, and
- fertiliser market rules will remain consistent over the project life.

8.4.2 Governance Models Observed Across Comparator Countries

a) Morocco – Strategic State Stewardship with Commercial Discipline

- The state provides long-term strategic direction.
- OCP operates with clear commercial objectives and autonomy.
- Infrastructure and policy support are predictable.

Lesson for Uganda: State participation can succeed when it is **strategic, not operationally intrusive**, and when commercial discipline is enforced.

b) Jordan – Corporate Governance and Risk Sharing

- JPMC combines state ownership with private participation and joint ventures.
- Governance structures allow risk sharing in downstream investments.

Lesson for Uganda: Joint ventures and PPP structures can reduce fiscal risk, provided governance roles are clearly defined.

c) South Africa – Lessons from Governance Challenges

- Foskor's performance has been affected by:
    - governance instability,
    - inconsistent strategic direction,
    - exposure to commodity cycles without sufficient downstream buffering.

Lesson for Uganda: Weak governance undermines even resource-rich projects. **Institutional clarity and accountability are non-negotiable**.

d) India – Policy-Led Demand Governance

- Fertiliser demand is stabilised through:
    - predictable subsidy frameworks,
    - clear nutrient-based pricing rules,
    - strong quality regulation.

Lesson for Uganda: While Uganda may not replicate subsidies at scale, **predictable demand-side policy instruments** (e-voucher systems, public procurement, extension programmes) materially improve investor confidence.

e) China – Coordinated, Multi-Level Governance

- Strong coordination across:
    - central government,
    - provincial authorities,
    - industrial park administrations.
- Clear alignment between industrial, environmental, and agricultural policy.

Lesson for Uganda: Phosphate industrialisation requires **cross-ministerial coordination**, not siloed decision-making.

8.4.3 Governance Mechanisms Relevant for Uganda

Based on cross-country experience, Uganda's phosphate programme requires the following governance mechanisms:

a) Clear Institutional Roles

- **National Mining Company (NMC):**
    - Upstream resource management,
    - Mining and beneficiation oversight,
    - JV interface with private partners.
- **Line ministries and agencies:**
    - MEMD (mining and energy),
    - MAAIF (fertiliser policy and demand),
    - UIA (industrial facilitation),
    - NEMA and UNBS (environmental and quality regulation),
    - MoFPED (fiscal oversight and guarantees).

Lesson: Role clarity reduces delays, duplication, and investor uncertainty.

b) Stable Fiscal and Regulatory Framework

- Predictable royalties and taxes,
- Transparent environmental permitting,
- Consistent fertiliser quality and market regulation.

Lesson: Frequent policy changes increase risk premiums and financing costs.

c) Long-Term Offtake and Market Assurance

Comparator countries often use:

- anchor buyers,
- long-term supply agreements,

- public procurement or programme-based offtake.

Lesson for Uganda: Structured offtake arrangements (especially in early years) materially improve bankability without distorting markets.

d) Cluster-Level Governance

Where industrial clusters succeed, they have:

- a single cluster management authority,
- shared infrastructure governance,
- coordinated environmental compliance.

Lesson: A phosphate industrial cluster requires **centralised governance**, not fragmented oversight.

8.4.4 Risks of Policy and Governance Failure

Comparator experience also highlights risks Uganda should actively avoid:

- Policy reversals that undermine investor confidence,
- Over-politicisation of operational decisions,
- Fragmented approvals across agencies,
- Weak enforcement of quality standards, which erodes farmer trust and demand.

These risks are often more damaging than technical or market risks.

Cross-country evidence confirms that **policy stability and strong governance mechanisms are foundational to successful phosphate industrialisation**. Countries that combined:

- predictable policy frameworks,
- clear institutional roles,
- commercial governance of state participation, and
- coordinated industrial and agricultural policies have achieved resilient, competitive phosphate industries.

For Uganda, establishing such governance arrangements is not a peripheral issue—it is a **core competitiveness requirement** that directly affects project viability, financing terms, and long-term development impact.

8.5 Feasible Model(s) for Sukulu

The international lessons reviewed—on integration (Morocco), beneficiation discipline (South Africa), demand-led blending and market support (India), clustering (China), and strategic governance (Morocco/Jordan)—translate into a set of **feasible, Uganda-appropriate models** for developing the Sukulu phosphate resource. These models reflect Uganda's realities:

landlocked geography, **rapidly expanding domestic and regional fertiliser demand** (supported by ANRM and EDPM), and the need to build an investment that is **bankable, phased, and resilient**.

8.5.1 Decision Principles for Uganda's Sukulu Model

Applying the cross-country lessons yields five practical design principles for Sukulu:

1. **Domestic and regional market first (not export-led mining):** Uganda has rising demand; a mining-to-export strategy like Jordan is less suitable.
2. **Integration is necessary but should be phased:** Morocco's full integration is the goal, but Uganda should sequence it to reduce risk.
3. **Beneficiation is mandatory as an enabling step:** Like Foskor, upgrading ore quality determines downstream efficiency and competitiveness.
4. **Cluster-based development improves cost and compliance:** China shows clustering reduces unit cost and improves environmental management.
5. **Governance must combine state anchoring with commercial discipline:** NMC leadership is essential, but private expertise is needed for complex chemical operations.

8.5.2 Feasible Models for Sukulu

Based on the above, three feasible models emerge for Sukulu, each with a clear logic and sequencing pathway.

8.5.2.1 Model 1: Integrated Fertiliser Industrial Cluster (Recommended Core Model)

Description: A phased, integrated cluster anchored at Sukulu/Tororo that links:

- mining → beneficiation → sulphuric acid → phosphoric acid → fertiliser products → distribution.

Key products (Phase 1):

- SSP (for acid soils/legumes),
- TSP and/or DAP (for high-analysis demand and blending),
- NPK blending using locally produced intermediates.

Why it fits Uganda:

- EDPM confirms a strong and rising domestic P_2O_5 demand pathway through 2030–2040, supporting sustained off-take.

- Delivered-cost analysis shows domestic production displaces import logistics costs (freight/port/long-haul inland transport), which is Uganda's strongest competitiveness lever.
- Cluster model supports shared utilities (power, water, waste management), lowering unit costs and strengthening compliance.

Governance and delivery structure:

- **NMC leads upstream** mining and beneficiation (resource stewardship and JV interface).
- Private partner(s) lead acidulation and manufacturing under **PPP/JV** with clear risk allocation.
- UIA/industrial park framework supports cluster services and investor facilitation.

Strengths:

- Highest value addition and domestic spillovers.
- Strongest alignment with NDP IV and import substitution.
- Best platform for future diversification (feed phosphates, industrial products).

Risks and mitigation:

- Higher capital intensity → mitigated through phasing and PPP/JV structures.
- Input risk (sulphur, ammonia imports) → mitigated through bulk contracts and storage systems.

Verdict: This is the **most feasible and most transformative Sukulu model**, balancing competitiveness, bankability, and national value capture.

8.5.2.2 Model 2: Beneficiation + Superphosphate (SSP/TSP) First-Stage Model (Transitional / Early-Ramp Model)

Description: A simpler initial model prioritising:

- mining and beneficiation,
- sulphuric acid plant,
- SSP production (and/or TSP where feasible),
- distribution expansion.

Why it fits Uganda:

- SSP is structurally competitive for domestic production because importing low-analysis, bulky fertilisers is inherently expensive.
- Requires less complexity than full DAP/MAP integration.
- Builds market presence, distribution networks, and farmer confidence while generating early revenue.

Strengths:

- Lower technical complexity and lower initial capital requirements than full integration.
- Fastest route to visible import substitution impacts.
- Strong fit for Uganda's acidic soils and legumes where SSP and sulphur have agronomic value.

Constraints:

- Lower margins per tonne compared to DAP.
- Does not fully capture value from high-analysis fertiliser markets unless TSP or DAP are added later.

Verdict: A strong **Phase 1 entry model** if staged capital mobilisation is required, provided it is explicitly designed to scale into a full integrated model.

8.5.2.3 Model 3: Domestic Production + Blending Hub Model (India-style Adaptation)

Description:
A model that prioritises creating a **national blending and distribution backbone**, supported by domestic phosphate intermediates (TSP or phosphoric acid), to produce customised NPKs for Uganda's soils.

Why it fits Uganda:

- Uganda's fertiliser market increasingly demands crop- and soil-specific formulations.
- Blending reduces product mismatch and raises adoption.
- Works effectively when paired with demand-side measures (extension, soil testing, credit).

Strengths:

- Strong demand responsiveness (EDPM confirms demand is price- and income-sensitive).
- Lower complexity than manufacturing multiple high-analysis products in early years.
- Rapid scalability and flexibility.

Constraints:

- Requires reliable upstream phosphate supply and quality assurance.
- Still depends on imported nitrogen and potash components.

Verdict: A feasible **parallel demand-development model**, best implemented alongside Model 1 or Model 2—not as a replacement for beneficiation and acidulation.

8.5.3 Recommended Feasible Pathway for Sukulu

The most feasible adaptation for Sukulu is a **phased integrated cluster pathway**:

Phase 1 (Market entry and stability):

- Beneficiation + sulphuric acid + SSP/TSP + blending readiness
- Establish strong domestic distribution and quality assurance.

Phase 2 (Value capture and scale-up):

- Add phosphoric acid and DAP/TSP production capacity
- Expand NPK product lines and regional exports.

Phase 3 (Diversification and upgrading):

- Introduce feed phosphates (DCP/MCP)
- Develop industrial phosphate chemicals and by-product utilisation (phosphogypsum valorisation).

This pathway aligns with:

- EDPM demand trajectory (market-based uptake),
- ANRM agronomic need (long-run ceiling),
- competitiveness screening (most competitive product lines first),
- and governance lessons (commercial discipline with state anchoring).

8.5.4 Recommendation

Feasible model for Sukulu: A **phased, integrated phosphate fertiliser industrial cluster**, anchored by NMC upstream and implemented through PPP/JV arrangements for midstream processing, combined with a national blending and distribution strengthening programme.

This model maximises:

- delivered-cost competitiveness,
- demand responsiveness,
- bankability,
- and national development impact under NDP IV.

8.6 Institutional and Infrastructural Requirements

The successful development of a phosphate-to-fertiliser value chain at Sukulu requires a tightly coordinated **institutional architecture** and **fit-for-purpose infrastructure platform**. International experience shows that phosphate projects fail less from geology and more from weak governance, fragmented mandates, and under-provisioned shared infrastructure. For

Uganda, the requirements must support **phased integration**, **private participation**, and **long-term competitiveness**, in line with NDP IV and the Tenfold Growth Strategy

8.6.1 Institutional Requirements

8.6.1.1 Lead Institutional Anchor

The **National Mining Company (NMC)** should act as the **upstream anchor** for phosphate resource development, including:

- Holding mining rights on behalf of Government,
- Entering structured joint ventures or PPPs with private operators,
- Safeguarding national interest in pricing, beneficiation obligations, and downstream supply security.

This mirrors Morocco (OCP) and Tunisia (CPG/GCT), where a strong state anchor de-risks private capital while preserving commercial discipline.

8.6.1.2 Clear Institutional Separation of Roles

To avoid conflicts of interest and inefficiency:

- **NMC**: mining and resource stewardship;
- **Private operators**: beneficiation, acidulation, fertiliser manufacturing, and logistics;
- **Sector MDAs** (MEMD, MAAIF, MoWT): policy, regulation, and coordination;
- **NPA / MoFPED**: project appraisal, PIM compliance, fiscal oversight, and PPP approvals.

This separation is explicitly required under the DC Guidelines to ensure accountability across the project cycle

8.6.1.3 Dedicated Project Governance Structure

Establish a **Sukulu Phosphate Project Steering Committee**, chaired by MoFPED/NPA, with representation from:

- MEMD, MAAIF, MoWT,
- NMC,
- NEMA,
- UIA and PPP Unit.

A **ring-fenced Project Management Unit (PMU)** is required during pre-feasibility, feasibility, and early implementation to coordinate studies, land acquisition, ESIA, and transaction preparation.

8.6.1.4 PPP and Contracting Framework

Where private participation is pursued, contracts must:

- Mandate **minimum beneficiation and local value addition** thresholds,
- Include **off-take obligations** for the domestic market,
- Clearly allocate risks (resource, price, demand, FX, environmental).

The PPP Act and DC Guidelines require early screening of PPP suitability and fiscal exposure before feasibility approval

8.6.1.5 Regulatory and Standards Institutions

UNBS and **MAAIF** must be institutionally prepared to:

- Enforce fertiliser quality standards,
- Certify phosphate-based products (DAP, TSP, SSP, NPK),
- Prevent market distortion from sub-standard imports.

Weak enforcement has undermined fertiliser uptake in several African markets; comparator countries that invested early in standards achieved higher adoption and returns.

8.6.2 Infrastructural Requirements

8.6.2.1 Mining and Beneficiation Infrastructure

- Open-pit mining infrastructure including:
    - Access roads, haulage systems, overburden handling,
    - Beneficiation plant (washing, screening, flotation as required).
- Beneficiation is **non-negotiable**: international benchmarks show that exporting or processing unbeneficiated rock sharply erodes downstream competitiveness (South Africa, Tunisia)

8.6.2.2 Chemical Processing Infrastructure

For the preferred integrated cluster model:

- **Sulphuric acid plant** (anchor utility),
- **Phosphoric acid plant** (wet-process),
- Granulation units for **SSP/TSP/DAP/NPK**. Shared utilities (steam, power, water, effluent treatment) materially reduce unit costs, as demonstrated in Morocco and India.

8.6.2.3 Energy Infrastructure

- Reliable grid power with industrial-scale redundancy is essential.
- Acidulation and beneficiation are energy-intensive; outages materially increase operating risk.
- Alignment with the Tenfold Growth Strategy's emphasis on industrial energy readiness is critical

8.6.2.4 Water Supply and Waste Management

- Secure raw and process water sources (surface/groundwater).
- **Zero-liquid-discharge (ZLD)** or advanced wastewater treatment systems for fluorides, phosphates, and effluents are required to meet environmental standards and social licence requirements

8.6.2.5 Transport and Logistics Infrastructure

- Upgraded **mine-to-plant access roads**,
- Rail connectivity to eastern corridor markets,
- Efficient linkages to Mombasa/Dar es Salaam ports for sulphur, ammonia, and potential exports.
  Delivered-cost competitiveness hinges more on logistics than on plant-gate costs.

8.6.2.6 Industrial Cluster Infrastructure

Sukulu should be developed as a **dedicated phosphate industrial cluster**, not a stand-alone plant:

- Shared power, water, waste treatment, labs, storage,
- Space for future downstream users (feed phosphates, industrial chemicals). Cluster models consistently outperform isolated investments (China, Morocco).

8.6.3 Capacity and Skills Infrastructure

Technical skills in mineral processing, chemical engineering, and plant operations are scarce domestically.

Early investment is required in:

- Operator training programmes,
- Partnerships with experienced international operators,

- Local universities and TVET institutions. Comparator cases show that human-capital constraints are binding within 3–5 years if not addressed upfront.

8.6.4 Summary Implications for Sukulu

For Sukulu to succeed:

- **Institutionally**: a strong state anchor (NMC), clear role separation, DC-compliant governance, and PPP-ready frameworks are mandatory.
- **Infrastructurally**: beneficiation, acidulation, energy, water, and logistics must be planned as an integrated platform, not as incremental add-ons.
- **Strategically**: cluster-based development provides the lowest cost, highest resilience, and strongest alignment with NDP IV and the Tenfold Growth Strategy.

These requirements form **pre-conditions** for advancing the preferred Sukulu model from value-chain analysis into pre-feasibility and full feasibility under the DC Guidelines.

9 Value Chain Constraints and Bottlenecks

9.1 Introduction

Despite the strong agronomic need, positive market outlook, and competitiveness of domestic phosphate-based products, Uganda's phosphate value chain faces several **structural constraints and bottlenecks** across upstream, processing, and downstream segments. Addressing these constraints is essential to unlock the full economic and financial potential of the Sukulu phosphate resource.

This section identifies the **binding constraints**, explains their implications, and provides a basis for targeted interventions in subsequent recommendations.

9.1.1 Upstream Constraints (Mining and Beneficiation)

9.1.1.1 Limited Commercial Mining Readiness

While the Sukulu phosphate resource is well known, **large-scale commercial mining readiness remains limited**, including:

- ✓ incomplete mine development infrastructure,
- ✓ limited modern mining equipment deployment,
- ✓ lack of operating-scale beneficiation facilities.

Historical extraction was not configured for modern, integrated fertiliser production.

Implication: Without upfront investment in mine development and beneficiation, downstream processing costs rise and supply reliability is compromised.

9.1.1.2 Ore Quality Variability and Beneficiation Requirements

Sukulu phosphate exhibits **variable P_2O_5 grades and impurities**, requiring:

- ✓ consistent beneficiation,
- ✓ strict grade control,
- ✓ reliable water and energy inputs.

Absence of beneficiation increases acid consumption and processing costs.

Implication: Beneficiation is not optional; failure to invest in it creates a structural cost disadvantage relative to imports.

9.1.1.3 Institutional and Licensing Complexity

- ✓ Multiple institutions are involved in licensing, land access, environmental approvals, and mineral rights.
- ✓ Fragmented processes increase transaction costs and timelines.

Implication: Delayed approvals and unclear role allocation deter private investment and delay project execution.

9.1.1.4 Infrastructure Gaps at Mine Site

Inadequate access roads, power reliability, and water infrastructure increase:

- ✓ operating risk,
- ✓ capital costs,
- ✓ downtime during early operations.

Implication: Mine-site infrastructure gaps directly affect downstream plant utilisation and financial viability.

9.1.2 Processing and Industrial Constraints

9.1.2.1 8.2.1 High Capital Intensity of Chemical Processing

Sulphuric acid and phosphoric acid plants require:

- ✓ large upfront capital,
- ✓ specialised equipment,
- ✓ stringent environmental controls.

Financing such assets is challenging without strong demand certainty and risk-sharing structures.

Implication: Without phased development and PPP/JV structures, capital intensity becomes a major bottleneck.

9.1.2.2 Dependence on Imported Critical Inputs

Key inputs such as **sulphur and ammonia** are not produced domestically.

Import dependency exposes operations to:

- ✓ global price volatility,
- ✓ logistics disruptions,
- ✓ foreign exchange risk.

Implication: Unmanaged input risk can erode margins and reduce competitiveness of domestic production.

9.1.2.3 Energy and Utilities Reliability

- ✓ Phosphate beneficiation and acidulation are energy-intensive.
- ✓ Power interruptions or unstable supply increase unit costs and operational risk.

Implication: Energy reliability is a precondition for processing competitiveness and bankability.

9.1.2.4 Environmental Compliance and By-Product Management

Processing generates by-products such as **phosphogypsum**, acidic effluents, and emissions.

Inadequate waste management systems increase:

- ✓ environmental risk,
- ✓ regulatory exposure,
- ✓ community opposition.

Implication: Environmental infrastructure must be integrated from the outset; retrofitting is costly and risky.

9.1.2.5 Skills and Technical Capacity Constraints

Limited domestic experience in:

- ✓ chemical plant operations,
- ✓ fertiliser quality control,
- ✓ integrated acidulation processes.

High dependence on expatriate expertise in early years.

Implication: Skills shortages increase operating costs and constrain scale-up unless addressed proactively.

9.1.3 Downstream Distribution and Market Constraints

9.1.3.1 High Cost and Fragmentation of Distribution Networks

Fertiliser distribution in Uganda remains:

- ✓ fragmented,
- ✓ dominated by small-scale dealers,

- ✓ constrained by limited storage and seasonal logistics.

Long inland transport distances increase farm-gate prices.

Implication: Even cost-competitive domestic fertiliser can fail to reach farmers affordably without distribution reform.

9.1.3.2 Limited Fertiliser Adoption and Demand Realisation

EDPM results confirm that fertiliser demand is **price- and income-sensitive**.

Constraints include:

- ✓ affordability,
- ✓ limited access to credit,
- ✓ weak extension and soil testing services.

Implication: Demand does not automatically materialise with supply; complementary demand-side measures are required.

9.1.3.3 Competition from Imported and Substandard Products

Imported fertilisers often benefit from:

- ✓ economies of scale,
- ✓ inconsistent quality enforcement.

Substandard or adulterated products undermine farmer confidence.

Implication:
Weak market regulation reduces effective demand and penalises compliant domestic producers.

9.1.3.4 Regional Market Access Constraints

While regional markets exist (EAC/COMESA), exporters face:

- ✓ non-tariff barriers,
- ✓ inconsistent standards,
- ✓ logistics inefficiencies.

Implication: Regional export potential cannot be assumed without targeted trade facilitation and standards harmonisation.

9.1.4 Policy and Institutional Bottlenecks

Despite strong alignment with national priorities under NDP IV and agro-industrialisation objectives, several **policy and institutional bottlenecks** constrain the effective development of the phosphate value chain at Sukulu.

9.1.4.1 Fragmented Policy Mandates

Responsibilities for mining, fertiliser production, agriculture, industrial development, and trade are spread across multiple MDAs.

Coordination mechanisms remain weak, resulting in:

- ✓ delayed approvals,
- ✓ overlapping requirements,
- ✓ inconsistent policy signals to investors.

Implication: Fragmentation increases transaction costs and undermines investor confidence, particularly for integrated mining-to-fertiliser projects that cut across sectors.

9.1.4.2 Weak Integration Between Mineral and Agricultural Policy

Mining policy traditionally focuses on extraction and royalties, while agricultural policy focuses on productivity and food security.

Limited operational linkage exists between:

- ✓ phosphate resource development, and
- ✓ fertiliser availability, pricing, and distribution strategies.

Implication: Without explicit integration, domestic phosphate development risks becoming disconnected from fertiliser uptake and agricultural impact.

9.1.4.3 Unclear Role of State-Owned Entities and Private Sector

The National Mining Company is expected to anchor upstream development, but:

- ✓ governance frameworks for joint ventures and PPPs are still evolving,
- ✓ risk allocation between state and private partners is not yet fully standardised.

Private investors face uncertainty regarding:

- ✓ offtake guarantees,
- ✓ pricing policy,
- ✓ long-term regulatory stability.

Implication: Ambiguity in institutional roles delays financial close and raises the cost of capital.

9.1.4.4 Limited Enforcement of Fertiliser Standards

Weak enforcement of quality standards allows:

- ✓ substandard or adulterated fertilisers into the market,
- ✓ erosion of farmer trust,
- ✓ unfair competition against compliant domestic producers.

Implication: Quality enforcement gaps directly suppress effective demand and weaken the business case for domestic manufacturing.

9.1.4.5 Policy Predictability and Investment Risk

Large phosphate projects require long-term certainty on:

- ✓ fiscal regimes,
- ✓ environmental compliance requirements,
- ✓ trade and pricing policies.

Policy reversals or inconsistent implementation increase perceived risk.

Implication: Policy instability translates into higher risk premiums, reducing financial viability even where demand fundamentals are strong.

9.1.5 Environmental and Social Constraints

Phosphate mining and chemical processing pose **significant environmental and social challenges** that must be addressed proactively to ensure sustainability, regulatory compliance, and social licence to operate.

9.1.5.1 Environmental Risks from Mining and Processing

Potential impacts include:

- ✓ land disturbance and habitat loss from open-pit mining,
- ✓ water contamination from beneficiation and acidulation effluents,
- ✓ air emissions (SO_2, fluorides, dust),
- ✓ generation of phosphogypsum and other solid wastes.

Implication: Inadequate environmental management increases regulatory risk, community opposition, and long-term remediation costs.

9.1.5.2 Water Resource Constraints

Beneficiation and acidulation are water-intensive.

Competing demands from:

- ✓ agriculture,
- ✓ communities,
- ✓ ecosystems, may constrain sustainable abstraction.

Implication: Water insecurity can become a binding constraint unless recycling, efficiency, and integrated water management are incorporated from the outset.

9.1.5.3 Land Acquisition and Community Impacts

Expansion of mining and industrial facilities may affect:

- ✓ customary landholders,
- ✓ livelihoods dependent on agriculture and grazing,
- ✓ community access to land and resources.

Implication: Poorly managed land acquisition risks conflict, delays, and reputational damage.

9.1.5.4 Social Acceptance and Trust Deficits

Historical experiences with extractive projects have created scepticism in some communities.

Limited early engagement and benefit-sharing mechanisms exacerbate mistrust.

Implication: Without credible community engagement and benefit-sharing, project implementation faces high social risk.

9.1.6 8.6 Gender, Equity, and Climate Resilience Issues

The phosphate value chain intersects with broader **equity and resilience challenges** that affect its sustainability and development impact.

9.1.6.1 Gender Gaps in Fertiliser Access and Benefits

Women constitute a significant share of Uganda's agricultural labour force but:

- ✓ have lower access to fertiliser,
- ✓ face constraints in credit, land ownership, and extension services.

Fertiliser distribution systems often favour larger, male-dominated commercial farms.

Implication: Without targeted measures, domestic fertiliser production may disproportionately benefit better-resourced farmers, limiting inclusive growth.

9.1.6.2 Equity and Smallholder Inclusion

Smallholders dominate Uganda's agriculture but:

- ✓ face affordability constraints,
- ✓ have limited bargaining power in input markets,
- ✓ are vulnerable to price volatility.

Implication: If distribution and pricing strategies do not explicitly consider smallholders, demand growth may fall below potential levels identified in the EDPM.

9.1.6.3 Climate Vulnerability of Fertiliser Demand

Fertiliser use is sensitive to:

- ✓ rainfall variability,
- ✓ droughts and floods,
- ✓ climate-induced production risks.

Climate shocks can reduce farmers' willingness to invest in inputs.

Implication: Climate risk can suppress fertiliser uptake unless paired with climate-resilient agricultural practices and risk-mitigation instruments.

9.1.6.4 Climate and Environmental Performance of the Industry

Phosphate processing is energy-intensive and contributes to emissions.

Without mitigation, the industry may:

- ✓ increase carbon intensity,
- ✓ face future regulatory or financing constraints.

Implication: Failure to integrate climate resilience and low-emission strategies undermines long-term competitiveness and access to concessional finance.

9.1.7 Cross-Cutting Bottlenecks

Several constraints cut across the entire value chain:

- ✓ **Financing and risk perception:** High upfront costs and perceived demand risk raise financing costs.

- ✓ **Policy coordination gaps:** Misalignment between mining, agriculture, industry, and trade policies weakens implementation.
- ✓ **Timing and sequencing challenges:** Simultaneous development of mining, processing, and markets is complex and requires strong coordination.

9.1.8 Implications for the Sukulu Phosphate Project

The constraints and bottlenecks identified across the upstream, processing, and downstream segments of the phosphate value chain have clear and material implications for the design, sequencing, and governance of the Sukulu Phosphate Project.

9.1.8.1 Need for Integrated and Phased Development

The analysis demonstrates that Sukulu's development **cannot succeed through isolated or piecemeal interventions**. Mining, beneficiation, chemical processing, and market development are strongly interdependent, and weaknesses in any one segment undermine the viability of the entire value chain.

Accordingly:

Integrated, phased, and institutionally coordinated solutions are required to align supply capacity with realistic market absorption.

Failure to address **upstream and downstream constraints simultaneously** creates a high risk of:

- ✓ underutilised processing and manufacturing capacity,
- ✓ weak financial performance and cash-flow instability, and
- ✓ loss of investor and lender confidence.

These findings directly underpin the recommended **phased development model**, in which beneficiation and fertiliser production capacity are scaled in line with demand growth, supported by enabling infrastructure and logistics investments.

9.1.8.2 Institutional, Environmental, and Social Risk Considerations

The policy, institutional, environmental, and social constraints identified above indicate that **technical feasibility alone is insufficient** to ensure the success of the Sukulu Phosphate Project.

Specifically:

- ✓ **Institutional coordination failures** increase transaction costs, delay implementation, and weaken private-sector participation.

- ✓ **Environmental safeguards** and by-product management are central to regulatory compliance, financing eligibility, and long-term operational sustainability.
- ✓ **Social acceptance and community trust** are essential to securing and maintaining a social licence to operate.

These dimensions represent **core project risks**, not peripheral considerations, and must be addressed explicitly in project design and implementation.

9.1.8.3 Inclusion, Gender, and Climate Resilience Implications

The analysis further shows that **gender, equity, and climate resilience considerations are integral to demand realisation and sustainability**.

In particular:

Addressing gender and smallholder inclusion is necessary to ensure that fertiliser availability translates into effective and broad-based uptake.

Climate resilience measures are required to reduce demand volatility arising from rainfall variability and climate shocks.

Inclusive and resilient design strengthens:

- ✓ realisation of projected fertiliser demand,
- ✓ long-term social licence and community support, and
- ✓ the sustainability and credibility of the investment over its life cycle.

9.1.8.4 Link to Strategic Interventions and Recommendations

The constraints and implications outlined in this section directly inform:

- ✓ the proposed **institutional reforms**,
- ✓ the **infrastructure investment priorities**, and
- ✓ the **sequenced, cluster-based development strategy** for Sukulu.

They provide the analytical basis for the **strategic interventions and policy recommendations** presented in the next section, which are designed to systematically remove the identified bottlenecks and de-risk the Sukulu Phosphate Project for both public and private stakeholders.

10 Opportunities and Strategic Entry Points

The analysis of demand (ANRM and EDPM), competitiveness, benchmarking, and value-chain constraints indicates that the Sukulu phosphate resource presents **multiple, sequenced investment opportunities** across the upstream, midstream, and downstream segments. These opportunities differ in **capital intensity, risk profile, and development impact**, and therefore require **targeted entry strategies** rather than a single, uniform investment approach.

This section identifies **strategic entry points** that are commercially viable, policy-aligned, and suitable for phased development.

10.1 Upstream Investment Opportunities

Upstream investments form the **foundation of the Sukulu phosphate value chain** and determine the reliability, cost, and quality of downstream processing.

10.1.1 Phosphate Mining Development

There is a clear opportunity for investment in:

- Commercial-scale open-pit phosphate mining,
- Mine development infrastructure (access roads, haulage systems),
- Modern mining equipment and operational systems.

Strategic rationale:

- Uganda currently lacks an operating commercial phosphate mine supplying domestic fertiliser production.
- Secure, long-term access to phosphate rock is a prerequisite for midstream and downstream investments.
- Mining investment underpins national resource sovereignty and reduces dependence on imported phosphate inputs.

Entry model:

- Mining investment is best structured through **joint ventures or PPP arrangements** anchored by the National Mining Company, with private partners providing capital, technology, and operational expertise.

10.1.2 Beneficiation Facilities

Beneficiation is a **critical upstream entry point**, rather than an optional add-on.

Investment opportunities include:

- Crushing, washing, screening, and flotation facilities,
- Grade control and stockpiling systems,
- Water recycling and tailings management infrastructure.

Strategic rationale:

- Beneficiation improves ore quality and reduces downstream chemical and energy consumption.
- International benchmarking confirms beneficiation as a key competitiveness lever.
- Beneficiated rock enables flexibility across multiple downstream product lines.

Entry model:

- Can be co-developed with mining or structured as a dedicated beneficiation JV supplying multiple processors within an industrial cluster.

10.1.3 Upstream Infrastructure and Utilities

There are investable opportunities in:

- Dedicated power supply infrastructure,
- Industrial water supply and recycling systems,
- Mine-to-plant logistics and internal transport systems.

Strategic rationale:

- Upstream infrastructure failures directly constrain processing plant utilisation.
- Shared infrastructure lowers unit costs and improves bankability.

Entry model:

- Public investment or blended finance for shared infrastructure,
- User-fee recovery from private operators.

10.2 Midstream Processing and Manufacturing

Midstream processing offers the **highest value-addition potential** and is central to import substitution and industrialisation objectives.

10.2.1 Sulphuric and Phosphoric Acid Production

Key midstream opportunities include:

- Sulphuric acid plants (anchor chemical infrastructure),
- Wet-process phosphoric acid production.

Strategic rationale:

- Acidulation is the gateway to multiple fertiliser and industrial products.
- Domestic acid production reduces exposure to imported intermediates.
- Phosphoric acid enables product diversification over time.

Entry model:

- Capital-intensive investments best suited for **PPP or JV structures** with experienced chemical operators.
- Long-term supply contracts with fertiliser plants within the cluster improve bankability.

10.2.2 Fertiliser Manufacturing (DAP, TSP, SSP)

There is strong market-backed opportunity for:

- DAP and TSP granulation units,
- SSP production for domestic and regional markets.

Strategic rationale:

- EDPM confirms strong and growing market demand through 2030–2040.
- Delivered-cost advantages over imports are structurally significant.
- Fertiliser manufacturing provides stable cash flows to anchor the entire value chain.

Entry model:

- Private-led manufacturing within a regulated cluster framework,
- Phased capacity expansion aligned with demand growth.

10.2.3 NPK Blending and Custom Formulation

Investment opportunities exist in:

- NPK blending plants,
- Soil- and crop-specific fertiliser formulation.

Strategic rationale:

- Uganda's fertiliser demand is increasingly differentiated by crop and soil.
- Blending enhances market responsiveness and fertiliser adoption.
- Lower capital intensity compared to full chemical integration.

Entry model:

- Private blending operators supplied with domestic phosphate intermediates,
- Integration with extension and soil-testing services.

10.3 Downstream Fertiliser and Industrial Product Markets

Downstream investments determine **market reach, demand realisation, and revenue stability**.

10.3.1 Domestic Fertiliser Distribution and Storage

Opportunities include:

- Regional fertiliser warehouses,
- Bulk and bagged product distribution systems,
- Seasonal inventory management solutions.

Strategic rationale:

- Distribution inefficiencies currently inflate farm-gate prices.
- Improved logistics directly support fertiliser uptake and demand realisation.

Entry model:

- Private logistics operators,
- Cooperative and agro-dealer partnerships,
- Possible public support for strategic storage nodes.

10.3.2 Feed-Grade Phosphates

There is a growing opportunity for:

- Dicalcium phosphate (DCP),
- Monocalcium phosphate (MCP).

Strategic rationale:

- Expanding poultry, dairy, and feed milling industries.
- Higher margins per tonne compared to fertiliser products.
- Reduced exposure to fertiliser price cycles.

Entry model:

- Second-phase investment once phosphoric acid production stabilises,
- Partnerships with established feed manufacturers.

10.3.3 Industrial Phosphate Products

Longer-term opportunities include:

- Industrial-grade phosphoric acid,

- Detergent and water-treatment phosphates,
- By-product utilisation (phosphogypsum-based materials).

Strategic rationale:

- High value addition and industrial upgrading potential.
- Strong alignment with cluster-based industrial development.

Entry model:

- Export-oriented or niche-market investments,
- Best pursued after fertiliser-led cash-flow stabilisation.

10.4 Opportunities for Export and Regional Leadership

Beyond serving domestic demand, the Sukulu phosphate value chain presents clear opportunities for **regional market integration and leadership**, particularly within the **EAC and COMESA** regions. While exports should not displace domestic market priorities, they offer a strategic pathway for **scale, risk diversification, and foreign-exchange earnings**.

10.4.1 9.4.1 Regional Fertiliser Supply Gap

Many neighbouring countries remain:

- Highly import-dependent for phosphate fertilisers,
- Exposed to long international supply chains,
- Constrained by high delivered costs.

Uganda's geographic position allows Sukulu-based production to competitively serve:

- Eastern DRC,
- South Sudan,
- Rwanda and Burundi,
- Northern Tanzania and western Kenya (in selected products).

Strategic implication: Uganda can position itself as a **regional fertiliser hub**, supplying landlocked markets that face even higher delivered costs than Uganda.

10.4.2 Export-Ready Product Segments

The most viable export-oriented phosphate products include:

- **SSP and TSP**, where bulkiness penalises long-distance imports,
- **DAP and blended NPKs**, for organised commercial farming zones,

- **Feed-grade phosphates (DCP/MCP)**, which have regional demand but limited local production.

Export competitiveness is strongest where:

- products are produced within an integrated cluster,
- logistics corridors are reliable,
- quality standards are harmonised regionally.

10.4.3 Regional Leadership and Policy Influence

By developing a reliable domestic phosphate industry, Uganda gains:

- Influence in **regional fertiliser standards and quality regulation**,
- Ability to shape regional fertiliser trade norms,
- Strategic leverage in regional food-security initiatives.

Implication: Sukulu can anchor Uganda's transition from a fertiliser importer to a **regional supplier and standards-setter**, consistent with broader regional integration goals.

10.5 PPP and Private Sector Opportunities

Given the scale, capital intensity, and technical complexity of phosphate development, **public–private collaboration is essential**. The Sukulu project presents multiple entry points for private capital and expertise under structured PPP and JV arrangements.

10.5.1 PPP Opportunities Across the Value Chain

Table 10- 1: PPP Opportunities Across the Value Chain

Segment	PPP Opportunity	Rationale
Mining & beneficiation	JV with NMC	Risk-sharing, technology transfer
Sulphuric & phosphoric acid	PPP/JV	Capital-intensive, specialist expertise
Fertiliser manufacturing	Private-led	Market-driven efficiency
Blending & distribution	Private operators	Flexibility and responsiveness
Shared infrastructure	Public or blended finance	Natural monopoly characteristics

10.5.2 Role of the National Mining Company

The National Mining Company provides a **credible public anchor** by:

- Holding strategic mineral rights,
- Entering JVs with qualified private operators,
- Ensuring beneficiation and domestic supply obligations.

This structure:

- De-risks upstream investment,
- Preserves national interest,
- Improves bankability for downstream private investors.

10.5.3 Private Sector Incentives and Risk Allocation

Private investors are most responsive when:

- Demand risk is supported by EDPM-validated market outlook,
- Regulatory and fiscal frameworks are stable,
- Infrastructure risks are shared or mitigated publicly.

PPP structuring should clearly allocate:

- Resource risk (public),
- Construction and operational risk (private),
- Market and price risk (shared),
- Environmental compliance (shared, with strong oversight).

10.5.4 Circular Economy and Green Growth

Phosphate beneficiation and processing generate significant by-products and waste streams. International experience shows that **circular economy approaches** can transform these liabilities into **cost savings, revenue streams, and environmental gains**.

10.5.5 Phosphogypsum Reuse

Phosphogypsum is the principal by-product of phosphoric acid production. Rather than long-term storage, opportunities exist for:

- Use in cement and construction materials,
- Road base and embankment stabilisation,
- Controlled agricultural soil conditioning (subject to standards).

Strategic implication: Early planning for phosphogypsum reuse reduces environmental risk and long-term storage costs, improving social acceptance and ESG performance.

10.5.6 Waste Heat and Energy Recovery

Acidulation and granulation processes generate substantial waste heat. Opportunities include:

- Heat recovery for internal power generation,
- Steam reuse within the plant,
- Reduced external energy demand.

Implication: Waste-heat recovery lowers operating costs and carbon intensity, strengthening competitiveness.

10.5.7 Water Recycling and Resource Efficiency

Circular water management options include:

- Closed-loop process water systems,
- Effluent treatment and reuse,
- Reduced freshwater abstraction.

Implication: Water efficiency is critical for environmental compliance, climate resilience, and community acceptance.

10.5.8 Green Growth and Financing Implications

Circular economy integration:

- Improves eligibility for **green and climate finance**,
- Reduces long-term environmental liabilities,
- Aligns the project with Uganda's climate and sustainability commitments.

This enhances:

- bankability under PPP structures,
- access to concessional and blended finance,
- long-term competitiveness.

10.6 Strategic Sequencing of Entry Points

The opportunity assessment confirms that the Sukulu phosphate value chain should be developed through a **sequenced entry strategy**, rather than a single, large-scale intervention.

Phasing is essential to manage capital intensity, reduce implementation risk, and align production capacity with realistic market absorption as confirmed by the EDPM.

10.6.1 Phase 1: Market Entry and Capacity Stabilisation

- Commercial mining and beneficiation
- Production of **SSP and/or TSP**
- Establishment of **NPK blending readiness** and domestic distribution systems

This phase focuses on rapid import substitution, early revenue generation, and market development, while building operational capability and stakeholder confidence.

10.6.2 Phase 2: Value Capture and Product Expansion

- Development of **phosphoric acid production**
- Introduction of **DAP** manufacturing
- Entry into **feed-grade phosphates** (DCP/MCP)

This phase deepens value addition, improves margins, and expands the product portfolio once demand, logistics, and institutional systems are stabilised.

10.6.3 Phase 3: Industrial Upgrading and Circular Economy Integration

- Production of **industrial phosphate chemicals**
- **By-product valorisation**, including phosphogypsum reuse and waste-heat recovery
- Expansion into higher-value regional and niche export markets

This phase positions Sukulu as a broader chemical and industrial platform rather than a single-product fertiliser facility.

10.6.4 Rationale for Sequencing

The proposed sequencing:

- **Aligns capital deployment with risk**, avoiding over-investment ahead of demand;
- **Matches EDPM-confirmed demand growth**, ensuring high utilisation and financial resilience;
- **Maximises domestic value capture and employment**, with progressively higher skill and technology intensity.

The Sukulu phosphate value chain presents **multiple, complementary investment entry points**, each with distinct risk–return profiles. When strategically sequenced and institutionally coordinated, these entry points provide a clear and credible pathway for:

- domestic manufacturing,
- fertiliser import substitution, and
- long-term industrial upgrading.

Taken together, the opportunity assessment demonstrates that the Sukulu phosphate value chain offers **reinforcing strategic opportunities**, including:

- strong **domestic market anchoring**,
- scope for **regional export leadership**,
- **structured PPP and private-sector participation**, and
- **circular-economy-driven efficiency and sustainability**.

When pursued in a **phased, integrated, and institutionally coordinated manner**, these opportunities position Sukulu not merely as a fertiliser project, but as a **platform for regional industrial leadership, private investment mobilisation, and green growth**, fully aligned with Uganda's development priorities.

11 Recommendations and Investment Scenarios

The value chain analysis confirms that Sukulu presents a **strategic, bankable opportunity** for domestic phosphate beneficiation and fertiliser manufacturing, provided that interventions are **sequenced, coordinated, and risk-aware**. This section sets out **recommended intervention areas**, **priority policy reforms**, and **investment scenarios**, and explains how these feed directly into the **pre-feasibility options analysis**.

11.1 Intervention Areas Across the Value Chain

The recommended interventions span the full phosphate value chain and are designed to remove binding constraints while maximising value addition and competitiveness.

11.1.1 Upstream (Mining and Beneficiation)

Key interventions:

- Develop **commercial-scale phosphate mining** at Sukulu under a structured JV/PPP anchored by the National Mining Company (NMC).
- Establish **mandatory beneficiation facilities** to stabilise ore quality and reduce downstream processing costs.
- Invest in **mine-site infrastructure** (access roads, power, water, grade control, tailings management).

Purpose:

- Secure reliable, cost-competitive phosphate feedstock.
- De-risk downstream processing and manufacturing investments.

11.1.2 Midstream (Chemical Processing and Manufacturing)

Key interventions:

- Develop **sulphuric acid production** as an anchor chemical utility.
- Phase in **phosphoric acid capacity** to unlock higher-value fertiliser and feed-grade products.
- Establish flexible manufacturing lines for **SSP, TSP, DAP**, and blending-ready intermediates.
- Embed **shared utilities and environmental infrastructure** within a cluster framework.

Purpose:

- Capture value currently embedded in imports.

- Improve margins and resilience through integration.

11.1.3 Downstream (Distribution, Markets, and Demand Realisation)

Key interventions:

- Strengthen **domestic fertiliser distribution and warehousing**, particularly in high-demand regions.
- Support **NPK blending capacity** linked to soil and crop requirements.
- Promote **regional market access** through standards harmonisation and trade facilitation.
- Integrate fertiliser production with **extension, credit, and soil-testing services**.

Purpose:

- Ensure that supply translates into effective demand.
- Expand market size beyond simple import substitution.

11.2 Strategic Policy Reforms

The analysis shows that infrastructure and capital alone are insufficient without **supportive and stable policy frameworks**.

11.2.1 Institutional and Governance Reforms

- Clarify and formalise the **role of NMC** as upstream anchor and JV interface.
- Establish a **Sukulu Phosphate Steering Committee** to coordinate MDAs across mining, agriculture, industry, environment, and finance.
- Ring-fence project governance during pre-feasibility and feasibility stages to meet DC and PIM requirements.

11.2.2 Fertiliser Market and Quality Regulation

- Strengthen **fertiliser quality enforcement** to protect farmers and compliant producers.
- Align mineral development with **agricultural input policy**, ensuring domestic phosphate supports fertiliser availability and affordability.

11.2.3 Demand-Side and Inclusion Policies

- Use targeted instruments (e-voucher systems, public procurement for strategic crops, credit facilitation) to:
    - accelerate fertiliser uptake,
    - support smallholders and women farmers,
    - stabilise early-stage demand.
- Integrate **climate-resilient agriculture** and risk-management tools to reduce demand volatility.

11.2.4 Environmental and Green Growth Policies

- Mandate **early integration of circular-economy measures**, including phosphogypsum reuse, water recycling, and waste-heat recovery.
- Align Sukulu development with **green finance and ESG requirements** to improve bankability.

11.3 Investment Scenarios

Based on demand, competitiveness, and risk analysis, three investment scenarios are recommended for consideration.

11.3.1 Scenario 1: Phased Integrated Fertiliser Cluster (Preferred Scenario)

Features:

- Mining + beneficiation + sulphuric acid + SSP/TSP + blending readiness (Phase 1).
- Phosphoric acid + DAP + feed-grade phosphates (Phase 2).
- Industrial phosphates + by-product valorisation (Phase 3).

Strengths:

- Strong alignment with EDPM demand growth.
- Highest domestic value addition and employment.
- Best balance between risk, return, and resilience.

11.3.2 Scenario 2: Beneficiation and Superphosphate-First Model (Transitional Scenario)

Features:

- Mining, beneficiation, sulphuric acid, SSP/TSP.

- Delayed phosphoric acid and DAP.

Strengths:

- Lower initial capital requirements.
- Fast import substitution impact.

Limitations:

- Lower margins and slower upgrading.
- Must be explicitly designed to scale into Scenario 1.

11.3.3 Scenario 3: Mining and Export-Led Model (Non-Preferred Scenario)

Features:

- Mining and limited beneficiation.
- Export of phosphate rock and intermediates.

Limitations:

- High exposure to global price volatility.
- Weak alignment with Uganda's domestic demand and landlocked geography.
- Limited industrial and employment spillovers.

Conclusion:

- Suitable only as a short-term or residual option, not a development anchor.

11.4 Linkage to Pre-Feasibility Options Analysis

The recommendations and scenarios above directly inform the **pre-feasibility options analysis** as follows:

- **Option structuring:** The phased integrated cluster scenario provides the technical and economic basis for defining **preferred pre-feasibility options**.
- **Options screening:** Scenarios are screened against:
    - demand realism (EDPM),
    - economic viability,
    - financial bankability,
    - policy alignment,
    - environmental and social risk.
- **Option selection:** The analysis supports advancing the **phased integrated fertiliser cluster** as the **preferred option** into pre-feasibility, with clear expansion pathways.

- **Investment packaging:** Each phase can be packaged as a distinct but linked investment, enabling:
    - staged financing,
    - PPP participation,
    - reduced fiscal exposure.

The Sukulu Phosphate Project should proceed to pre-feasibility on the basis of a **phased, integrated, and cluster-based fertiliser manufacturing model**, anchored by strong institutional coordination and supported by targeted policy reforms. This approach offers the most credible pathway for **import substitution, regional leadership, private investment mobilisation, and long-term green industrial growth** in Uganda.

12 Conclusion

12.1 Summary of Major Insights

The Value Chain Analysis demonstrates that the Sukulu phosphate resource represents a **strategic opportunity for Uganda's agro-industrialisation and industrial upgrading**, provided it is developed through an **integrated, phased, and institutionally coordinated approach**.

The key insights are as follows:

Demand fundamentals are strong and credible. The ANRM confirms a large agronomic phosphorus requirement, while the EDPM shows a realistic and rising market-based demand trajectory through 2030 and 2040. Together, these models establish a robust demand foundation for domestic phosphate beneficiation and fertiliser manufacturing.

Delivered-cost competitiveness favours domestic production. Uganda's dependence on imported fertilisers imposes high logistics, freight, and financing costs. Domestic production at Sukulu offers a clear delivered-cost advantage, particularly for bulky phosphate products such as SSP and TSP, and increasingly for DAP and blended fertilisers as scale improves.

Integration and clustering are decisive success factors. Cross-country benchmarking confirms that competitiveness is achieved not through isolated plants, but through **integrated value chains and industrial clusters** that reduce unit costs, improve utilisation, and strengthen environmental and social compliance.

Phasing is essential to manage risk and capital intensity. A sequenced development pathway—starting with mining, beneficiation, and core fertiliser products, and progressing toward phosphoric acid, feed-grade, and industrial phosphates—aligns investment with demand growth and reduces the risk of underutilised capacity.

Institutional, environmental, and social factors are central, not peripheral. Governance clarity, policy stability, environmental safeguards, gender inclusion, and climate resilience directly affect demand realisation, bankability, and long-term sustainability.

12.2 Implications for the Pre-Feasibility Study

The findings of the VCA provide a **clear and focused basis for advancing to the pre-feasibility stage**.

Specifically, the pre-feasibility study should:

Prioritise a phased integrated fertiliser cluster model at Sukulu as the preferred development option.

Translate demand analysis into plant sizing and phasing assumptions, anchored on EDPM-confirmed market absorption rather than agronomic ceilings alone.

Assess technical configurations for beneficiation, sulphuric acid, phosphoric acid, and fertiliser manufacturing that allow modular expansion.

Evaluate PPP and JV structures that align public resource stewardship with private capital, technology, and operational efficiency.

Incorporate early environmental, social, and circular-economy design choices to reduce future compliance and remediation risks.

Explicitly test sensitivity to logistics, energy, input prices, and demand growth, reflecting the key risk drivers identified in the VCA.

The VCA therefore narrows the option space and ensures that pre-feasibility resources are focused on **viable, competitive, and policy-aligned configurations**.

12.3 Transition to Technical and Financial Modelling

This Value Chain Analysis marks the transition from **strategic diagnosis** to **investment-grade assessment**.

The next phase of work will build directly on the VCA by:

- converting value-chain insights into **technical layouts and process flow diagrams**,
- translating cost and competitiveness analysis into **CAPEX and OPEX estimates**,
- integrating demand projections into **financial cash-flow models**,
- evaluating **economic and financial viability** under alternative phasing and financing structures, and
- preparing the project for **formal pre-feasibility submission and Development Committee review**.

In this sense, the VCA does not conclude the analysis; it **sets the foundation** for a technically sound, financially robust, and development-impact-oriented pre-feasibility study for the Sukulu Phosphate Project.

13 Annex1 – Explaining Modelling

13.1 Agronomic Nutrient Requirement Model (ANRM)

13.1.1 What the ANRM is doing

The **Agronomic Nutrient Requirement Model (ANRM)** estimates **how much phosphorus fertiliser (P_2O_5)** Uganda needs in total, based on:

1. **What crops people grow**
2. **How many hectares** those crops cover
3. **How much phosphorus each crop needs per hectare**
4. **How many farmers are likely to actually use fertiliser** (adoption)

Think of it like: **"How much fertiliser does Uganda need if farms were properly fertilised?"** and then: **"How much is likely to be used by 2030 and 2040?"**

13.1.2 Step 1: List the main crops and their land area

In the sheet **INPUTS_CROP_BASKET**, the model lists major crops (maize, beans, bananas, coffee, rice, tea, horticulture) and the land area for each crop in hectares.

Example idea:

- If maize covers 1,150,000 hectares, that's a lot of land that could need fertiliser.

So this step answers: **"How big is each crop in Uganda?"**

13.1.3 Step 2: Give each crop a recommended phosphorus rate

Still in **INPUTS_CROP_BASKET**, each crop is assigned a recommended **P_2O_5 rate (kg per hectare)**.

Example idea:

- If maize needs 40 kg of P_2O_5 per hectare, then every maize hectare "should" receive that much P_2O_5 for good yields.

So this step answers: **"How much P_2O_5 should be applied per hectare for each crop?"**

13.1.4 Step 3: Calculate the total "ideal" national P_2O_5 requirement

In **CALCS_ANRM**, the model multiplies:

Crop area (ha) × recommended P_2O_5 rate (kg/ha) for each crop, then adds them all up.

That produces the "optimal" national requirement.

Example in simple terms:

- Maize: 1,150,000 ha × 40 kg/ha = 46,000,000 kg P_2O_5
- Convert kg to tonnes: 46,000,000 ÷ 1,000 = 46,000 tonnes

It repeats this for all crops and adds everything.

So this step answers: **"If Uganda fertilised properly, how many tonnes of P_2O_5 would be needed per year?"**

13.1.5 Step 4: Adjust for real life: not everyone will use fertiliser

Even if agronomists recommend a lot of fertiliser, farmers don't all buy it (due to cost, access, knowledge, etc.).

So in **INPUTS_ADOPTION**, the model uses adoption rates for:

- **Baseline** (slow growth)
- **Moderate** (policy-aligned growth)
- **Accelerated** (strong uptake) for both **2030** and **2040**.

Example idea:

- If Uganda needs 200,000 tonnes of P_2O_5 in theory, but only 50% of farmers adopt by 2030, then expected demand is:
  - 200,000 × 50% = 100,000 tonnes

So this step answers: **"How much of the ideal requirement will actually be demanded in 2030 and 2040?"**

- **Step 5: Convert P_2O_5 demand into real fertiliser products**

Farmers don't buy "P_2O_5", they buy products like **DAP, TSP, SSP**.

So in **OUTPUTS_SUMMARY**, the model converts P_2O_5 tonnes into product tonnes using nutrient content:

- **DAP is 46% P_2O_5** → DAP tonnes = P_2O_5 tonnes ÷ 0.46
- **TSP is 46% P_2O_5** → TSP tonnes = P_2O_5 tonnes ÷ 0.46
- **SSP is 16% P_2O_5** → SSP tonnes = P_2O_5 tonnes ÷ 0.16

Example:

- If demand is 100,000 tonnes of P_2O_5:
  - DAP needed ≈ 100,000 / 0.46 = 217,391 tonnes
  - SSP needed ≈ 100,000 / 0.16 = 625,000 tonnes

So this step answers: **"How many tonnes of actual fertiliser products does that P_2O_5 demand translate into?"**

13.1.6 Step 6: Show the results and a simple chart

The model then presents:

- Demand for P_2O_5 in 2030 and 2040 under each scenario
- Equivalent DAP/TSP/SSP quantities
- A chart that compares 2030 vs 2040 across scenarios

So quickly see:

- "Demand grows by 2040"
- "Accelerated scenario is highest"
- "SSP volumes are much larger because it's lower strength"

13.1.7 Why this model is useful

It gives a realistic, structured way to answer:

- How big Uganda's phosphate fertiliser demand could be
- How much fertiliser a Sukulu plant could sell domestically
- What product mix (DAP vs SSP vs TSP) could make sense

13.2 Econometric Demand Projection Model (EDPM)

13.2.1 What the EDPM model does

The **Econometric Demand Projection Model (EDPM)** estimates **future fertiliser demand in Uganda** based on how fertiliser demand has historically behaved as the economy changes.

It answers this policy question:

"Given Uganda's economic growth, credit conditions, and exchange-rate environment, how much fertiliser is Uganda likely to use by 2030 and 2040?"

It then converts that projected fertiliser demand into:

- **P_2O_5 demand (tonnes)**, and
- fertiliser product equivalents: **DAP / TSP / SSP**.

The EDPM complements the ANRM:

- **ANRM** estimates *what Uganda should use* (agronomic need).
- **EDPM** estimates *what Uganda is likely to use* (real market behaviour).

That distinction matters for:

- feasibility sales forecasts,

- plant sizing and utilisation,
- financial viability,
- sensitivity testing and risk analysis.

13.2.2 How the EDPM model was developed

13.2.2.1 Step 1: Use actual Uganda WDI data (not "made-up" series)

From a WDI extract, and the model uses Uganda's actual time-series data from it.

The model uses **only the years where all selected indicators overlap** so the econometric estimation remains consistent. In your dataset, the overlap produced a usable annual window of:

2011–2021 (11 observations)

The model uses the following WDI indicators:

1. **Fertilizer consumption (kilograms per hectare of arable land)**
   - Code: **AG.CON.FERT.ZS**
2. **Arable land (hectares)**
   - Code: **AG.LND.ARBL.HA**
3. **Agriculture, forestry, and fishing value added (constant 2015 US$)**
   - Code: **NV.AGR.TOTL.KD**
4. **Domestic credit to private sector (% of GDP)**
   - Code: **FS.AST.PRVT.GD.ZS**
5. **Real effective exchange rate index (2010 = 100)**
   - Code: **PX.REX.REER**

These are documented in the workbook's **GUIDE** and **DATA_QUALITY** sheets.

13.2.2.2 Step 2: Convert WDI fertiliser intensity into national fertiliser tonnage

WDI provides fertiliser use as **kg per hectare** of arable land. But for project planning, we need **tonnes per year**.

So the model converts as follows:

$$\text{Total fertiliser (tonnes)} = \frac{\text{Fertiliser use (kg/ha)} \times \text{Arable land (ha)}}{1{,}000{,}000}$$

That gives a national fertiliser tonnage series for Uganda (for 2011–2021).

This is important because it aligns the EDPM with:

- volume-based demand forecasting,
- plant sizing and product supply planning.

13.2.2.3 Step 3: Transform variables into a form suitable for econometric estimation

To estimate elasticities (how demand responds to changes in drivers), the EDPM uses a log-log format.

The model computes:

- $\ln(F_t)$: log of fertiliser tonnage
- $\ln(AgVA_t)$: log of agricultural value-added
- $\ln(Credit_t)$: log of credit proxy
- $\ln(REER_t)$: log of real effective exchange rate

A log-log model is used because:

- coefficients become elasticities (easy to interpret),
- relationships become closer to linear,
- the model behaves better for long-run projections.

13.2.2.4 Step 4: Estimate the econometric relationship (the demand equation)

The EDPM then estimates a regression relationship of the form:

$$\ln(F_t) = \alpha + \beta_1 \ln(REER_t) + \beta_2 \ln(AgVA_t) + \beta_3 \ln(Credit_t) + \varepsilon_t$$

Where:

- F_t= Uganda fertiliser use (tonnes)
- $AgVA_t$= agricultural value-added (proxy for farm income and sector scale)
- $Credit_t$= domestic credit proxy (captures financing capacity)
- $REER_t$= real effective exchange rate index (proxy for import cost pressures and macro-price transmission)

The outputs of that estimation are placed in the workbook's **COEFFICIENTS** sheet and used directly in the projection formulas.

Important note for reviewers: This is a **WDI-calibrated model**. It is a valid approach for macro demand estimation, but it is not a substitute for a full URA import-based model if URA annual fertiliser imports are available.

13.2.3 How the projection part works (2025–2040)

13.2.3.1 Step 5: Define three future scenarios

The EDPM does not assume one future. It projects demand under three scenarios:

- **Baseline:** modest growth, weaker credit expansion, limited price advantage
- **Moderate:** aligned to national planning ambition (NDPIV-like trajectory)
- **Accelerated:** higher growth + stronger credit + more favourable conditions

Scenario drivers are stored in the **SCENARIOS** sheet:

- AgVA growth rate
- Credit growth rate
- REER trend (appreciation/depreciation path)

These parameters are editable inputs in the Excel model.

13.2.3.2 Step 6: Project each driver forward from an anchor year

The model anchors projections using the last observed year in the WDI overlap series (anchor year), and grows forward:

$$AgVA_t = AgVA_{anchor} \times (1 + g_{AgVA})^{(t-anchor)}$$
$$Credit_t = Credit_{anchor} \times (1 + g_{Credit})^{(t-anchor)}$$
$$REER_t = REER_{anchor} \times (1 + g_{REER})^{(t-anchor)}$$

This produces annual driver series from 2025 to 2040 for each scenario.

13.2.3.3 Step 7: Forecast fertiliser demand using the estimated equation

For each year and scenario, the model calculates:

$$\ln(F_t) = \alpha + \beta_1 \ln(REER_t) + \beta_2 \ln(AgVA_t) + \beta_3 \ln(Credit_t)$$

Then converts back to tonnes:

$$F_t = \exp(\ln(F_t))$$

This is exactly what you see in the **PROJECTIONS** sheet: driver cells, ln-demand, and converted tonnage.

13.2.4 Converting fertiliser demand to P_2O_5 and fertiliser products

13.2.4.1 Step 8: Convert total fertiliser tonnes to P_2O_5 tonnes

Because the phosphate project needs phosphorus demand, we convert:

$$P2O5_t = F_t \times s$$

Where $s$is the P_2O_5 share of total fertiliser demand (a policy parameter). The model currently uses a default **share of 0.40**, which is editable.

This step is clearly shown in the projections and outputs.

13.2.4.2 Step 9: Convert P_2O_5 demand into product equivalents

For practical planning, the model translates P_2O_5 into product volumes:

- **DAP equivalent:**

$$DAP = \frac{P2O5}{0.46}$$

- **TSP equivalent:**

$$TSP = \frac{P2O5}{0.46}$$

- **SSP equivalent:**

$$SSP = \frac{P2O5}{0.16}$$

These appear in the **OUTPUTS** sheet for 2030 and 2040.

13.2.5 How the model avoids misleading "zeros"

The model should not produce zeros that hide real issues.

So the model was built to:

- generate **positive values** using calibrated coefficients and scenario drivers, and
- avoid false "0" outputs that come from missing inputs.

Instead:

- if a required driver is missing, the sheet shows blanks and flags the issue (data quality approach), and
- when drivers exist, the model always produces meaningful projections.

13.2.6 What the OUTPUTS and DATA_QUALITY sheets do

- **OUTPUTS sheet**

Summarises for each scenario (Baseline/Moderate/Accelerated):

- **2030 P_2O_5 demand**
- **2040 P_2O_5 demand**
- DAP/TSP/SSP equivalents

This is the policy-facing summary for inclusion in the report.

- **DATA_QUALITY sheet**

Shows:

- which WDI indicators were used,
- their codes,
- the year range used (2011–2021),
- count of years (completeness confirmation).

This is important for transparency and DC review.

13.2.7 Limitations (transparent and important for the report)

1. **Short estimation window (2011–2021).** The regression is based on 11 annual observations because that's the overlap in the WDI extract for the required indicators. This is usable for initial calibration but improves significantly with longer time series.
2. **REER and credit are proxies.** They are macro proxies, not direct fertiliser price series or agricultural credit series. This is acceptable for macro modelling but should be strengthened with:
   - URA import unit values (price proxy),
   - Uganda-specific input price indices,
   - BoU credit-to-agriculture data.
3. **P_2O_5 share is an assumption.** The conversion from total fertiliser tonnes to P_2O_5 tonnes uses a parameter. This should be validated using Uganda's import composition (DAP/NPK/TSP shares).

14.1 Scenario 1: Phased Integrated Fertiliser Cluster (Preferred Scenario)

14.1.1 Technical Layout – Phased Integrated Fertiliser Cluster (Preferred Scenario)

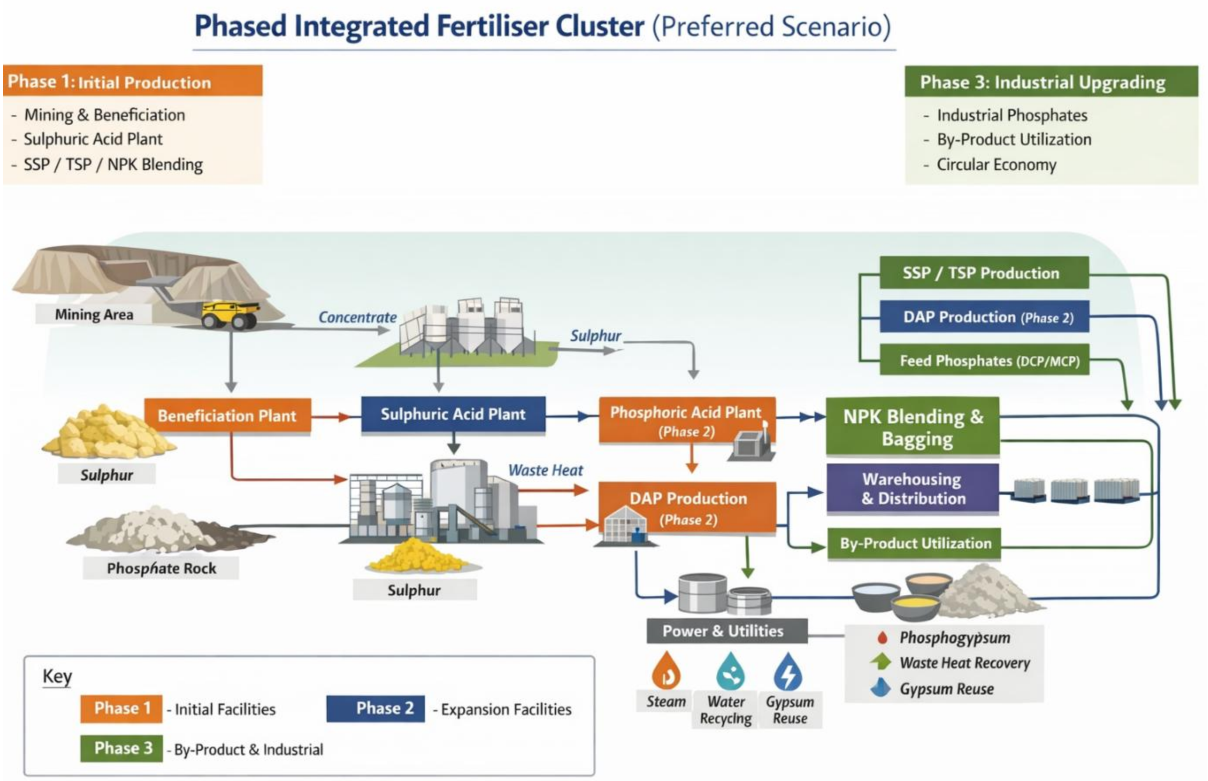
14.1.1.1 Overall Cluster Concept

The Sukulu development is structured as a co-located, modular industrial cluster that integrates mining, beneficiation, chemical processing, fertiliser manufacturing, blending, and shared utilities. The layout minimises internal transport, enables shared infrastructure, and allows incremental expansion without disrupting operations.

Design principles

- ✓ Co-location of high-mass flows (rock, acids, fertilisers)
- ✓ Single-direction material flow (mine → beneficiation → acids → products)
- ✓ Shared utilities and environmental systems
- ✓ Modular units sized for phased expansion
- ✓ Segregated zones for safety, environment, and logistics

Figure 14- 1: Technical Layout – Phased Integrated Fertiliser Cluster



14.1.1.2 Spatial Zoning of the Cluster

The cluster is organised into **five functional zones**, arranged to reflect process flow and safety requirements.

Zone 1: Mining and Raw Materials Zone

- Open-pit phosphate mine
- Overburden dumps and rehabilitation areas
- Mine workshops and fuel storage
- Run-of-mine (ROM) stockpiles

Key interface: short-haul road or conveyor to beneficiation.

Zone 2: Beneficiation and Ore Preparation Zone

- Crushing and screening
- Washing / flotation (as required by ore characteristics)
- Tailings thickener and lined tailings storage
- Process water recovery and recycling

Outputs:

- Beneficiated phosphate concentrate
- Fines management stream

Key interface: covered conveyor or sealed truck transfer to acidulation zone.

Zone 3: Chemical Processing and Utilities Zone

This is the **core industrial zone** of the cluster.

Core units

- Sulphur handling and storage (bulk import, covered storage)
- Sulphuric acid plant
- Phosphoric acid plant (added in Phase 2)

Shared utilities

- Industrial power substation (with redundancy)
- Steam and heat recovery systems
- Industrial water intake, treatment, and recycling
- Effluent treatment and neutralisation
- Central control room

Safety: buffer distances, acid containment, emergency response corridors.

Zone 4: Fertiliser Manufacturing and Blending Zone

- SSP granulation unit (Phase 1)

- TSP and/or DAP granulation units (Phase 2)
- NPK blending and formulation plant
- Bagging, palletising, and bulk loading systems

Design note: granulation lines are parallel and modular, allowing new lines to be added without shutting down existing production.

Zone 5: Logistics, Storage, and Distribution Zone

- Finished product warehouses (bagged and bulk)
- Bulk loading bays
- Truck staging and dispatch yards
- Rail siding provision (where feasible)
- Quality assurance laboratory (UNBS-aligned)

Key interface: national and regional distribution corridors.

14.1.1.3 Phased Technical Configuration

Phase 1: Market Entry and Operational Stabilisation

Installed process units

- Mining and beneficiation plant
- Sulphur handling and sulphuric acid plant
- SSP granulation unit
- NPK blending readiness (infrastructure + equipment)

Products

- SSP
- TSP-ready intermediates (optional)
- Custom NPK blends (using domestic phosphate intermediates)

Technical objective

- Achieve minimum economic scale
- Replace imported bulky phosphate fertilisers
- Establish logistics, QA, and environmental systems

Phase 2: Value Capture and Product Expansion

Additional process units

- Wet-process phosphoric acid plant
- DAP granulation line
- Feed-grade phosphate line (DCP/MCP)

Products

- DAP
- TSP (full scale)
- DCP / MCP

Technical objective

- Improve margins through higher-value products
- Increase flexibility across fertiliser markets
- Strengthen cash flows and utilisation rates

Phase 3: Industrial Upgrading and Circular Economy Integration

Additional units

- Industrial phosphate chemical units (e.g. STPP, water-treatment phosphates)
- Phosphogypsum processing / reuse facilities
- Expanded waste-heat recovery and power integration

Products

- Industrial phosphates
- Construction or industrial by-products

Technical objective

- Long-term value upgrading
- Export-oriented diversification
- Enhanced environmental performance

14.1.1.4 Material and Energy Flow Logic

Primary material flows

- Phosphate rock → beneficiation → acids → fertilisers
- Sulphur → sulphuric acid → phosphoric acid / SSP
- Acids → fertiliser granulation / feed phosphates

Energy integration

- Waste heat from acidulation recovered for steam and internal power
- Centralised energy management reduces operating cost

Water management

- Closed-loop process water system
- Minimal freshwater abstraction
- Compliance with NEMA discharge standards

14.1.1.5 Environmental and Safety Integration

- Lined phosphogypsum stacks with future reuse pathways
- Centralised effluent treatment plant
- Stack emissions control (SO_2, fluorides, particulates)
- Safety buffer zones between acid plants and other units
- Emergency response and spill containment systems

Environmental systems are designed **once**, then scaled with production.

14.1.1.6 Expandability and Modularity

The layout explicitly allows:

- Addition of parallel granulation lines
- Expansion of acid capacity without relocation
- Introduction of new downstream units within the same cluster

This avoids stranded infrastructure and preserves capital efficiency.

14.1.1.7 Why This Layout Is Preferred

Compared to stand-alone or export-led configurations, this layout:

- Minimises delivered cost to market
- Maximises domestic value addition and employment
- Reduces logistics and operational risk
- Supports PPP participation through clear asset boundaries
- Aligns directly with EDPM demand phasing and NDP IV objectives

14.1.1.8 Output to Pre-Feasibility Stage

This technical layout provides the basis for:

- Block flow diagrams and site plans
- Preliminary CAPEX/OPEX estimation by phase
- Environmental and social screening
- PPP packaging and risk allocation
- Option comparison in the pre-feasibility study

14.2 Scenario 2: Beneficiation and Superphosphate-First Model (Transitional Scenario)

14.2.1 Overall Concept

Scenario 2 establishes a **lean, beneficiation-led fertiliser complex** focused on **SSP and TSP production**, supported by **sulphuric acid as the anchor chemical**, without immediate phosphoric acid or DAP units. The layout is designed to:

- achieve early import substitution,
- stabilise domestic supply,
- build operational capability, and
- preserve physical space and interfaces for later expansion into Scenario 1.

This scenario is particularly suitable where **capital mobilisation is staged** or where early demonstration of commercial viability is required.

14.2.2 Functional Zoning and Site Layout

The site is organised into **four functional zones**, with reserved corridors for future expansion.

14.2.2.1 Zone 1: Mining and Raw Materials Zone

- Open-pit phosphate mining area
- Overburden dumps and progressive rehabilitation
- Mine workshops, fuel storage, and weighbridges
- Run-of-mine (ROM) stockpiles

Interface: short-haul road or conveyor to beneficiation.

14.2.2.2 Zone 2: Beneficiation and Ore Preparation Zone

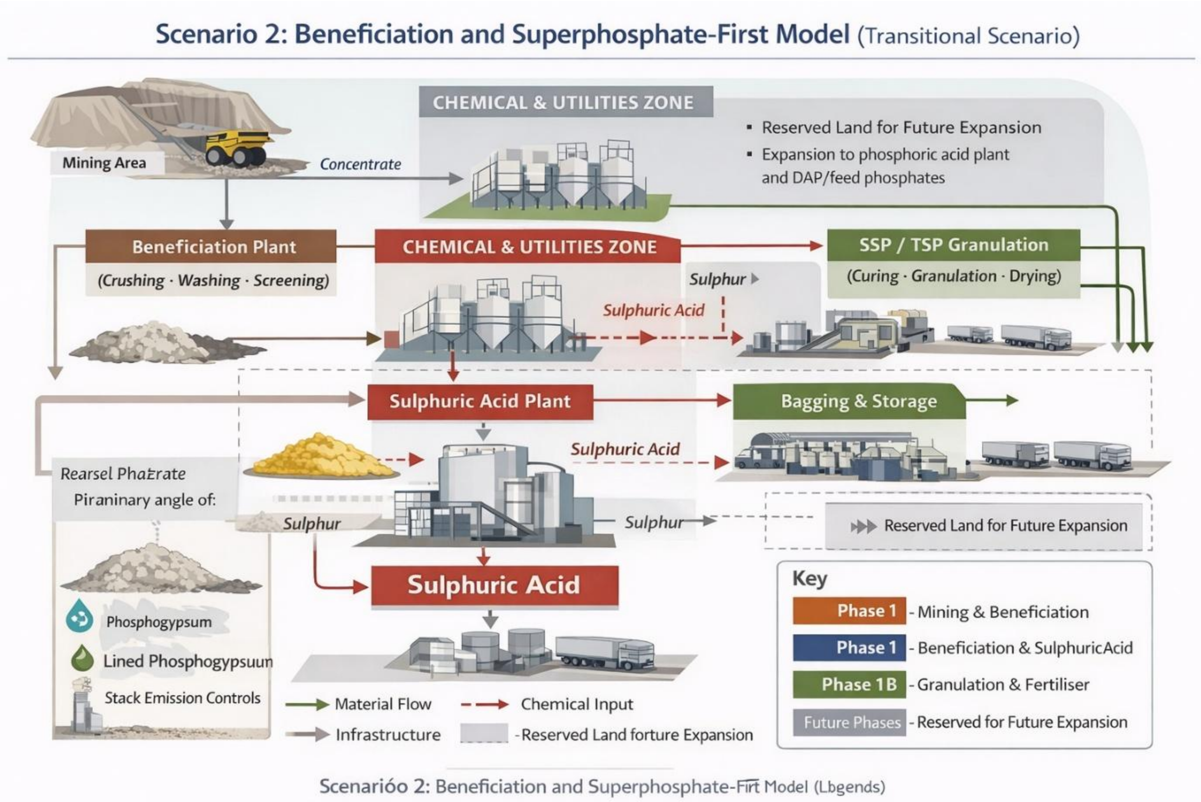
- Primary crushing and screening
- Washing and classification (as dictated by ore characteristics)
- Tailings thickener and lined tailings storage facility
- Process water recovery and recycling system

Outputs:

- Beneficiated phosphate rock suitable for acidulation
- Fines stream managed within tailings system

Design note: beneficiation capacity is sized to support SSP/TSP demand plus contingency for future phosphoric acid feed.

Figure 14- 2: Scenario 2: Beneficiation and Superphosphate-First Model



14.2.2.3 Zone 3: Chemical Processing and Utilities Zone (Anchor Zone)

This is the **technical core** of Scenario 2.

Installed units:

- Bulk sulphur receiving and covered storage
- Sulphur melting and handling systems
- **Sulphuric acid plant** (single-train, modular design)

Shared utilities:

- Power substation with redundancy
- Industrial water intake, treatment, and recycling
- Effluent neutralisation and fluoride control
- Central control room and laboratory

Design note: the sulphuric acid plant is deliberately designed as a **future shared utility** for phosphoric acid and DAP units.

14.2.2.4 Zone 4: Fertiliser Manufacturing, Storage, and Dispatch

- **SSP granulation unit** (primary product line)
- **TSP granulation unit** (optional or Phase-1B addition)
- Cooling, screening, and product sizing
- Bagging and palletising lines
- Finished-goods warehouses and truck dispatch yards

Products:

- Single Super Phosphate (SSP)
- Triple Super Phosphate (TSP)

14.2.3 Phase Configuration and Expansion Logic

14.2.3.1 Phase 1A – Initial Operations

Installed:

- Mining and beneficiation
- Sulphuric acid plant
- SSP granulation and bagging

Objective:

- Rapid import substitution for bulky phosphate fertilisers
- Early revenue generation
- Operational learning and market confidence building

14.2.3.2 Phase 1B – Product Deepening (Optional within Scenario 2)

Added:

- TSP granulation line (using sulphuric acid and beneficiated rock)
- Expanded storage and distribution capacity

Objective:

- Serve higher-analysis fertiliser market segments
- Improve margins without full acidulation complexity

14.2.3.3 Future Upgrade Path (Transition to Scenario 1)

Scenario 2 preserves:

- land parcels,
- pipe racks,
- utility corridors,
- control-system capacity,

to allow seamless addition of:

- phosphoric acid plant,
- DAP granulation,
- feed-grade phosphates.

14.2.4 Material, Energy, and Utility Flows

Material flow:

- Phosphate rock → beneficiation → sulphuric acid reaction → SSP/TSP
- Sulphur → sulphuric acid → fertiliser granulation

Energy integration:

- Heat recovery from sulphuric acid plant used for:
    - internal steam,
    - drying and granulation support,
    - utility heating.

Water management:

- Closed-loop process water system
- Minimal freshwater abstraction
- Compliance with NEMA discharge requirements

14.2.5 Environmental and Safety Systems

- Lined phosphogypsum storage areas (from SSP/TSP production)
- Acid containment bunds and spill-response systems
- Stack emissions control (SO_2, particulates)
- Separation buffers between chemical units and storage/logistics areas

Environmental infrastructure is sized to **accommodate later expansion** without redesign.

14.2.6 Technical Advantages of Scenario 2

- **Lower initial capital intensity** than full integration

- **Simpler process control** and reduced operational risk
- Strong delivered-cost advantage for SSP/TSP
- Faster implementation timeline
- Clear and low-risk pathway to Scenario 1

14.2.7 Limitations and Design Mitigation

Limitations:

- Lower margins than DAP-based systems
- Partial value capture relative to full integration

Mitigation:

- Maximise plant utilisation and logistics efficiency
- Integrate blending readiness and distribution scale
- Design sulphuric acid plant as a future multi-user utility

14.2.8 Output to Pre-Feasibility Stage

This technical layout enables the pre-feasibility study to:

- Develop block flow diagrams and preliminary site plans
- Prepare Phase-1 CAPEX and OPEX estimates
- Conduct environmental and social screening
- Test financial viability under conservative demand scenarios
- Compare Scenario 2 objectively against Scenario 1 in the options analysis

Scenario 2 provides a **credible transitional technical configuration** for Sukulu: one that is **technically robust, financially cautious, and strategically reversible**. It delivers early market impact and operational credibility while keeping Uganda firmly on the pathway toward a **fully integrated fertiliser industrial cluster** when demand, capital, and institutional readiness converge.

14.3 Scenario 3: Mining and Export-Led Model (Non-Preferred Scenario)

14.3.1 Overall Concept

Scenario 3 focuses on **phosphate rock extraction, limited beneficiation, and bulk export** with minimal downstream processing. The configuration is designed to:

- maximise early export volumes,

- minimise initial processing complexity and capital expenditure, and
- rely on international markets for value realisation.

This model resembles the **Jordan (JPMC)** approach and parts of **South Africa's Foskor export streams**, but is structurally disadvantaged in Uganda due to **landlocked geography and strong domestic fertiliser demand**.

14.3.2 Functional Zoning and Site Layout

The site is organised into **three main zones**, with logistics dominating the configuration.

Zone 1: Mining and Raw Materials Zone

- Open-pit phosphate mine
- Overburden dumps and progressive rehabilitation
- Run-of-mine (ROM) stockpiles
- Mine workshops, weighbridges, and internal haul roads

Interface: short-haul transport to beneficiation plant.

Zone 2: Beneficiation and Export Preparation Zone

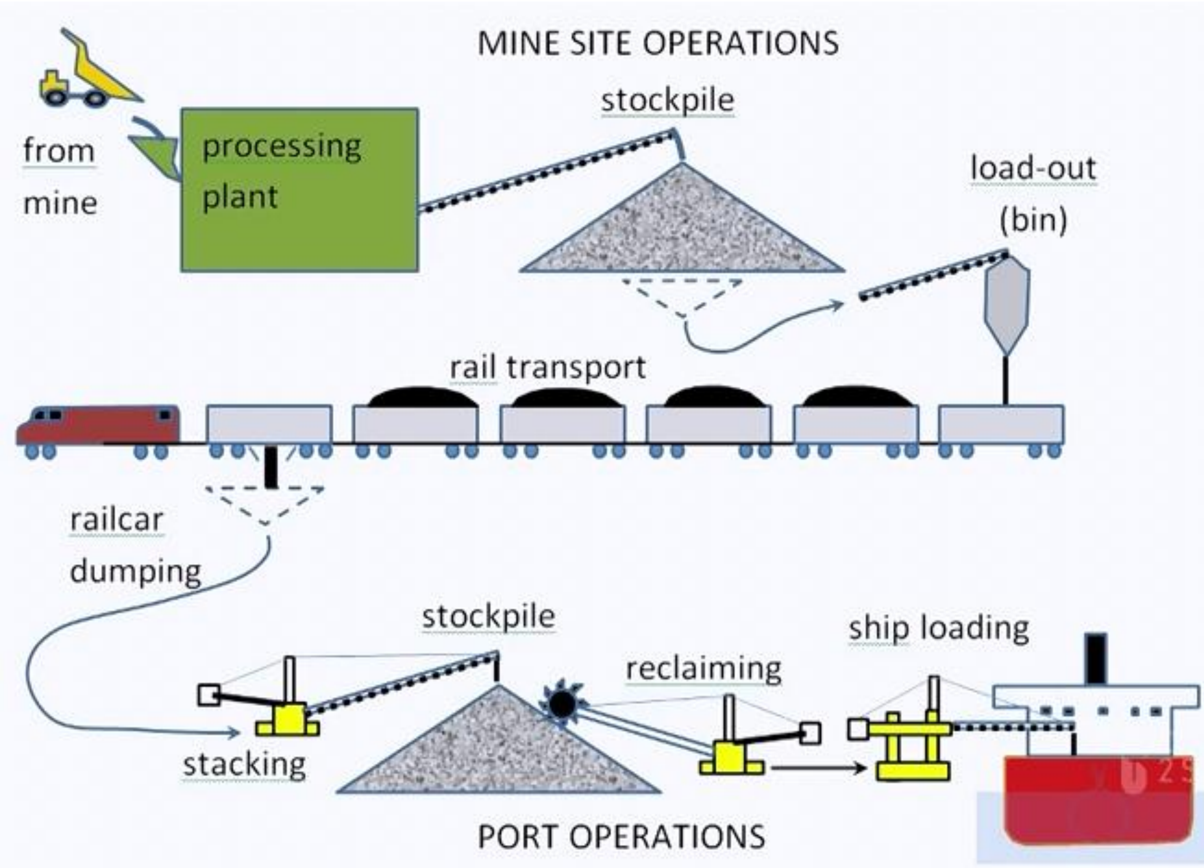
- Crushing and screening
- Basic washing/classification to meet export specifications
- Limited grade control and blending of rock
- Dewatering and drying (as required for export handling)

Outputs:

- Export-grade phosphate rock (beneficiated)
- Tailings managed in lined storage facilities

Design note: beneficiation is sized only to meet **minimum export quality**, not optimised for acidulation efficiency.

Figure 14- 3: Scenario 3: Mining and Export-Led Model



Zone 3: Export Logistics and Bulk Handling Zone

- Covered bulk stockpiles for export rock
- Truck or rail loading facilities
- Weighbridges and customs inspection areas
- Logistics interface to:
    - Mombasa or Dar es Salaam ports, or
    - regional cross-border bulk markets

No acidulation, fertiliser manufacturing, or blending facilities are installed in this scenario.

14.3.3 Utilities and Infrastructure

Installed infrastructure:

- Mine access roads and haulage routes
- Basic power supply for mining and beneficiation
- Limited water supply for washing and dust suppression
- Tailings management facilities

Not installed:

- Sulphuric acid plant
- Phosphoric acid plant
- Fertiliser granulation or blending
- Advanced waste-heat or circular-economy systems

14.3.4 Environmental and Safety Systems

- Lined tailings storage facilities
- Dust suppression systems at mine and beneficiation
- Progressive mine rehabilitation
- Basic stormwater and runoff management

Note: environmental footprint per tonne of value added is higher than in integrated scenarios due to limited downstream value capture.

14.3.5 Technical Advantages of Scenario 3

- Lowest initial processing complexity
- Lower upfront CAPEX than integrated scenarios
- Faster start-up for mining operations
- Technically straightforward mining and beneficiation

14.3.6 Technical and Strategic Limitations

Despite technical feasibility, Scenario 3 has major limitations in the Ugandan context:

- **High delivered-cost disadvantage** due to long inland transport to ports
- **High exposure to global phosphate rock price volatility**
- **No domestic fertiliser supply impact**, despite strong ANRM/EDPM demand
- **Limited employment and industrial spillovers**
- **No platform for downstream upgrading** unless re-investment occurs

14.3.7 Upgrade and Transition Constraints

While theoretically upgradeable, Scenario 3 creates **path-dependency risks**:

- Site layouts prioritise export logistics, not chemical processing
- Retrofitting acidulation and fertiliser plants later is costlier

- Early export contracts may crowd out domestic supply

14.3.8 Why This Scenario Is Non-Preferred for Sukulu

Based on:

- EDPM-confirmed domestic demand growth,
- delivered-cost disadvantages of landlocked exports, and
- benchmarking lessons from Jordan and South Africa,

Scenario 3 is **not recommended as the core development model** for Sukulu. It may only be justified:

- as a **temporary early-stage option**, or
- as a **residual outlet for surplus rock** once domestic demand is met.

14.3.9 Output to Pre-Feasibility Options Analysis

This technical layout allows Scenario 3 to be:

- clearly defined and costed,
- objectively compared against Scenarios 1 and 2, and
- screened out or downgraded on economic, financial, and development criteria.

Scenario 3 represents a **technically feasible but strategically weak** configuration for Sukulu. While it minimises early processing complexity, it **fails to leverage Uganda's domestic fertiliser demand**, delivers limited value addition, and exposes the project to higher price and logistics risk. As such, it is appropriately classified as a **non-preferred reference scenario** for options analysis rather than a development anchor.

14.4 Comparative Assessment of Development Scenarios for Sukulu

14.4.1 Overview of the Three Scenarios

Table 14- 1: Overview of the Three Scenarios

Scenario	Core Description	Strategic Intent
Scenario 1: Phased Integrated Fertiliser Cluster (Preferred)	Mining + beneficiation + sulphuric acid + phosphoric acid + fertilisers + blending + circular economy	Maximise domestic value addition, competitiveness, and long-term industrialisation

Scenario	Core Description	Strategic Intent
Scenario 2: Beneficiation and Superphosphate-First (Transitional)	Mining + beneficiation + sulphuric acid + SSP/TSP, with future upgrade path	Lower-risk market entry with clear pathway to full integration
Scenario 3: Mining and Export-Led (Non-Preferred)	Mining + limited beneficiation + bulk export	Short-term monetisation with minimal value addition

14.4.2 Technical Scope Comparison

Table 14- 2: Technical Scope Comparison

Element	Scenario 1	Scenario 2	Scenario 3
Mining	✓	✓	✓
Beneficiation	✓ (full)	✓ (full)	✓ (limited/export grade)
Sulphuric acid	✓	✓	✕
Phosphoric acid	✓ (Phase 2)	✕ (reserved only)	✕
Fertiliser manufacturing	✓ (SSP, TSP, DAP, NPK)	✓ (SSP, TSP)	✕
Blending	✓	Readiness only	✕
Circular economy systems	✓ (Phase 3)	Limited	✕

14.4.3 Demand Alignment (ANRM & EDPM)

Table 14- 3: Demand Alignment (ANRM & EDPM)

Criterion	Scenario 1	Scenario 2	Scenario 3
Alignment with agronomic need (ANRM)	High	Medium	Low
Alignment with market demand (EDPM)	High	Medium–High	Low
Domestic fertiliser supply impact	Very high	High	None

Criterion	Scenario 1	Scenario 2	Scenario 3
Demand risk	Low (phased)	Low–Medium	High (export volatility)

Interpretation: Scenarios 1 and 2 align with Uganda's **confirmed domestic fertiliser demand growth**. Scenario 3 ignores this demand despite strong evidence from EDPM.

14.4.4 Cost and Competitiveness

Table 14- 4: Cost and Competitiveness

Dimension	Scenario 1	Scenario 2	Scenario 3
Delivered-cost competitiveness	Strong	Strong (SSP/TSP)	Weak (long logistics)
Exposure to freight & port costs	Low	Low	Very high
Exposure to global price volatility	Low–Medium	Medium	High
Ability to stabilise domestic prices	High	Medium	None

Interpretation: Scenario 3 is structurally disadvantaged in a **landlocked country**. Scenarios 1 and 2 exploit delivered-cost advantages.

14.4.5 Capital Intensity and Risk Profile

Table 14- 5: Capital Intensity and Risk Profile

Aspect	Scenario 1	Scenario 2	Scenario 3
Initial CAPEX	High (phased)	Medium	Low
Technical complexity	High (phased)	Medium	Low
Implementation risk	Medium	Low–Medium	Low
Upgrade flexibility	High	High	Low
Risk of stranded assets	Low	Low	High

Interpretation: Scenario 2 offers the **lowest-risk entry**, while Scenario 1 offers the **best long-term return**. Scenario 3 creates **path-dependency risks**.

14.4.6 Economic and Development Impact

Table 14- 6: Economic and Development Impact

Impact Area	Scenario 1	Scenario 2	Scenario 3
Domestic value addition	Very high	Medium	Low
Employment (direct & indirect)	High	Medium	Low
Skills and industrial upgrading	High	Medium	Minimal
Import substitution	High	Medium–High	None
Regional leadership potential	High	Medium	Low

14.4.7 Environmental, Social, and ESG Performance

Table 14- 7: Environmental, Social, and ESG Performance

Dimension	Scenario 1	Scenario 2	Scenario 3
Environmental control systems	Strong	Adequate	Basic
Circular economy potential	High	Medium	None
Social licence & inclusion	Strong	Medium	Weak
Green / climate finance eligibility	High	Medium	Low

14.4.8 Governance and Bankability

Table 14- 8: Governance and Bankability

Criterion	Scenario 1	Scenario 2	Scenario 3
Suitability for PPP/JV	High	High	Low
Fiscal risk management	Good (phased)	Good	Poor
Long-term policy alignment (NDP IV)	Very high	High	Weak

Criterion	Scenario 1	Scenario 2	Scenario 3
Investor confidence	High	Medium–High	Low

14.4.9 Overall Ranking and Recommendation

Table 14- 9: Overall Ranking and Recommendation

Rank	Scenario	Conclusion
1st (Preferred)	**Scenario 1**	Best balance of competitiveness, demand alignment, value addition, and long-term industrial transformation
2nd (Transitional)	**Scenario 2**	Strong low-risk entry point with clear upgrade path to Scenario 1
3rd (Non-Preferred)	**Scenario 3**	Technically feasible but strategically weak for Uganda's context

Scenario 1 should be taken forward as the **preferred option** into pre-feasibility and feasibility, with phased implementation.

Scenario 2 is a **credible transitional option** where capital or implementation risk needs to be staged, provided it is contractually locked into an upgrade pathway.

Scenario 3 should be retained **only as a reference or residual outlet**, not as a development anchor.

This comparison provides a **clear, defensible basis** for options screening and selection under the DC Guidelines.